

Information and Financial Stability in Money Markets*

Levent Altinoglu[†] Jin-Wook Chang[‡]

March 6, 2026

Abstract

We develop a theory of financial crises in which liquidity conditions in money markets generate endogenous uncertainty about the quality of underlying assets. Agents can only partially infer their counterparties' private information from the availability of funding in the money market. This gives rise to information spillovers which imply that a fall in funding liquidity increases perceptions of downside risk in the asset payoff. Information spillovers both improve the allocation of liquidity during normal times, but also amplify market fragility. During financial stress, extraordinary central bank lending facilities reduce uncertainty about asset quality by reducing the informativeness of asset prices.

JEL classification: D53, D83, E58, G01, G18

Keywords: financial stability, money markets, learning, fire sales, lender of last resort, repos

*This paper was previously circulated under the title “Information Externalities, Funding Liquidity, and Fire Sales.” We would like to thank Danny Barth, Stefano Battiston, Francesca Carapella, Ding Dong, Jeongmin “Mina” Lee, Carlos Ramírez, Joseph Stiglitz, and Alexandros Vardoulakis for helpful discussions and comments. We also thank conference and seminar participants at the Sydney Banking and Financial Stability Conference, University of Sydney, the Reserve Bank of Australia, the INET Workshop at Columbia University, the Festschrift in honor of Joseph Stiglitz at the Università Cattolica del Sacro Cuore in Milan, and the Federal Reserve Board. The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as the views of the Board of Governors of the Federal Reserve System or of anyone else associated with the Federal Reserve System.

[†]Board of Governors of the Federal Reserve System. Email: levent.altinoglu@gmail.com

[‡]Board of Governors of the Federal Reserve System. Email: jin-wook.chang@frb.gov.

1 Introduction

Money markets—markets for short-term collateralized debt, such as repo, wholesale funding, and other forms of private money—are central to the modern financial system. In normal times, money markets are a stable source of funding, facilitating an efficient allocation of liquidity (Chen et al., 2019; Sinclair, 2023). But occasionally these markets are subject to highly disruptive bouts of sudden and extreme stress, such as those during the 2008 financial crisis or in March 2020, which trigger fire sale dynamics and an adverse interaction between funding and market illiquidity (Brunermeier and Pedersen, 2009; Acharya and Merrouche, 2013; Pérignon et al., 2018; Gorton et al., 2020).

These episodes also typically feature an acute rise in market participants’ uncertainty about the quality of underlying assets. For example, during the 2008 financial crisis, reduced repo market liquidity was accompanied by a sudden but temporary rise in the *perceived riskiness* of the mortgage-backed securities (MBS) held as underlying collateral. A key feature was the spike in uncertainty about the quality of residential MBS being traded in repo markets (Gorton and Metrick, 2012).¹ During such episodes, central banks have resorted to extraordinary interventions to stabilize money markets, including lending directly against a broad range of risky assets.² Moreover, the mere announcement of such policy interventions is often sufficient to restore confidence in these assets and stability, even after the interventions expire.³ Together, these historical episodes suggest that a deterioration in money market liquidity can worsen *perceptions* of risk in underlying assets.

The literature has made substantial progress in understanding these dynamics by studying the role of beliefs in financial markets. For instance, there is a vibrant literature on the intersection of dispersed information and financial stability, including Gorton and Ordonez (2014), Asriyan and Vanasco (2014), Asriyan (2021), Asriyan et al. (2022), Dávila and Walther (2023), Garcia-Macia and Villacorta (2023), and Bostanci and Ordoñez (2024). There is also a literature studying in more general terms asset markets with information frictions and learning, including Simsek (2013), Kurlat (2016), Asriyan et al. (2017), Vanasco (2017), Dávila and Parlato (2021), and

¹These dynamics are exacerbated by the limited price transparency across segmented repo markets (Kahn et al., 2023; Chang et al., 2025), making it difficult for market participants to observe liquidity conditions broadly or determine the underlying shocks that give rise to spikes in repo rates. Evidence suggests that this lack of information contributes to an environment of heightened uncertainty.

²For example, while standard liquidity injections by the central bank, such as the discount window and open market operations, involve lowering short-term interest rates and/or lending against a relatively narrow set of securities with minimal credit risk, the facilities implemented by central banks in 2020 lent against a broader set of risky assets, such as corporate bonds, sometimes requiring emergency authority to do so.

³For example, during COVID-19, the announcement of the Secondary Market Corporate Credit Facility (SMCCF) stabilized asset prices with minimal actual take-up. Evidence shows that this announcement effectively reduced the risk aversion of market participants (Haddad et al., 2021; Gilchrist et al., 2024; Haddad et al., 2025).

Asriyan et al. (2021). Finally, a large literature studies asset prices and fire sales in a context of financial frictions, including Lorenzoni (2008), Brunnermeier and Pedersen (2009), Acharya et al. (2011), and Malherbe (2014).

Yet open questions remain: How are money markets a stable source of funding during normal times yet so fragile at the same time? If the price-signal mechanism functions properly, why do perceptions of risk in fundamentals swing so wildly during episodes of stress? Why do central banks resort to extraordinary lending facilities to stabilize markets during such episodes, and why are the announcement effects of such interventions so powerful and persistent?

To help shed light on these questions, we develop a theory of financial crises in which liquidity conditions in money markets shape *perceptions* about downside risk in asset payoffs. In the model, short-term collateralized debt plays two roles: It allocates liquidity between lenders and borrowers, and it conveys agents' private information about underlying assets through the price-signal mechanism—a mechanism we refer to as an *information spillover*.

However, because markets are incomplete, the price of debt cannot perform both roles perfectly—for any asset, there is only one price to perform two roles. As a result, agents cannot fully disentangle whether a change in the availability of funding, given by the price of debt at date 1, reflects news about the asset or orthogonal liquidity shocks. That is, information spillovers are imperfect. Consequently, funding illiquidity increases agents' *perceived* downside risk in asset payoffs, as agents cannot rule out that a fall in the availability of funding is due to bad news about the underlying asset. Perceptions of risk are therefore fundamentally linked to the availability of liquidity, even when liquidity conditions are orthogonal to asset fundamentals. In turn, perceptions of risk affect the likelihood and severity of fire sales of the underlying assets.

The model yields three over-arching results. First, during normal times, money markets function well. The use of short-term collateralized debt gives rise to (imperfect) information spillovers through which the availability of funding from the lender shapes the borrower's beliefs about the quality of the asset. These information spillovers facilitate an efficient reallocation of liquidity between the borrower and lender in response to liquidity shocks and private news about the asset, provided they are not too adverse.

Second, money markets exhibit fragility in part due to how imperfect information spillovers shape endogenous perceptions of risk. In response to adverse liquidity shocks, the imperfect information spillover worsens the borrower's perception of downside risk in the asset fundamental. This effect increases the likelihood and severity of fire sales of assets in response to adverse liquidity shocks. Thus, the information spillover has a destabilizing effect on the market for such shocks. In response to adverse news about asset quality, however, the imperfect information spillover reduces perceptions of downside risk, and thus has a stabilizing effect on the market.

Taken together, these results suggest that information spillovers are a double-edged sword,

serving to both improve the allocation of liquidity during normal times, but also amplify market fragility in response to some shocks through their effects on perceptions of downside risk.

Third, the model shows that, during financial distress, standard liquidity injections by the central bank—through lower interest rates or discount window lending—may be insufficient to restore stability to the money market. In contrast, extraordinary liquidity facilities in which the central bank lends directly against risky assets are a more powerful tool to stabilize the market because of how they affect the information transmission of prices through a new ‘confidence channel’. By lending against risky assets at a government-administered interest rate, such liquidity facilities reduce the informativeness of the availability of funding about asset quality, disrupting the price-signal mechanism in the money market. Counterintuitively, this can have a stabilizing effect on the market by cutting the link between tight funding liquidity and rising perceptions of risk.

Our theory is motivated in part by the nature of various financial markets, such as repo markets, markets for asset-backed commercial paper (ABCP), and over-the-counter (OTC) markets for securities and derivatives, where short-term collateralized debt is used to finance investment in a longer-maturity asset. These markets have grown in importance in recent years and have featured occasional episodes of severe market disruption featuring liquidity dry-ups and panic-driven asset sales.

We build a three-period model with dispersed information, learning, and fire sales. In the initial period (date 0), a borrower issues short-term debt to a lender in order to finance investment in a long-term risky asset, which matures at the final period (date 2). Agents have a common prior belief about the date 2 return of the risky asset. The debt matures at the interim period (date 1) and is collateralized by the borrower’s holdings of the risky asset. At date 1, the borrower must refinance this debt with the lender, potentially under new terms, or liquidate some of its holdings of the risky asset to cover any shortfall in its funding.

At date 1, each agent receives a private, noisy signal about the asset’s future (date 2) return. In addition, the agent is simultaneously subject to an idiosyncratic shock to the opportunity cost of its funds at date 1 (a ‘cost shock’), which is orthogonal to the asset’s future return and which affects its willingness to roll over the debt at date 1. Both the cost shock and the signal received by the agent are private information. For simplicity, we assume that only one agent receives both a cost shock and a private signal, while the other agent receives neither.⁴ While much of our exposition here focuses on the case in which the lender receives the signal and shock, we show in section 7 that the same insights apply for the other case even though the mechanics differ slightly.

⁴This assumption makes it more tractable to solve for agents’ equilibrium beliefs. If, in contrast, both agents received private signals, then the equilibrium would depend not only on how one agent, e.g., the borrower, updates its beliefs in response to the actions of the other, but also on an infinite feedback loop in which the borrower’s actions also affect the lender’s beliefs, etc. This would make an analytical characterization of equilibrium belief formation extremely difficult.

This general setup can be applied to a number of real-world settings. For example, hedge funds often fund their holdings of risky assets, such as mortgage-backed securities, by borrowing from prime brokers in the repo market—that is, by effectively issuing short-term debt to the broker using the risky asset as collateral. Typically, there is dispersed information about the fundamental value of the risky asset being traded: The hedge fund and prime broker may each receive private information about conditions in various housing markets, which may lead them to have different valuations of the underlying mortgage-backed securities at a given point in time. Moreover, market segmentation across repo markets prevents a market participant from observing the private information of other participants. Nevertheless, there is evidence that hedge funds try to infer the private information of their brokers through their dealings with them (Duffie, 2010), similar to the information spillover in our model, and that this mechanism contributed to the failure of Bear Stearns in 2008.⁵ We explore this and other relevant applications in section 1.1.

In response to its private signal, an agent updates its prior belief about the risky asset’s date 2 return (the fundamental) according to Bayes’ Rule, and can reallocate its portfolio at date 1 given its posterior beliefs. In response to either an adverse cost shock or negative news about the risky asset’s return, the lender is less willing to refinance the borrower’s debt at date 1, since the debt is collateralized by the risky asset.⁶ This results in a contraction in the supply of credit at date 1. Therefore, the equilibrium price of debt conveys information to the borrower about the lender’s private information.

However, asset prices only *partially* convey private information in this environment. The borrower cannot perfectly disentangle the lender’s private news about the fundamental and its cost shock by looking at the price of debt, because both news and the cost shock affect the price of debt simultaneously.⁷ We refer to this as the ‘identification problem’ faced by the borrower.⁸

This environment leads to the key mechanism of the model: Beliefs about fundamentals are endogenously shaped by the availability of liquidity in funding markets due to an information spillover whereby asset prices affect agents’ beliefs about the fundamental. To illustrate this, suppose the lender’s opportunity cost of funds rises at date 1, causing it to reduce the supply of new

⁵See [Financial Crisis Inquiry Commission \(2011\)](#) and <https://elischolar.library.yale.edu/cgi/viewcontent.cgi?article=5929&context=ypfs-documents>.

⁶Similarly, for the borrower, an adverse cost shock or negative news reduces its willingness to hold the risky asset as opposed to cash at date 1.

⁷Each agent faces two layers of uncertainty when forming its posterior beliefs about the asset’s fundamental: its own noisy signal about the fundamentals, and asset prices as a noisy signal about its counterparty’s private signal. To form beliefs about the counterparty’s signal, it computes the likelihood of all possible signal and cost shock realizations consistent with observed asset prices.

⁸This identification problem derives from the incompleteness of financial markets, which implies that the price of each asset reflects two unknowns. The underlying assumption is that agents can neither observe one another’s private information, nor trade securities contingent on private information. This assumption is a stand-in for the various frictions which imply that financial markets may not be informationally efficient and risk markets are incomplete.

credit to the borrower. (In other words, the shock makes it more expensive at date 1 for the borrower to roll over its initial debt.) Because the borrower cannot perfectly infer both the cost shock and the lender’s private signal from the price of its new debt, it does not know whether the lender reduced its supply of credit because it faces a higher cost of funds or because it received bad news about the future return of the risky asset, which ultimately serves as collateral for the loan. Therefore, the borrower puts some weight on this latter possibility, and assigns a greater subjective probability to the possibility that the asset’s future payoff will be low. Thus, the identification problem implies that perceptions of risk, and beliefs about asset fundamentals more generally, are shaped by the availability of liquidity in funding markets.⁹

To study the financial stability implications of this mechanism, we introduce the possibility of fire sales of the risky asset at date 1, similar to [Lorenzoni \(2008\)](#): The borrower may sell its holdings of the risky asset to an outside traditional sector which has a lower marginal valuation of the asset.

The equilibrium at date 1 features two regimes: a ‘normal regime’ in which the risky asset continues to be held only by the borrower, and a ‘fire sale regime’ in which the borrower sells some of its holdings of the risky asset to the traditional sector at a fire sale price. The normal regime obtains when cost shocks and news about the risky asset received by the lender are not too bad, so that the lender does not contract the supply of credit too much.

In the normal regime, the money market facilitates a constrained efficient reallocation of liquidity in response to news and cost shocks. As the supply of credit falls, either due to bad news or a bad cost shock, the borrower’s perceived downside risk in the asset worsens due to the imperfect information spillover. Although this pessimism reduces desired holdings of the risky asset, it is not sufficient to cause the borrower to sell the asset in the normal regime. As a result, there is simply a reallocation of cash between the lender and borrower at date 1 owing to the reduced supply of credit. A planner who does not observe the lender’s private information cannot improve upon the date 1 allocation in the normal regime.

The fire sale regime obtains when the contraction in the lender’s supply of credit is sufficiently severe, owing to a severely adverse cost shock or very bad news about the risky asset. Fire sales can be of two types. In a *liquidity-driven fire sale*, the credit crunch at date 1 forces the borrower to fire-sell some of its holdings to the traditional sector in order to repay its debt. This is essentially the standard form of fire sales in the literature, in which fire sales arise from insufficient funding to service debt due to binding liquidity and borrowing constraints.¹⁰

In contrast to the literature, however, our model shows that fire sales can also be driven by (en-

⁹In section 7, we show that similar insights hold for the opposite case in which the borrower receives a cost shock and signal. In particular, market illiquidity, defined as a low risky asset price at date 1, may cause the lender to become over-optimistic.

¹⁰E.g., [Lorenzoni \(2008\)](#).

ogenous) perceptions of risk. In general, tighter funding liquidity (i.e., the contraction in lending at date 1) increases the borrower's perception of downside risk in the asset's future payoff due to the imperfect information spillover. This higher perceived risk lowers the borrower's valuation of the risky asset. If the borrower's perceptions of downside risk deteriorate sufficiently, this triggers a *belief-driven fire sale* of the risky asset in which the borrower is willing to fire-sell some of its holdings of the risky asset to the traditional sector in order to reduce its exposure to the risky asset. Because of the imperfect information spillover, a belief-driven fire sale can occur even if the underlying shock to the lender was orthogonal to the asset payoff. The nature of such fire sales is different from much of the literature in that they are driven by endogenous perceptions of downside risk.

To isolate the role of the identification problem and the imperfect information spillover, we compare the equilibrium to that of a counterfactual economy in which all information is *publicly* observed by all agents, including the underlying cost shock to the lender and the noisy signal about the asset's date 2 payoff.¹¹ In this counterfactual, which we call the 'common information benchmark,' there is no identification problem, so the borrower's perceptions of downside risk in the asset payoff are in line with the true signal. Therefore, comparing our baseline economy to this counterfactual allows us to isolate how the imperfect information spillover affects financial stability by altering agents' perceptions of downside risk.

Comparing the equilibrium allocation of the baseline economy to that in the common information benchmark reveals that, while the imperfect information spillover increases financial fragility in response to certain shocks, it reduces it for others. In response to severely adverse liquidity shocks, the imperfect information spillover increases the fragility of the money market by amplifying both the likelihood and the severity of fire sales relative to the common information benchmark. This is because the information spillover leaves the borrower more pessimistic about the asset's quality in response to an adverse liquidity shock. As a result, it enlarges the region of the state space in which a fire sale occurs, making fire sales more likely compared to the common information benchmark.¹² Moreover, conditional on a fire sale occurring at date 1, this greater pessimism results in more of the risky asset being liquidated and at a lower price.

By contrast, in response to very bad news about the quality of the risky asset, the imperfect information spillover dampens the likelihood and severity of a fire sale by lowering perceptions of downside risk. Borrowers are less likely to fire-sell the risky asset, and if they do, they sell less

¹¹Note that information is still incomplete in this benchmark: Agents still receive noisy signals about the asset's date 2 return. The difference is that signals are publicly observable rather than private, implying that agents have a common information set.

¹²More precisely, the likelihood of a fire sale conditional on a particular cost shock, and evaluated over the possible news shocks, is higher under the model with imperfect information spillovers than in the common information benchmark.

and at a higher price. Thus, financial markets overreact to negative liquidity shocks but are less responsive to bad news about fundamentals, because of how these shocks differentially affect the borrower's beliefs relative to the common information benchmark.

The confluence of these results yields the key positive takeaway from the theory: The information spillovers which arise from the use of short-term collateralized debt in the money market are a double-edged sword. On the one hand, they facilitate the smooth functioning of the money market in normal times, efficiently reallocating liquidity between the borrower and lender in response to shocks and news which are not too adverse. In addition, they dampen the effects of adverse news about the quality of the asset on the likelihood and severity of socially costly fire sales. On the other hand, the imperfect information spillovers increase the fragility of the money market in response to adverse liquidity shocks, through their effects on market participants' perceptions of downside risk in the asset payoff.

The imperfect information spillovers may exacerbate allocative inefficiencies in the fire sale regime owing to an interaction between beliefs and fire sales. In response to shocks or news, the imperfect information spillover can increase market participants' perceptions of downside risk, which may exacerbate fire selling of the risky asset relative to a benchmark in which there is no information spillover. This additional fire-selling depresses the price of the risky asset and exacerbates the pecuniary externality which arises during a liquidity-driven fire sale.

Turning to the model's policy implications, we show that during financial distress, the central bank's standard toolkit may be insufficient to restore market stability when the underlying assets are risky. Lowering interest rates or lending against very safe assets, for example via the discount window, increases funding liquidity but does not change the fact that the availability of funding conveys private information about the quality of the underlying risky assets. Therefore, standard liquidity injections do not prevent market participants' perceptions of risk about asset quality from deteriorating markedly. By contrast, direct lending against risky assets, such as the extraordinary facilities implemented during 2020, stabilize prices at government-administered rates, rendering those prices uninformative about agents' private information about the quality of risky assets. In that sense, the policy intervention *destroys* information, impairing the ability of prices to convey private information through the information spillover.

Nevertheless, such interventions may be socially desirable. Reducing the informativeness of prices can prevent perceptions of risk from deteriorating excessively due to the imperfect information spillover. Intuitively, if agents know that the availability of funding is pinned down by the government rather than private information about the quality of the risky asset, they are less likely to become pessimistic about asset quality in response to a tightening of private funding. Indeed, policymakers have frequently cited the effect of their interventions on a vague notion of investor confidence as a stabilizing force. Our model thus provides a conceptual framework for thinking

about these forces.

Finally, we show that the insights of our model obtain for the alternative case in which it is the borrower who receives a cost shock and a private signal rather than the lender. In this alternative case, which we analyze in section 7, the information spillover goes upstream from the borrower to the lender rather than downstream as in our baseline model. The risky asset is then priced by the borrower, implying that the price of the risky asset—that is, market liquidity—serves as a noisy signal to the lender about the borrower’s beliefs. Therefore, cost shocks to the borrower shape the lender’s beliefs through the availability of market liquidity. We discuss an application of this alternative case in section 1.1.

1.1 Discussion of real-world applications

Our model is designed to capture salient features of various financial markets in a parsimonious way. With respect to the structure of lending, for example, repo markets, markets for ABCP, and OTC markets for securities and derivatives typically feature collateral to protect lenders against counterparty risk. Moreover, at their core, the debt in these markets is short-term and is used to finance investment in longer maturity assets. As such, this debt either explicitly needs to be rolled over, is renegotiable, or is subject to margin requirements which effectively render it short-term. The collateralized, short-term debt in our model is intended to capture such characteristics.

Another salient feature of these markets is that they typically feature private information. In certain cases, lenders may be better-informed about the assets backing the debt. For example, consider a repo contract between a broker-dealer and a hedge fund in which the hedge fund invests in a risky asset and borrows from a dealer using the asset as collateral. Dealers typically have additional information about the return of the asset. Even if the collateral asset is a Treasury security, the value of the asset at the maturity of the repo contract may vary depending on changes in the yield curve driven by various macroeconomic factors about which dealers, by virtue of their expertise, may have more information. Similarly, in markets for ABCP, the issuers of the securities may have better information on the value of the underlying collateral securities but would not hold the securities themselves and instead lend funds to leveraged buyers.

Moreover, lenders in this situation may be unable to credibly reveal their private information to their counterparties for two reasons. On the one hand, they may have an incentive to claim that the asset is worth less than it actually is in order to be able to lend against the asset at a higher rate. On the other hand, they may have incentive to inflate the collateral value of the asset in order to prevent borrowers in the market from liquidating assets at fire sale prices.

In other cases, however, the borrower may be better-informed about the fundamental of the risky asset. For example, consider the secondary market for corporate bonds in which an asset

manager finances its purchase of a corporate bond by borrowing from a dealer in an OTC market. The asset manager may be able to monitor the issuer of the corporate bond more closely than the dealer, who is one step further removed from the corporate issuer, and therefore may obtain private information about the credit risk of the bond that the dealer does not have. Moreover, borrowers in this situation may be unable to credibly reveal their private information, as they have an incentive to inflate the collateral value of their asset holdings. As we show in section 7, the core insights of our model still obtain in this case, although the mechanism differs slightly.

1.2 Related literature

Our paper relates to several strands of the literature. The paper is similar in spirit to the theoretical literature studying liquidity freezes in financial markets, including [Brunnermeier and Pedersen \(2009\)](#), [Gale and Yorulmazer \(2013\)](#), [Garcia-Macia and Villacorta \(2023\)](#), [Plantin \(2009\)](#), and [Malherbe \(2014\)](#).¹³ Relative to these papers, our focus is on how liquidity conditions affect the formation of beliefs about fundamentals, leading to belief disagreement. Our model emphasizes how the interaction between counterparty risk and fire sale dynamics may contribute to the fragility of such markets.

[Acharya et al. \(2011\)](#) develop a model of debt capacity in which assets serve as collateral and demonstrate that when debt is short-term and must be rolled over frequently, even small changes in an asset’s fundamental value can trigger a sudden collapse in debt capacity. Our paper complements their framework by providing an alternative mechanism through which financial instability can arise without requiring changes in fundamental asset values or outright *default*. By demonstrating how liquidity shocks can generate endogenous uncertainty about asset quality through information spillovers, our paper offers a complementary perspective that enriches our understanding of financial fragility and provides insights into policy-relevant issues surrounding central bank interventions and market stability.

In the context of bank runs, [Chari and Jagannathan \(1988\)](#) show that panic can arise due to information, which is orthogonal to the health of a bank. Building on [Diamond and Dybvig \(1983\)](#), [Goldstein and Pauzner \(2005\)](#) and [Allen et al. \(2006\)](#) study the role of beliefs and financial stability in the context of global games, while [Infante and Vardoulakis \(2021\)](#) study runs on collateral. In contrast to this literature, our focus is on the role played by asset prices, particularly the price of debt, in disseminating private information and the formation of beliefs. Moreover, we study the implications of this information structure in for financial stability in the context of fire sales rather than coordination failures.

¹³These mechanisms are supported by the empirical facts documented in [Gorton and Metrick \(2012\)](#) and [Copeland et al. \(2014\)](#).

Our model is also related to a more recent literature on information production in credit markets. [Gorton and Ordonez \(2014\)](#), [Dang et al. \(2020\)](#), and [Asriyan et al. \(2022\)](#) show how collateralized debt can alleviate problems of asymmetric information by making debt informationally-insensitive and how boom-bust cycles emerge due to the effect of asset prices on the incentive to produce information. In contrast, we abstract from the decision to produce information and instead focus on how prices convey private information due to market forces, albeit imperfectly. A paper closely related to ours is [Bostanci and Ordoñez \(2024\)](#), which studies how the volume of trade in equity markets affects the informativeness of prices. The authors study the aggregate implications of the mechanism in a quantitative macroeconomic model. In contrast, we focus on the informativeness of prices in markets for collateralized debt, and on isolating the theoretical mechanisms by which prices affect beliefs and financial stability.

Several papers study information aggregation in general equilibrium. [Simsek \(2013\)](#) provides a general characterization of the effect of belief disagreement on asset prices in the presence of collateral constraints. [Asriyan et al. \(2017\)](#), [Vanasco \(2017\)](#), [Babus and Kondor \(2018\)](#), [Banerjee et al. \(2018\)](#), [Asriyan et al. \(2021\)](#), [Asriyan \(2021\)](#), [Dávila and Parlatore \(2021\)](#), and [Dávila and Walther \(2023\)](#) study how information frictions and market structure—including dispersed information, decentralized trading, and imperfect competition—shape the aggregation of private information through prices and real economic outcomes such as investment, leverage, and the emergence of multiple equilibria. Our focus is on the implications of for asset price fluctuations in a model of fire sales. The broad insights from these papers largely apply for our paper as well. Indeed, many of these papers build on the insights about the informational inefficiency of financial markets from [Grossman and Stiglitz \(1980\)](#). Our paper can be interpreted as an application of these general frameworks to the specific context of collateralized debt markets and fire sales.

Methodologically, our paper borrows from the literature on fire sales to capture a notion of financial crises, especially [Lorenzoni \(2008\)](#) and [Kurlat \(2016\)](#). Such fire sales imply the presence of pecuniary externalities, discussed in [Dávila and Korinek \(2018\)](#). In our model, the pecuniary externality interacts with an additional information externality.

Finally, our paper is related to the literature on leverage cycles developed by [Geanakoplos \(1997\)](#), [Geanakoplos \(2003\)](#), [Geanakoplos \(2010\)](#), and [Fostel and Geanakoplos \(2015\)](#). Similarly to these papers, we build a general equilibrium model with heterogeneous agents and collateralized debt contracts, but our focus is on the role of asymmetric information in amplifying leverage cycles.

2 Model setup

There are three dates (0, 1, and 2), a single good, and three representative agents (a borrower, lender, and traditional sector) indexed by $i \in \{B, L, T\}$, respectively. The borrower and lender are

risk-neutral and consume only at date 2 according to $U_i(C_2^i) = C_2^i$. Each agent is endowed with e_0^i of the consumption good at the beginning of date 0. At date 0, a representative borrower issues short-term debt to a representative lender to finance investment in risky asset, which serves as collateral for the loan. The risky asset is in positive net supply A and yields a gross rate of return of $R \in \{\underline{R}, \bar{R}\}$ at date 2, where $\underline{R} < 1$ and $\bar{R} > 1$. At date 0, agents have a common prior belief about R such that $R = \underline{R}$ with probability π_0 . We further assume that $\pi_0 \underline{R} + (1 - \pi_0) \bar{R} = 1$.

The risky asset is illiquid at date 1: It cannot be converted into the consumption good at date 1, nor can new investment in the risky asset be initiated after date 0. In each period, there is a spot market for the risky asset where p_t denotes the price of the risky asset at time t . There is also a traditional sector along the lines of [Lorenzoni \(2008\)](#), whom we include to introduce possibility of fire sales at date 1. The traditional sector participates in the spot market for the asset at date 1 (along with the borrower), and values the asset at a discount.

The borrower's date 0 debt matures after one period. Therefore, to maintain its holdings of the risky asset at date 1, the borrower must issue new debt at date 1, potentially under different terms. By financing long-term risky assets with short-term debt, the borrower will be exposed to liquidity risk at date 1.¹⁴ Debt is issued at date $t = 0, 1$ in a competitive market at unit price q_t . The debt promises a gross interest rate of 1 at date $t + 1$, and is collateralized by the borrower's holdings of the risky asset. As we show later, the borrower chooses to default at date $t + 1$ if the return on the risky asset R is less than the promised gross interest rate 1. We therefore define the effective gross interest rate on the loan as $R_2^d \equiv \min\{R, 1\}$. Assumption 1 we introduce later ensures that there is no default at date 1.¹⁵ This contracting environment is formalized as a competitive market for repo contracts in appendix D.1. Because the debt is backed by the risky asset, the lender's willingness to lend at t will depend on its beliefs about the fundamental, R .

Between periods, each agent can store the consumption good in a storage technology we call 'cash'. The gross return to holding cash is τ_t^i where we normalize the rate of return between dates 0 and 1 at $\tau_0^i = 1$ for $i \in B, L$. At date 1, the return to cash τ_1^i will control each agents' opportunity cost of funds.

Shocks At date 1, agents are simultaneously subject to two shocks. This is illustrated in figure 1. First, they receive a shock to the opportunity cost of their funds τ_1^i , which we refer to as a 'cost shock'. The cost shock is independent of R , and is drawn from a continuous and smooth distribution with probability density function $\lambda_T(\tau)$ with mean 1 and support $[\underline{\tau}^L, \bar{\tau}^L]$. We further

¹⁴One may wonder why the lender lends at date 0 at all instead of holding cash and lending at date 1. The reason is that investment in the risky asset can only be initiated at date 0. Therefore, any equilibrium with positive real investment must involve lending at date 0.

¹⁵If the borrower defaults on its debt, the lender seizes the collateral and liquidates it to the traditional sector. Both of these actions are payoff-equivalent, so we ignore the confiscation of collateral by the lender and instead assume that the borrower liquidates to the traditional sector and uses the proceeds to pay its debt.

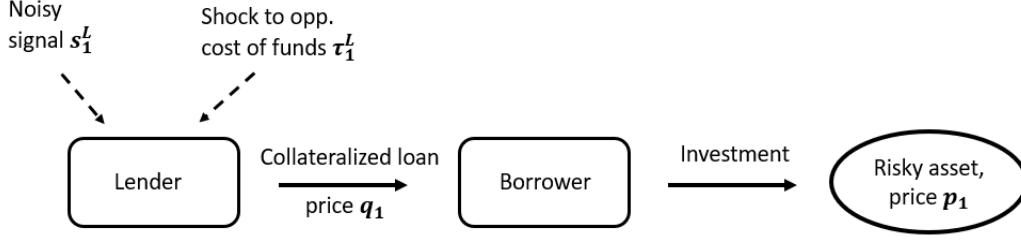


Figure 1: Model Setup

Note: This figures illustrates the setup of the model and the shocks at date 1.

assume that $\lambda_T(\tau)$ is decreasing in $|\tau - 1|$.

In addition to the cost shock, each agent receives private news about the fundamental R at date 1. In particular, at date 1, agent i gets a private noisy signal s_1^i about R , where $s_1^i = R + \varepsilon^i$ and where ε^i is drawn from a continuous and smooth distribution with cumulative density function $\Lambda_\varepsilon(\varepsilon)$, which has mean zero and variance σ_ε^2 , and probability density function $\lambda_\varepsilon(\varepsilon)$ with support $(-\infty, \infty)$. We assume that $\lambda_\varepsilon(\varepsilon)$ is decreasing in $|\varepsilon|$, i.e., $\lambda_\varepsilon(\varepsilon)$ is single-peaked.

For simplicity, we assume that only one agent (the lender) receives both a cost shock and a private signal, while the other agent (the borrower) receives neither. Consistent with this case, we assume that τ_1^B is a fixed parameter such that $\tau_1^B = \tau_0^B = 1$, while τ_1^L is a random variable such that $E[\tau_1^L] = \tau_0^L = 1$. In addition, only the lender receives a private signal. These assumptions ensure the model is tractable and that we can solve for agent's equilibrium beliefs.¹⁶ While much of our exposition focuses on the case in which the lender receives the signal and shock, as illustrated in figure 1, we show in section 7 the same insights apply for the case in which the borrower receives the signal and shock, although the mechanics differ slightly.

Information sets and beliefs Agents have the same information set and beliefs at date 0. At date 1, however, neither the lender's cost shock τ_1^L nor its signal about s_1^L are observable or verifiable to the borrower. Therefore, while the borrower's information set at date 1, I_1^B , consists only of publicly observable variables, including asset prices p_1, q_1 , the information set of the lender, I_1^L , additionally includes the private shocks τ_1^L, s_1^L . Formally, the lender's information set at date 1 is $I_1^L = \{\Theta_1, \tau_1^L, s_1^L\}$, while the borrower's information set is $I_1^B = \{\Theta_1\}$, where Θ_1 denotes the set of all variables observable to all agents, including prices p_1, q_1 and the terms of the contract. At date 1, each agent updates their beliefs π_1^L, π_1^B that $R = \underline{R}$ in response to new information at

¹⁶If, in contrast, both agents received private signals, then the equilibrium would depend not only on how one agent, e.g. the borrower, updates its beliefs in response to the actions of the other, but also on an infinite feedback loop in which also the borrower's actions affect the lender's beliefs, etc. This would make an analytical characterization of equilibrium belief formation extremely difficult.

date 1 according to Bayes' rule. We denote agent i 's expectation of variable x , conditional on its information set at time date t , as $E_t^i[x] \equiv E_t[x|I_t^i]$.

At date 1, after the realization of the cost shocks and signals, agents simultaneously settle the date 0 contracts and enter into another one-period contract at date 1. Uncertainty is fully resolved for all agents at date 2.

We make the following assumption about the endowments of each agent.

Assumption 1. $e_0^B \tau_1^B < A(E_0^B[R] - 1)$, $e_0^L > A\bar{R}/\tau_1^L - e_0^B$, and $e_0^T > \max_{a \in [0, A]} aF'(a)$.

The first part of the assumption ensures that borrowers cannot fund their asset purchases without debt, while the second part ensures that the lender's endowment is sufficiently large to satisfy the borrower's demand. Finally, the third part of the assumption ensures that the traditional sector always has enough cash to purchase any of the risky asset amount sold to them.¹⁷

We also make the following assumptions on the traditional sector.

Assumption 2. $F'(A) \geq E_0^B[R] - e_0^B/A$ and $E_0^B[R]/\tau_1^B > F'(0)$.

The first part of the assumption ensures that, at date 1, the liquidation value of the borrower's holdings of the risky asset is sufficient to repay all of its date 0 debt d_0 so that the borrower never defaults in equilibrium at date 1. (The borrower's liquidation value of its holdings A at price $p_1 = F'(A)$ is $AF'(A)$, while the maximum date 0 debt the borrower can have in equilibrium is $AE_0^B[R] - e_0^B$.) The second part of the assumption ensures that, ex ante, the borrower always values the asset more than the traditional sector does.

2.1 Real-world applications

Before proceeding further, we build intuition for the model by considering how this general setup can be applied to real-world settings. Generally, the model captures the dynamics of various money markets which feature short-term collateralized debt, some degree of market segmentation, and dispersed information about asset quality.

For example, consider prime brokers and hedge funds. Hedge funds borrow from their prime brokers in repo markets to invest in various securities: The hedge fund will invest in a security and finance it with a repo contract in which it sells the security to an asset manager with the promise to buy it back at a later date – a form of short-term collateralized borrowing.

Often there is dispersed information about the fundamental value of the securities being

¹⁷This assumption is almost exactly the same as the assumptions in [Simsek \(2013\)](#) and [Gottardi et al. \(2019\)](#), serving the same purposes.

traded.¹⁸ For instance, suppose the hedge fund is repo'ing a mortgage backed security. In general, the hedge fund and prime broker may receive different information about the conditions in various housing markets, at different points in time, which cause them to have different valuations of the security. In fact, prime brokers typically have significant information advantages due to their position in the market, as they execute trades for multiple hedge funds, have visibility across various investment strategies, and likely possess valuable market intelligence.¹⁹ These features of the market are captured in our model by the assumption that the lender receives a private signal about the quality of the risky asset.

Moreover, segmentation across repo markets prevents the hedge funds from directly observing the private information of their brokers and other hedge funds (Kahn et al., 2023; Chang et al., 2025). Nevertheless, there is evidence that hedge funds try to infer the private information of their brokers through their dealings with them.²⁰ Our model will capture these dynamics in the form of (imperfect) information spillovers, in which the terms on which the lender is willing to extend credit to the borrower affect the borrower's assessment of the quality of the underlying asset.²¹

Another real-world setting which maps into our model setup is the market for asset-backed commercial paper, in which the issuers of these debt securities may have better information on the value of the underlying collateral securities but would not hold the securities themselves and instead lend funds to leveraged buyers.

In other cases, however, the borrower may be better-informed about the fundamental of the risky asset. For example, consider the secondary market for corporate bonds in which an asset manager finances its purchase of a corporate bond by borrowing from a dealer in an over-the-counter market. The asset manager may be able to monitor the issuer of the corporate bond more closely than the dealer, who is one step further removed from the corporate issuer, and therefore may obtain private information about the credit risk of the bond that the dealer does not get. Moreover, the borrowers in this situation may be unable to credibly reveal their private information, as they have an incentive to inflate the collateral value of their asset holdings. As we show in section

¹⁸This may even be true for securities like Treasuries. While the fundamental value of Treasuries depends on publicly-observable information like fiscal deficits and inflation, the prices also involve liquidity or risk premia which are not publicly observed. For example, off the run treasuries, which are treasuries which are not the most recent issue, typically trade at a discount because they're less liquid. That 'liquidity premium' component of the price depends on the appetite of market participants to hold such securities, and that information is not publicly observed. Hence, even for US treasuries, there is typically dispersed information about fundamental values of the underlying securities being traded in the repo market.

¹⁹For example, Botsch and Vanasco (2019) find that banks learn through repeated interactions with borrowers and adjust loan prices and amounts.

²⁰For example, evidence points to this as a factor in the failure of Bear Stearns. See <https://elischolar.library.yale.edu/cgi/viewcontent.cgi?article=5929&context=ypfs-documents>.

²¹If market participants have speculative opportunities and can learn about the beliefs of others, the information spillovers could be further amplified as shown by Banerjee et al. (2018). In such a case, reducing transparency (information) can increase informational efficiency.

7, the core insights of our model still obtain in this case, although the mechanism differs slightly.

3 Equilibrium

The equilibrium conditions can be partitioned into two blocks: a real block which determines the allocation conditional on agents' beliefs, and a set of conditions characterizing the formation of beliefs given the real block. We first characterize each block and then turn to general equilibrium, which is pinned down by the joint determination of both blocks.

3.1 Real block conditional on beliefs

3.1.1 Lender's problem

Each period, the lender solves a portfolio choice problem to maximize its date 2 expected consumption. In particular, at $t = 0, 1$, it chooses between cash κ_t^L and the borrower's debt d_t , taking the unit price of debt q_t as given, and conditional on its opportunity cost of funds (the return to cash τ_t^L) and its beliefs π_t^L . The lender's budget constraints at dates $t = 0, 1$ are

$$\underbrace{\kappa_0^L}_{\text{date 0 cash holdings}} + \underbrace{q_0 d_0^L}_{\text{value of date 0 loan}} \leq \underbrace{e_0^L}_{\text{date 0 cash endowment}}$$

$$\underbrace{\kappa_1^L}_{\text{date 1 cash holdings}} + \underbrace{q_1 d_1^L}_{\text{value of date 1 loan}} \leq \underbrace{\kappa_0^L}_{\text{date 0 cash holdings}} + \underbrace{d_0^L}_{\text{value of date 0 loan}},$$

respectively. At date 2, the lender consumes its resources C_2^L , which consists of the proceeds of its date 1 holdings of debt and the return on its cash holdings.

$$\underbrace{C_2^L}_{\text{date 2 consumption}} = \underbrace{\tau_1^L \kappa_1^L}_{\text{proceeds from cash holdings}} + \underbrace{R_2^d d_1}_{\text{proceeds from date 1 loan}}.$$

We state and solve the lender's full problem in appendix A.1. Assumptions 1 and 2 ensure that lending takes place in equilibrium so that $d_1 > 0$ and that the lender holds a strictly positive quantity of cash in equilibrium at date 1, $\kappa_1^L > 0$. Therefore, given these assumptions, the lender's first-order condition (FOC) for d_1 is

$$q_t = \frac{E_t^L [R_t^d]}{\tau_t^L}. \quad (1)$$

This condition states that the expected marginal return from lending at date 1 equals the expected marginal return on its opportunity cost from holding cash at date 1. The lender's willingness to hold debt is lower the higher is its opportunity cost of funds τ_1^L and the more pessimistic its beliefs about the fundamental R , since the loan is backed by the risky asset. Because the lender is the marginal buyer of debt, the competitive spot price of the contract at date 1, q_1 , will ensure that this condition holds in equilibrium.

3.1.2 Borrower's problem

At dates 0 and 1, the borrower decides how much to borrow and how to allocate its portfolio between the risky asset and cash in order to maximize its expected date 2 consumption $E^B [C_2^B]$. At date 0, the borrower has a cash endowment and receives a cash loan from the lender, which it can allocate between cash holdings κ_0^B and investment in the risky asset a_0^B . Therefore, its date 0 budget constraint is

$$\underbrace{q_0 d_0}_{\text{date 0 loan}} + \underbrace{e_0^B}_{\text{date 0 cash endowment}} \geq \underbrace{p_0 a_0^B}_{\text{value of date 0 asset holdings}} + \underbrace{\kappa_0^B}_{\text{date 0 cash holdings}}.$$

At date 1, the borrower must repay its date 0 debt. (Recall that assumption 2 ensures that it is never optimal to default at date 1, so we neglect this case here for simplicity.) However, because the risky asset only pays out at date 2, the borrower has no cash flow from the risky asset at date 1 to repay its debt. Therefore, to repay its debt, it must either issue new debt (which is equivalent to rolling over some of its date 0 debt), use any cash holdings it may have carried over from date 0, or raise cash by selling some of its holdings of risky asset to the traditional sector in a competitive spot market at price p_1 . Accordingly, the borrower's date 1 budget constraint is

$$\underbrace{q_1 d_1 - d_0}_{\text{net increase in loan}} \geq \underbrace{p_1 (a_1^B - a_0^B)}_{\text{net increase in asset holdings}} + \underbrace{\kappa_1^B - \kappa_0^B}_{\text{net increase in cash holdings}}.$$

At date 2, the borrower chooses to default if the realized return on the risky asset R is less than 1, the promised gross interest on its debt. It then consumes its remaining resources.

$$\underbrace{C_2^B}_{\text{date 2 consumption}} \leq \underbrace{\tau_1^B \kappa_1^B}_{\text{proceeds from cash holdings}} + \underbrace{a_1^B R}_{\text{proceeds from asset holdings}} - \underbrace{d_1 R_2^d}_{\text{repayment of debt}},$$

We state and solve the borrower's full problem in appendix A.2. Combining the borrower's

optimality conditions for borrowing and investing in the risky asset yields a relation between the borrower's expected net return from buying the risky asset, $E_1^B[R] - \tau_1^B p_1$, and the cost of financing its holdings of the risky asset, $E_1^B[R_2^d] - q_1 \tau_1^B$, which in turn balances the benefit from being able to borrow more with the cost of a larger expected repayment.

$$\underbrace{E_1^B[R_2^d] - q_1 \tau_1^B}_{\text{cost of financing asset holdings}} + \underbrace{\xi_{\kappa_1}^B (p_1 - q_1)}_{\text{shadow value of cash}} = \underbrace{E_1^B[R] - \tau_1^B p_1}_{\text{expected net return of risky asset}} \quad (2)$$

This equation captures how the borrower's beliefs and the price of its debt q_1 affect its portfolio choice at date 1: The borrower chooses to hold less of the risky asset compared to cash if it is more pessimistic about the fundamental R (a higher π_1^B), or if the availability of funding is lower (lower price of its date 1 debt q_1).

Thus, the borrower chooses to liquidate part of its holdings of the risky asset when either (i) it becomes sufficiently pessimistic about the asset's date 2 payoff R , or (ii) it is sufficiently liquidity constrained at date 1 such that its cash holdings κ_0^B are insufficient to repay its debt at date 1. (In equilibrium, (i) will correspond to a belief-driven fire sale while (ii) will correspond to a liquidity-driven fire sale, as we discuss in section 3.3.) Note that the borrower's perceived downside risk—its subjective probability π_1^B that the risky asset's date 2 payoff will be low—lowers its expected return from holding the asset and therefore tilts its decision at the margin in favor of liquidating the asset.

Rearranging the optimality condition yields an expression for the borrower's marginal valuation for the risky asset, $p_1 = \frac{1}{\tau_1^B + \xi_{\kappa_1}^B} (E_1^B[R] - E_1^B[R_2^d]) + q_1$. For the purpose of exposition, we define $p_1^B \equiv \frac{1}{\tau_1^B} (E_1^B[R] - E_1^B[R_2^d]) + q_1$, which corresponds to the borrower's marginal valuation of the asset when the shadow value of cash is 0. By construction, $p_1 = p_1^B$ if and only if the borrower has positive cash holdings in equilibrium at date 1, $\xi_{\kappa_1}^B = 0$.

3.1.3 Traditional sector's problem

We incorporate a traditional sector which values the risky asset at a discount in order to introduce a notion of fire sales along the lines of [Lorenzoni \(2008\)](#). The traditional sector maximizes its date 2 consumption C_2^T . At date 0 it receives an endowment e_0^T with which it can buy existing risky assets in the spot market at date t , or store as cash between dates t and $t + 1$, for $t = 0, 1$. Existing assets need to be managed subject to some increasing costs. The date 2 return of holding a_1^T of the risky asset at date one, net of these management costs, is given by $F(a_1^T)$ units of the consumption good, where $F' > 0$ and $F'' < 0$. In appendix A.3, we show that the traditional sector's optimality condition at date 1 is

$$p_1 = F'(a_1^T) + \xi_{a_1}^T. \quad (3)$$

This describes the traditional sector's marginal valuation of the risky asset. We assume that the expected return of the asset satisfies $E_0^B[R]/\tau_1^B > F'(0)$ so that the traditional sector always values the risky asset less than other agents do at date 0. This implies that the traditional sector is relevant for the equilibrium only at date 1. For simplicity, and without loss of generality, we assume the traditional sector's endowment of cash is sufficiently large that, at date 1, the traditional sector could feasibly buy all existing assets in equilibrium.

3.1.4 Market clearing at date 1

Recall that no new risky assets can be created after date 0. Therefore, the market clearing condition for the date 1 spot market for the risky asset implies that the total quantity of the asset held by the borrower and the traditional sector at date 1 is equal to the total amount of the asset created at date 0.²²

$$a_1^B + a_1^T = a_0^B \quad (4)$$

The spot market for the risky asset is competitive, so the price p_1 will be pinned down in equilibrium at the marginal valuation of the marginal buyer. Since the lender cannot directly participate in the market for the risky asset, either the borrower or the traditional sector is the marginal buyer in equilibrium. The marginal buyer is the agent who has the maximum willingness to pay in equilibrium. Therefore, the risky asset price is given by

$$p_1 = \max \left\{ \frac{1}{\tau_1^B + \xi_{\kappa_1}^B} \left(E_1^B[R] - E_1^B[R_2^d] \right) + q_1, p_1^T \right\} \quad (5)$$

where the first term and p_1^T denote the marginal valuations of the borrower and traditional sector, respectively. We showed in the previous section that the traditional sector's marginal valuation is given by its optimality condition, $p_1^T \equiv F'(a_1^T) + \xi_{a_1}^T$.

3.2 Beliefs at date 1

We now turn to the evolution of agents' beliefs about the fundamental of the risky asset. Because agents have different information sets at date 1, their beliefs evolve differently.

Lender's beliefs at date 1 The lender's information set at date 1 consists of all commonly observed variables $\Theta_1 = \{d_1, q_1, p_1\}$ and its private cost shock and private signal τ_1^L, s_1^L . The lender's posterior beliefs about the fundamental of the risky asset R are determined by Bayes' Rule, where

²²For ease of exposition, we have already imposed a market clearing conditions for the market for debt at each date. These conditions are made explicit in appendix D.1.

the likelihood function depends on its private signal s_1^L .

$$\pi_1^L = \frac{\pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R})}{\pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R}) + (1 - \pi_0) \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \bar{R})}. \quad (6)$$

This is derived formally in appendix B.1. The lender's pessimism about the fundamental, measured by π_1^L , is decreasing in s_1^L by the assumption on the distribution from which ε is drawn is single-peaked. We denote the reduced form cumulative distribution function of π_1^L as $G_\pi(\cdot)$ and its probability density function as $g_\pi(\cdot)$.

Borrower's beliefs at date 1 The borrower observes all prices and contract terms, but does not observe the lender's liquidity shock or private signal. Therefore the borrower's date 1 information set consists only of commonly observed variables $I_1^B = \{\Theta_1\}$. The borrower updates its beliefs about the risky asset's return conditional on its date 1 information set, where its posterior belief is denoted as π_1^B .²³

While the borrower does not directly observe the lender's private signal about R , it can partially infer it from the lender's the willingness to supply credit at date 1, which is reflected in the equilibrium price of debt q_1 . Recall from (1) that the price of debt q_1 reflects the private information of lender τ_1^L, s_1^L through the equilibrium relationship

$$q_1 = \frac{1 - \pi_1^L + \pi_1^L \underline{R}}{\tau_1^L}.$$

The borrower is rational and therefore understands this mapping between the price of debt q_1 and the lender's beliefs π_1^L . Moreover, the borrower also understands how the lender's beliefs π_1^L reflect its private news s_1^L about the fundamental through the mapping (6). Therefore, the equilibrium price of debt q_1 conveys some information about the lender's private signal about the quality of the risky asset, consistent with the usual price-signal mechanism. This reflects an *information spillover* through which the price of debt conveys the lender's private information to the borrower. However, the information spillover is imperfect due to a kind of identification problem faced by the borrower.

Identification problem The observed q_1 is not sufficient for the borrower to separately identify the lender's private news s_1^L versus its cost shock τ_1^L . This *identification problem* arises because there is only one observable q_1 to infer two unobservables s_1^L, τ_1^L .²⁴ The identification

²³The price p_1 of the risky asset does not convey any additional information about the lender's beliefs about R . This is because, as discussed in section 3.3, p_1 is determined by the valuation of the marginal buyer of the risky asset. But because the lender cannot participate in the market for the risky asset, p_1 never reflects the lender's valuation directly. Therefore, in equilibrium, only the price of debt q_1 conveys information to the borrower about the lender's private information.

²⁴This identification problem arises because the borrower faces two layers of uncertainty about the asset's return R .

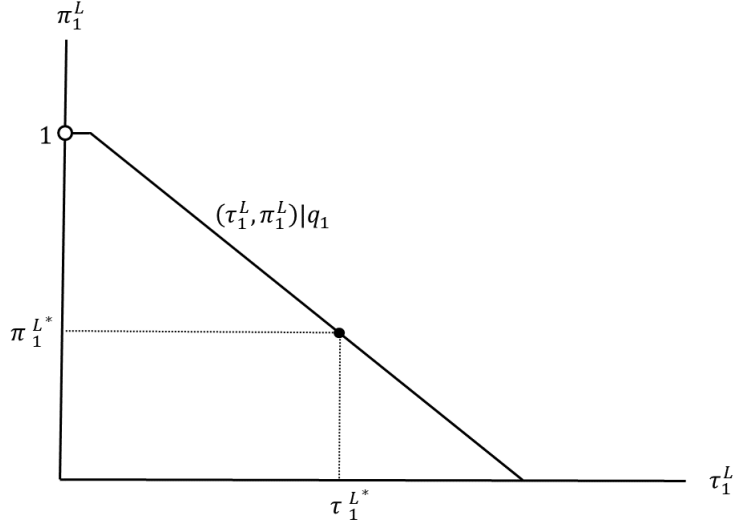


Figure 2: Identification Problem

Note: This figure illustrates the identification problem faced by borrowers at date 1. τ_1^{L*} denotes the realized cost shock and π_1^{L*} denotes the lender's equilibrium belief given the realized signal q_1 it receives. q_1 is the equilibrium price of debt consistent with $(\tau_1^{L*}, \pi_1^{L*})$. The line plots the set of all possible realizations of (τ_1^L, π_1^L) consistent with the equilibrium q_1 .

problem faced by the borrower is illustrated in figure 2. The curve in the figure plots the combinations of the lender's cost shock and beliefs, (τ_1^L, π_1^L) , which are consistent with the observed q_1 according to (6). Therefore, the observed q_1 is not sufficient to infer the true realizations $(\tau_1^{L*}, \pi_1^{L*})$ separately. As a result of this identification problem, the information spillover is *imperfect*.

The borrower's posterior beliefs evolve according to

$$E_1^B [\tilde{\pi}_1^L] = \int_{\underline{\tau}^L}^{\bar{\tau}^L} \int_0^1 \pi \mathbb{1} \left\{ \pi = \frac{1 - \tau q_1}{1 - \underline{R}} \right\} dG_\pi(\pi) d\Lambda_T(\tau), \quad (7)$$

where $\mathbb{1}\{\cdot\}$ is an indicator function, and $\Lambda_T(\cdot)$ is the distribution function of $\lambda_T(\cdot)$. By change of variables, the above equation can be rearranged as follows

$$\pi_1^B = \int_0^1 \pi \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q_1} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q_1} < \bar{\tau}^L \right\} dG_\pi(\pi). \quad (8)$$

The price of debt q_1 is effectively a noisy signal about the lender's private signal s_1^L , where the noise is introduced by the lender's idiosyncratic cost shock τ_1^L . The lender's private signal s_1^L itself is a noisy signal about R . The presence of two layers of uncertainty will give rise to endogenous belief disagreement, and will imply that the borrower's beliefs are shaped by liquidity conditions which affect the noisiness of q_1 . Fundamentally, this problem arises due to the absence of risk markets, as discussed in [Stiglitz \(1981\)](#).

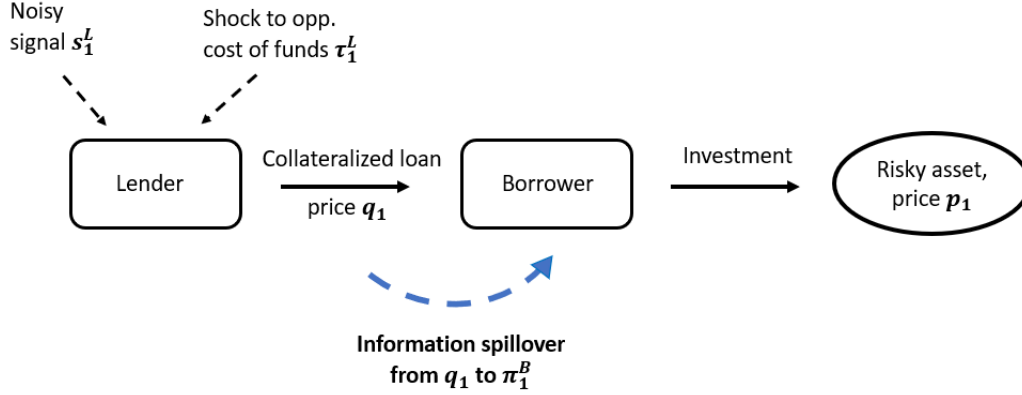


Figure 3: Downstream Information Spillover

Note: This figure illustrates the downstream information spillover in which the price of debt affects the borrower's belief about the fundamental of the asset.

Intuitively, the borrower computes the likelihood of all possible realizations of (τ_1^L, s_1^L) consistent with the observed value of q_1 , based on its knowledge of the joint distribution of τ_1^L and s_1^L , and given the mapping (6) between s_1^L and π_1^L . Based on the likelihood of these realizations, the borrower forms an expectation over the lender's private signal s_1^L and updates its own beliefs π_1^B accordingly.

Imperfect information spillover The *imperfect information spillover* implies that the availability of funding liquidity affects the borrower's perceptions of downside risk about asset quality. This mechanism is illustrated by the blue arrow in figure 3, and is formalized in the proposition below.

Lemma 1. *The borrower's posterior belief π_1^B has the following properties in equilibrium:*

- (A) *The borrower's belief is the same as the expected posterior of the lender, i.e. $\pi_1^B = E_1^B[\pi_1^L]$.*
- (B) *Perceptions of risk and funding liquidity: π_1^B is decreasing in q_1 .*

Proof. See appendix B.2. ■

Part (A) of the lemma shows that the borrower's posterior belief about the fundamental value of the asset is given by the borrower's expectation about the lender's belief. This is a direct result of Bayesian updating and simply reflects that the borrower knows the lender to be better informed due to its private signal.

Part (B) of the lemma shows how perceptions of downside risk are shaped by the availability of funding liquidity. In particular, tight funding liquidity, given by a low price of debt q_1 , increases the borrower's perception of downside risk in the risky asset's future payoff, given by π_1^B . Importantly,

this is true even if the low price of debt q_1 is driven entirely by an adverse cost shock τ_1^L which is orthogonal to R . The imperfect information spillover thus gives rise to belief disagreement in equilibrium whereby the borrower's beliefs about the quality of the risky asset diverge from those of the lender, $\pi_1^B \neq \pi_1^L$.

3.3 Characterization of equilibrium at date 1

Definition of date 1 equilibrium An equilibrium at date 1 is a set of prices p_1, q_1 , quantities $d_1, \kappa_1^L, a_1^B, \kappa_1^B, a_1^T, C_2^L, C_2^B, C_2^T$, and posterior beliefs π_1^L, π_1^B , satisfying the agents' optimality conditions and constraints, market clearing conditions, and the equations characterizing belief formation, taking as given variables determined at date 0 and the date 1 shocks τ_1^L, s_1^L . Note that a unique equilibrium always exists because of the linearity of the agents' objective functions and the constraints of the borrower's optimization problem, and the smoothness of the inverse demand function of the traditional sector, which determines the market clearing condition.

The date 1 equilibrium features two regimes, which we call the 'normal regime' and the 'fire sale regime'. In the normal regime, the borrower holds all of the risky asset date 1: The borrower's marginal valuation of the risky asset p_1^B , given by its optimality condition (2), exceeds the marginal valuation of the traditional sector $F'(0)$. In the fire sale regime, by contrast, the borrower sells some of its holdings of the risky asset to the traditional sector: The borrower's marginal valuation falls sufficiently such that $p_1^B < F'(0)$. The lemma below formally defines the two regimes.

Lemma 2 (Two regimes in date-1 equilibrium). *In equilibrium, the following holds:*

- (A) When $p_1 \geq F'(0)$, the economy is in the **normal regime** in which $p_1 = p_1^B$, $a_1^B = c_1 = a_0^B$, and $a_1^T = 0$.
- (B) When $p_1 < F'(0)$, the economy is in the **fire sale regime** in which $p_1 = F'(a_1^T)$, $a_1^B = c_1^B < a_0^B$, and $a_1^T > 0$.

Proof. See appendix B.5. ■

Because there is a one-to-one mapping from the date 1 price of debt q_1 to the price of the risky asset p_1 , there is a threshold $\hat{q}_1$ above which the normal regime applies and below which the fire sale regime applies. The lender's optimality condition defines the set of (τ_1^L, π_1^L) consistent with $\hat{q}_1$, given by

$$\pi_1^L = \frac{1 - \tau_1^L \hat{q}_1}{1 - R}. \quad (9)$$

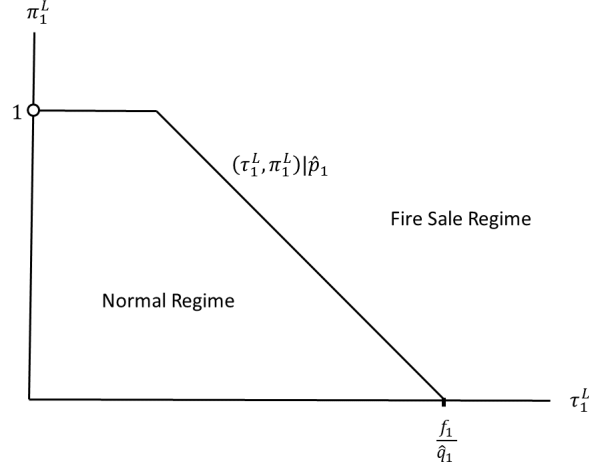


Figure 4: Two Regimes at Date 1

Note: This figure illustrates the bisection of the state space into two regimes at date 1. The equilibrium is in the fire sale regime if and only if the equilibrium value of p_1 is below a threshold value $\hat{p}_1$ (or, equivalently, the equilibrium value of q_1 is below a threshold value $\hat{q}_1$). The curve in the figure plots the combinations of the states (τ_1^L, π_1^L) consistent with the threshold $\hat{p}_1$ and marks the frontier between the two regimes.

This equation (9) defines the frontier partitioning the state space into the normal and fire sale regimes, and is illustrated in figure 4.²⁵

Intuitively, a fire sale of the risky asset occurs at date 1 if and only if funding liquidity conditions at date 1 are sufficiently tight – that is, if the lender’s supply of credit is sufficiently low that $q_1 < \hat{q}_1$. Accordingly, the normal regime obtains when cost shocks are not too large and the signal is not too bad. By contrast, the economy enters the fire sale regime when the cost shock or news are sufficiently severe.

Normal Regime In the normal regime, the borrower is at a corner solution in its portfolio choice and wants to hold only the risky asset at date 1. Because no new investment in the risky asset can be initiated after date 0, the borrower’s holdings of the risky asset remain unchanged from date 0, $a_1^B = a_0^B$. The asset price p_1 is relatively high and is pinned down by the borrower’s valuation.

How does the money market adjust in response to shocks in the normal regime? In response to a tightening in funding liquidity – that is, a relatively low equilibrium price of debt q_1 – the money market simply facilitates a reallocation of cash between lender and borrower: The borrower finances any funding shortfall $d_0 - q_1 d_0$ out of its cash holdings stored from date 0 rather than by liquidating the risky asset to the traditional sector. This is true regardless of whether the underlying

²⁵Note that the line plotted in this figure is conceptually different from that plotted in figure 2. Figure 2 plotted the set of all (τ_1^L, π_1^L) consistent with the equilibrium q_1 . By contrast, figure 4 plots such combinations consistent with the threshold price $\hat{q}_1$ which demarcates the two regimes. The equilibrium price q_1 is not, in general, equal to the threshold price $\hat{q}_1$.

shock is an adverse cost shock or bad news about the quality of the asset.

While the contraction in the supply of funds worsens the borrower's perceptions of downside risk π_1^B due to the imperfect information spillover, in the normal regime, these effects are not strong enough to lead to fire sales. In section 5, we show that this reallocation in the normal regime is welfare-improving.

Fire Sale Regime If funding conditions become sufficiently tight at date 1—that is, if q_1 is sufficiently low—this triggers a fire sale of the risky asset. To see how, recall the borrower's date 1 optimality condition (2)

$$\underbrace{E_1^B [R_2^d] - q_1 \tau_1^B}_{\text{cost of financing risky asset holdings}} = \underbrace{E_1^B [R] - \tau_1^B p_1}_{\text{expected net return of risky asset}},$$

where we have imposed the result that $\xi_{\kappa_1}^B = 0$, so that the borrower holds a positive amount of cash at date 1. Tighter funding liquidity (a fall in q_1) affects the optimality condition in two ways: (i) There is a liquidity effect through which a lower q_1 directly increases the effective cost to the borrower of financing its risky asset holdings; and (ii) an effect on the borrower's beliefs about the risky asset's date 2 return due to the imperfect information spillover. Through both effects, tighter funding liquidity (a lower q_1) tilts the borrower's portfolio choice in favor of holding cash rather than the risky asset at date 1, prompting the borrower to sell some its risky asset holdings to the traditional sector.

Equilibrium in the fire sale regime is reached once the risky asset price p_1 falls sufficiently to restore equality in the borrower's optimality condition. The strict concavity of the traditional sector's technology $F(\cdot)$ implies that this equilibrium price $p_1 = F'(a_1^T)$ occurs at an interior optimum in which the borrower holds both cash and the risky asset. The following proposition characterizes the fire sale regime.

Proposition 1. *In equilibrium, the following holds:*

- (A) **Fire sales are more severe the tighter is funding liquidity:** *In the fire sale regime, a lower realization of s_1^L or a higher realization τ_1^L causes a_1^T to be higher and p_1 to be lower at the margin.*
- (B) **Fire sales may be driven by liquidity needs or pessimism:** *In a liquidity-driven fire sale, $p_1^B > F'(0) > p_1$ obtains. In a belief-driven fire sale, we have $p_1^B < F'(0)$.²⁶*

Proof. See appendix B.6. ■

²⁶The price p_1 can be less than p_1^B in this case, if the borrower is extremely pessimistic and $p_1^B < F'(a_0^B)$. See appendix D.2 for the full characterization of this case.

Part (A) of the proposition states that the lower that q_1 is, the more severe the fire sale, and hence the lower the equilibrium price p_1 of the risky asset. For a given asset price p_1 , tighter funding liquidity (lower q_1) worsens the tradeoff to investing in the risky asset by both increasing the borrower's liquidity need and by making the borrower's beliefs more pessimistic.

Part (B) of the proposition shows that, depending on whether the proximate cause of the fire sale is the liquidity effect or the information effect of the lower q_1 on the borrower's optimality condition (2), a fire sale can be of either of two types. In a liquidity-driven fire sale, the borrower has insufficient funding from the lender to repay its date 0 debt. Put differently, the lender is unwilling at date 1 to rollover the full amount of the date 0 debt, and the borrower's spare cash holdings are insufficient to cover the funding shortfall $d_0 - q_1 d_0$. As a result, the borrower is forced to liquidate some of its risky asset holdings to generate the funds needed to reduce its debt position. This can occur even if the borrower believes the quality of the asset to be high. In much of the literature, fire sales take this form.²⁷

In a belief-driven fire sale, however, the borrower may have sufficient cash holdings at date 1 to cover the difference between its debt repayment and issuance of new debt, without having to liquidate its holdings of the risky asset. However, the fall in the lender's supply of credit causes the borrower to lose confidence in the quality of the risky asset, since the borrower cannot rule out that the contraction in lending occurred due to bad news. As a result of this pessimism, the borrower chooses to reduce its holdings of the risky asset by liquidating some of it to the traditional sector.

Unlike most of the literature, the fire sale in this case arises not due to a lack of liquidity *per se*, but rather arises due to the effect that the imperfect information spillover has on endogenous perceptions of risk. Such belief-driven fire sales have the flavor of asset market turmoil driven by investor panic, in which an initial tightening of financial conditions seem to spiral, triggering a spike in uncertainty and perceptions of risk.

3.4 Equilibrium at date 0

The date 1 equilibrium characterized above takes for granted that the borrower issues a strictly positive amount of debt at date 0 in order to finance an initial investment in the risky asset. To ensure that this is consistent with optimal date 0 behavior of agents, we briefly characterize the date 0 equilibrium. Given the equilibrium at date 1, we can solve recursively for the date 0 equilibrium. For tractability, we assume here that $F'(a) = \alpha$ for all $a \geq 0$.²⁸ The date 0 equilibrium is derived in appendix D.3, and the key properties are summarized in the following proposition. For our

²⁷For example, see Lorenzoni (2008) and Dávila and Korinek (2018).

²⁸Effectively, this assumption implies that, in a fire sale at date 1, the borrower liquidates all of its holdings of the risky asset. This assumption is imposed only for tractability of the date 0 equilibrium characterization. This assumption does not affect our main results, which are related to the equilibrium at date 1.

purposes, the key features of the date 0 equilibrium are that initial lending and investment in the risky asset are strictly positive.

Lemma 3. *Suppose that assumptions 1 and 2 hold. At date 0, the borrower is at a corner solution in its portfolio choice, maximizing its holdings of the risky asset at holding zero cash. That is, $(c_0, a_0, \kappa_0^B) = (A, A, 0)$ —and the prices of the asset and contracts are $(p_0, q_0) = (1 + e_0^B/A, 1)$.*

Proof. See appendix D.3. ■

The date 0 loan from the lender to the borrower is strictly positive, the borrower invests only in the risky asset, and uses all of its holdings of the risky asset as collateral for the loan.

4 Financial stability implications of information spillovers

In this section, we study the financial stability implications of imperfect information spillovers by comparing the date 1 equilibrium in our baseline economy to a counterfactual benchmark economy.

4.1 Counterfactual benchmark economy with common information

To isolate the role of the identification problem and information spillovers, we compare the equilibrium to that of a counterfactual economy in which agents have common information at date 1. This counterfactual economy is equivalent to one in which the information spillovers are *perfect*.

In this benchmark, the shocks (τ_1^L, s_1^L) are *publicly* observed by all agents. In particular, we assume that the borrower can directly observe both the lender’s signal s_1^L and the lender’s shock to the opportunity cost of funds τ_1^L . In this sense, neither agent has private information. In this counterfactual economy, there is no identification problem: Perceptions of downside risk in the asset payoff are in line with the true signal.

Note that information is still incomplete: Agents receive noisy signals about the asset’s date 2 return, but signals are publicly observable rather than being private. Agents therefore have a common information set. A corollary of lemma 1 is that, under this common information benchmark, the borrower and lender have identical posterior beliefs, $\pi_1^B = \pi_1^L$ in every state. The form of the equilibrium (see appendix D.5 for details) is otherwise the same as that in the baseline economy with private information.²⁹

First, we characterize how the partition of the state space between the normal and fire sale regimes in the common information benchmark and compare it with that in the baseline model.

²⁹In general, differences in the equilibrium allocations at date 0 across the baseline and benchmark may also manifest as differences in the date 1 allocation. We can show that the date 0 equilibrium allocations of the two cases would be the same under assumptions 1 and 2.

Recall that $\hat{q}_1$ is the threshold q_1 separating the two regimes in the baseline economy, defined implicitly by $(1 - \pi_1^B(\hat{q}_1))(\bar{R} - 1) + \tau_1^B \hat{q}_1 = \tau_1^B F'(0)$, and $\hat{p}_1 = F'(0)$ is the corresponding asset price of the frontier. As in the baseline economy, the frontier under the common information benchmark is defined by the set of states (τ_1^L, π_1^L) consistent with $p_1 = \tilde{p}_1 \equiv F'(0)$, which is given by

$$(\bar{R} - 1) - \left[(\bar{R} - 1) + \frac{\tau_1^B}{\tau_1^L} (1 - \bar{R}) \right] \pi_1^L + \frac{\tau_1^B}{\tau_1^L} = \tau_1^B F'(0). \quad (10)$$

This characterizes the partition $(\tau_1^L, \pi_1^L) | \tilde{p}_1$, between the fire sale and normal regimes in the common information benchmark. In appendix D.6, we show analytically that the frontier takes the form depicted in figure 5, and that the frontiers in the baseline economy and the common information benchmark intersect in the domain $\pi_1^L \leq 1$.³⁰

4.2 Effect of information spillovers on the likelihood of fire sales

How do imperfect information spillovers affect the likelihood of fire sales relative to the common information benchmark? This depends on the type of shocks hitting the economy at date 1.

When the lender receives severely adverse cost shocks, the imperfect information spillover leaves the borrower excessively pessimistic relative to the common information benchmark. This occurs in the region of figure 5 shaded with dotted red lines. Conditional on a severely adverse cost shock, it takes a less severe signal at date 1 to push the economy into the fire sale regime in our baseline economy than in the benchmark. Thus, the imperfect information spillover increases the likelihood of a fire sale for such shocks.

By contrast, when the lender receives severely adverse news about the asset's date 2 payoff, which occurs in the region of the figure shaded by solid blue lines, the borrower is over-optimistic relative to the benchmark. Conditional on very bad news, it takes a larger adverse cost shock to push the economy into a fire sale regime in the baseline economy compared to the benchmark. The imperfect information spillover reduces the likelihood of a fire sale for such news.

The overall effect of the information spillover on the unconditional likelihood of crises depends on the relative importance of cost shocks versus news in driving asset price fluctuations—that is, on the likelihood of entering the red-shaded region of the state space in figure 5 versus entering the blue shaded region. These results are formalized in the following proposition.

³⁰Note that, while the slope of $(\tau_1^L, \pi_1^L) | \hat{p}_1$ in the baseline economy is constant, the slope of $(\tau_1^L, \pi_1^L) | \tilde{p}_1$ in the common information benchmark is not. In the baseline economy, the frontier between the two regimes is determined entirely by a threshold price of debt q_1 (which maps one-for-one into a threshold asset price p_1). The lender's optimality condition implies that the relationship between the set of π_1^L and τ_1^L consistent with this threshold $\hat{q}_1$ is linear. In the common information benchmark, by contrast, the date 1 regime is determined not by q_1 , but by p_1 . The relationship between the set of π_1^L and τ_1^L consistent with this threshold $\tilde{p}_1$ is nonlinear.

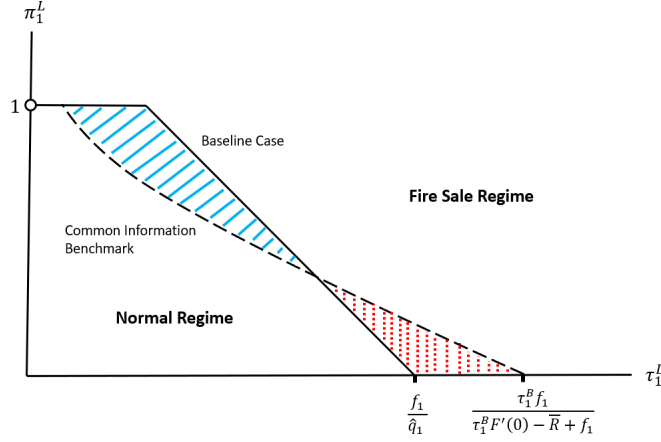


Figure 5: Comparison of Two Regimes under the Baseline Economy and the Common Information Benchmark

Note: This figure illustrates the bisection of the state space into two regimes at date 1 in the baseline case in which the lender's liquidity shock τ_1^L and beliefs π_1^L are private information, and under the common information benchmark case in which this information is directly observable by the borrower. The equilibrium is in the fire sale regime if and only if the equilibrium value of p_1 is below the threshold value $\hat{p}_1$. (For the baseline case, this corresponds to the threshold value of $\hat{q}_1$.) The solid curve in the figure plots the combinations of the states (τ_1^L, π_1^L) consistent the threshold $\hat{p}_1$, based on the lender's optimality condition for d_1^L , and denotes the frontier between the two regimes. The dashed curve plots the same frontier in the common information benchmark in which the borrower directly observes the lender's private information, and hence $\pi_1^B = \pi_1^L$ all along this curve. For both cases, the region to the southwest of these curves is the normal regime, while the northeast is the fire sale regime. The area shaded with dotted red (solid blue) lines marks the region of the state space in which fire sales occur only in the baseline economy (common information benchmark).

Proposition 2 (Effect of misinformation on the likelihood of fire sales).

- (A) *Conditional on a large adverse cost shock, the likelihood of entering a fire sale is higher in the baseline economy than in the common information benchmark.*
- (B) *Conditional on a severely adverse news shock, however, the likelihood of entering a fire sale is lower in the baseline economy.*
- (C) *The unconditional probability of a fire sale in the baseline economy relative to the common information benchmark is ambiguous.*

Proof. See appendix B.7. ■

4.3 Effect of information spillovers on the severity of fire sales

How do imperfect information spillovers affect the severity of a fire sale, conditional on one occurring at date 1? To help visualize this, we begin with Figure 4 and further partition the state space

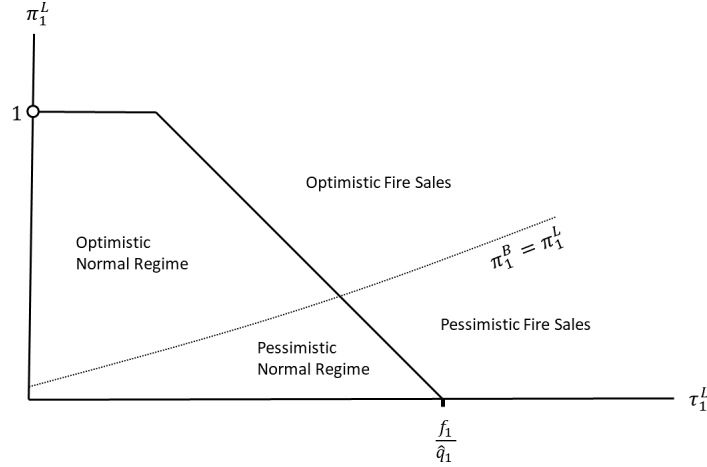


Figure 6: Four-Way Partition of the State Space

Note: This figure illustrates the demarcation of four regions of the state space by whether the equilibrium is in the normal regime or fire sale regime, and by whether the borrower is optimistic or pessimistic about the risky asset relative to the better-informed lender. The dotted curve demarcates the region of the state space in which $\pi_1^B < \pi_1^L$ in equilibrium to the northwest of the curve, and the region in which $\pi_1^B > \pi_1^L$ to the southeast. The dotted curve is constructed by finding, for a given q_1 , the borrower's posterior beliefs $(E_1^B[\tau_1^L], \pi_1^B) | q_1$ such that $\pi_1^B = \pi_1^L$, and then tracing this out at all possible realizations of q_1 . In the northwest region of the dotted curve, the borrower is optimistic about the risky asset relative to the lender (and relative to what its beliefs would be in the common information benchmark), while in the southeast region, the borrower is relatively pessimistic.

based on the relative optimism and pessimism of the lender and borrower about the asset date 1. This yields a four-way partition depicted in Figure 6. The dotted line in the figure demarcates the regions of the state-space in which the borrower is more- or less-pessimistic about the asset quality in the date 1 equilibrium than the lender. Which region the economy ends up in depends on the realization of the cost shock and news to the lender, and how this affects the borrower's perceptions of downside risk in the asset through the imperfect information spillover.

The region labeled 'Pessimistic Fire Sales' is characterized by over-pessimism on the part of the borrower. In response to a sufficiently adverse cost shock to the lender, the borrower becomes pessimistic about the quality of the asset relative to the lender due to the imperfect information spillover, and chooses to liquidate more of the risky asset as a result. The imperfect information spillover thus worsens the severity of the fire sale in this region.

In contrast, the region of the state space labeled 'Optimistic Fire Sales' features relative optimism by the borrower. When the lender receives sufficiently bad news about the asset quality, the borrower becomes relatively optimistic about the quality of the asset relative to the lender due to the imperfect information spillover, and chooses to liquidate less of the risky asset as a result. The

imperfect information spillover thus dampens the severity of the fire sale in this region.³¹

The following proposition formalizes these results by comparing the date 1 allocation in our baseline economy to that of the common information benchmark, holding date 0 variables constant.

Proposition 3 (Effect of information spillovers on the allocation in the normal and fire sale regimes). *Given the date 0 allocation, the following holds:*

- (A) *The allocation of the risky asset in the normal regime is identical in the baseline economy and the common information benchmark: $a_1^B = a_0^B$ in both economies.*
- (B) *The information spillover amplifies the severity of fire sales driven by cost shocks and dampens the severity of fire sales driven by bad news at date 1. Formally, for any state (τ_1^L, π_1^L) in the fire sale regime, a_1^B is lower in the baseline economy compared to the common information benchmark if τ_1^L is sufficiently high (exceeding some threshold $\bar{\tau}_1^L(\pi_1^L)$), and is lower if π_1^L is sufficiently high (exceeding some threshold $\bar{\pi}_1^L(\tau_1^L)$).*

Proof. See appendix B.8. ■

Part (A) shows that the imperfect information spillover does not affect the allocation of the risky asset in the normal regime at date 1; it simply affects the allocation of cash between the borrower and lender. Part (B) shows that the spillover has a destabilizing effect on the market when funding illiquidity is driven by severely adverse cost shocks, but a stabilizing effect when it is driven by bad private news about the asset quality.

4.4 Key positive takeaway

The confluence of these results suggests that the imperfect information spillovers which arise from the use of short-term collateralized debt in the money market are a double-edged sword. On the one hand, they facilitate the smooth functioning of the money market in normal times, reallocating liquidity between the borrower and lender in response to shocks and news which are not too adverse. (Indeed, the next section shows that this reallocation is constrained efficient.) In addition, they dampen the effects of adverse news about the quality of the asset on the likelihood and severity of socially costly fire sales.

On the other hand, the imperfect information spillovers increase the fragility of the money market in response to adverse cost shocks through their effects on market participants' perceptions of downside risk in the asset payoff.³² The next section shows that, if this latter effect is sufficiently

³¹The borrower's belief can be more optimistic or pessimistic compared to the lender's belief in the normal regime as well, corresponding to the other two regions shown in Figure 6.

³²Depending on the relative importance of these shocks, and the strength of these opposing effects, the information spillover may be a stabilizing or destabilizing force on net.

strong, the information spillover can worsen the allocative efficiency of the economy at date 1 on net.

Taken together, the model can help reconcile the dual nature of money markets: stable and effective at allocating liquidity during normal times, yet fragile and vulnerable to collapse under stress.

5 Welfare comparison

We define ex ante social welfare at date 1 as the sum of agents' expected utilities based on their own beliefs:

$$SW_1 \equiv E_1^L(C_2^L) + E_1^B(C_2^B) + E_1^T(C_2^T). \quad (11)$$

In appendix B.9, we discuss this in further detail. We then compare date 1 welfare in our baseline economy to a counterfactual economy in which debt contracts signed at date 0 cannot be changed at date 1 (i.e., long-term contracts without the possibility of renegotiation are in place, or equivalently, q_1 is fixed at $q_1 = q_0$). In this counterfactual economy, the borrower cannot adjust the supply of credit at date 1. This counterfactual isolates the role that short-term, as opposed to long-term, collateralized debt plays in the money market. It is therefore informative about the welfare effects of the information spillover and changes in funding liquidity, both of which result from the use of short-term debt.³³

The welfare analysis reveals how the double-edged nature of the information spillover identified in Section 4 affects welfare. While the imperfect information spillover facilitates an efficient reallocation of liquidity in normal times and the risky asset during fire sales, it also amplifies fire sales in response to adverse liquidity shocks through a pecuniary externality. The net welfare effect depends on which force dominates. This is formalized in the following proposition.

Proposition 4. *Consider a counterfactual economy in which q_1 is fixed at $q_1 = q_0$. Then the following holds:*

- (A) *Compared to the counterfactual economy with fixed q_1 , the reallocation of cash from the lender to the borrower in the normal regime of the baseline economy unambiguously increases welfare at date 1.*

³³To further isolate the role of learning, we also compare our baseline economy to an alternative benchmark in which agents receive private signals but do not update their beliefs—that is, beliefs remain fixed at their date-0 priors. Appendix D.4 shows that learning amplifies the severity of fire sales in our baseline economy relative to this stale-beliefs benchmark. This comparison confirms that it is not merely the presence of private information, but the active updating of beliefs in response to prices, that drives our results.

- (B) Compared to the counterfactual economy with fixed q_1 , the pecuniary externality from fire sales in the baseline economy unambiguously decreases welfare at date 1.
- (C) If the pecuniary externality is sufficiently severe (i.e., $F(\cdot)$ is sufficiently convex), expected welfare at date 1 in the baseline economy is lower than that in the counterfactual economy with fixed q_1 .

Proof. See appendix B.9. ■

The intuition is as follows. In the normal regime, the flexibility to adjust contract terms at date 1 allows cash to flow from agents with lower opportunity cost (the lender) to those with higher opportunity cost (the borrower). Moreover, although the information spillover cannot fully reveal the lender's private information about the asset quality, it nevertheless provides the borrower with some useful information about the quality of the asset, which changes the borrower's desired portfolio. Both of these effects improve allocative efficiency in the normal regime.

However, in the fire sale regime, this flexibility comes at a cost: A pecuniary externality arises when borrowers are forced to sell the risky asset at a fire sale price, depressing the price and forcing even more fire selling. As a result, there is an excessive transfer of the risky asset at date 1 from the borrower to the traditional sector relative to the counterfactual economy with fixed contract terms (long-term debt).

Moreover, in the fire sale regime, the imperfect information spillover worsens the severity of the pecuniary externality. The endogenous pessimism that arises due to the spillover can increase the borrower's fire sale of the risky asset, which depresses its price p_1 (i.e., it worsens market liquidity). In turn, the lower asset price forces atomistic borrowers who are fire selling the risky asset to sell even more to repay their debt.

If this pecuniary externality is sufficiently severe—captured by high convexity of the traditional sector's cost function F —this latter effect can dominate the efficiency gains from the reallocation of cash and the risky asset, making the economy worse off ex ante relative to the counterfactual economy with fixed contract terms.

6 Policy implications

Turning to the policy implications of the model we show that, during financial distress, the central bank's standard toolkit may be insufficient to restore stability to the money market. By contrast, extraordinary liquidity facilities in which the central bank lends directly against risky assets are a more powerful tool to stabilize the market because of how they affect the information transmission

of prices through a new ‘confidence channel’. We first state these results formally in the following proposition and derive them in appendix B.10.

Proposition 5.

- (A) *While standard monetary policy tools reduce the severity of fire sales by increasing liquidity, they have no effect on the formation of agents’ beliefs.*
- (B) *Collateralized central bank lending facilities can have stronger effects on reducing fire sales by eliminating the information spillover.*

Proof. See appendix B.10. ■

Standard monetary policy tools First, consider the standard toolkit of the central bank to increase the availability of liquidity in money markets, such as open market operations, lending via the discount window or a standing repo operations, or lowering the interest paid on central bank reserves. By lowering short-term interest rates or lending against very safe assets, these tools lower the opportunity cost of lending for participants in these markets. Therefore, these tools map into our setting as a change in the lender’s opportunity cost of funds τ_1^L . In doing so, these standard interventions have a stabilizing effect on the money market by increasing the date 1 supply of credit (increasing q_1) and reducing the severity of fire sales.

However, these standard tools do not alter informativeness of prices in markets for collateralized debt. Put differently, while lowering the opportunity cost of lending increases the level of credit and raises the price of debt q_1 , it does not change the sensitivity of q_1 to private news about the quality of the underlying asset. Therefore, the borrower still understands that a fall in the availability of funding may reflect bad news about the asset quality. As a result, the imperfect information spillover still operates at the margin. Stated formally, the lender’s optimality condition is still given by (1), and therefore the formation of the lender’s beliefs are still characterized by the process outlined in section 3.2.

Thus, while beneficial, standard policy tools may have a limited effect in stabilizing money markets, as they may not prevent market participants’ perceptions of risk about the quality of the underlying assets from deteriorating markedly.

Extraordinary central bank lending facilities In contrast, extraordinary central bank interventions, such as the lending facilities employed by central banks to support market functioning during the Global Financial Crisis or the onset of the COVID-19 pandemic, are a more powerful tool to stabilize money markets because of how they affect the information transmission of prices.³⁴ Under these facilities, central banks provided funding secured by the distressed risky

³⁴Examples of such facilities include the Federal Reserve’s Secondary Market Corporate Credit Facility and the European Central Bank’s Pandemic Emergency Purchase Programme.

assets to dealer banks (the lender, in our model). In turn, those dealer banks intermediated this secured funding to the hedge funds and other borrowers via money markets. In comparison to the standard toolkit, these extraordinary interventions are targeted specifically to the asset markets under stress and involve the central bank taking these distressed assets, themselves, as collateral rather than a narrower set of very safe assets.

To formalize this intervention, suppose the central bank offers collateralized loans to lenders at an exogenous price q_1^G , backed by the risky asset (see appendix B.10 for a more general setup and further details). The lender can borrow from the government by posting collateral $c_1^G \leq d_1^L$, where d_1^L is the lender's holdings of debt from the borrower. In effect, the lender intermediates collateralized debt from the borrower to the central bank.

The lender's optimization problem at date 1 becomes:

$$\begin{aligned} & \max_{\kappa_1^L, d_1^L, c_1^G} E_1^L [C_2^L] \\ \text{s.t. } & C_2^L = \tau_1^L \kappa_1^L + R_2^d d_1^L - R_2^d c_1^G \\ & \kappa_1^L - \kappa_0^L \leq d_0^L - q_1 d_1^L + q_1^G c_1^G \\ & c_1^G \leq d_1^L \end{aligned}$$

with non-negativity constraints. The second constraint shows that the lender receives $q_1^G c_1^G$ in cash from the government in exchange for posting collateral c_1^G . The third constraint is the collateral constraint: the lender can only borrow from the government using collateral it receives from the borrower.

The first-order conditions for d_1^L and c_1^G are, respectively:

$$q_1 = \frac{E_1^L [R_2^d] + \mu_1^L}{\tau_1^L} \quad (12)$$

$$q_1^G = \frac{E_1^L [R_2^d] + \mu_1^L}{\tau_1^L}, \quad (13)$$

where μ_1^L is the Lagrange multiplier on the collateral constraint. By competition among lenders, these conditions imply that the equilibrium price of private lending must equal the government-administered rate:

$$q_1 = q_1^G. \quad (14)$$

Because q_1^G is exogenous and independent of (τ_1^L, s_1^L) , (14) breaks the link between funding conditions and the lender's private information that drives the information spillover in Lemma 1.

This result is the key to understanding why extraordinary facilities are more powerful than standard policy tools. By stabilizing the price of debt at the government-administered rate q_1^G , the

lending facility not only provides liquidity, but also reduces the sensitivity of private lending to the lender's private information (τ_1^L, s_1^L) , including the private news about the quality of the underlying risky asset. By rendering those prices uninformative about the lender's private information, the policy intervention *destroys* information, impairing the ability of prices to convey the lender's private information through the information spillover.

Confidence channel of extraordinary lending facilities Counterintuitively, this destruction of information can have a stabilizing effect on the market due to a new *confidence channel* through which extraordinary central bank lending facilities affect market participants' perceptions of risk about the asset quality. During a fire sale, the extraordinary central bank facility temporarily eliminates the information spillover altogether. As a result, regardless of the underlying shock and news (τ_1^L, s_1^L) , the borrower's perceptions of risk about the asset quality remain anchored, $\pi_1^B = \pi_0^B$.

Intuitively, if borrowers know that the availability of funding is pinned down by the government rather than the lender's private information, they are less likely to become pessimistic about the asset quality in response to an additional tightening of private funding. This prevents market participants' perceptions of risk from deteriorating further in response to changes in the availability of funding liquidity, and thus mitigates the severity of the fire sale by preventing an excessive loss of confidence in the quality of the risky asset. Indeed, policymakers have frequently cited the effect of their interventions on a vague notion of investor confidence as a stabilizing force. Our model thus provides a conceptualization of this force. As shown in Proposition 4, stabilizing q_1 can increase welfare when the pecuniary externality from fire sales (captured by the convexity of F) is sufficiently severe.

The key distinction between standard and extraordinary interventions lies not in the amount of liquidity provided, but in their differential effects on information transmission. Standard tools preserve the information spillover while shifting funding conditions; extraordinary facilities eliminate the information spillover by making prices uninformative about private information. This difference explains why extraordinary facilities can be effective even with minimal actual usage, as observed during the COVID-19 crisis.

7 Alternative case with upstream information spillovers

Thus far, we have focused on the case in which the lender gets the cost shock and private signal, but the borrower does not. This implies that information spillovers flow downstream from the lender to the borrower. In this section, we briefly consider the alternative case in which the borrower gets a cost shock and private signal, but the lender does not.³⁵ As discussed in section 1.1, this case could

³⁵This assumption ensures tractability. With bilateral learning (i.e., both agents receive private signals), the equilibrium would involve a complex fixed-point in higher-order beliefs: each agent must infer the other's signal from

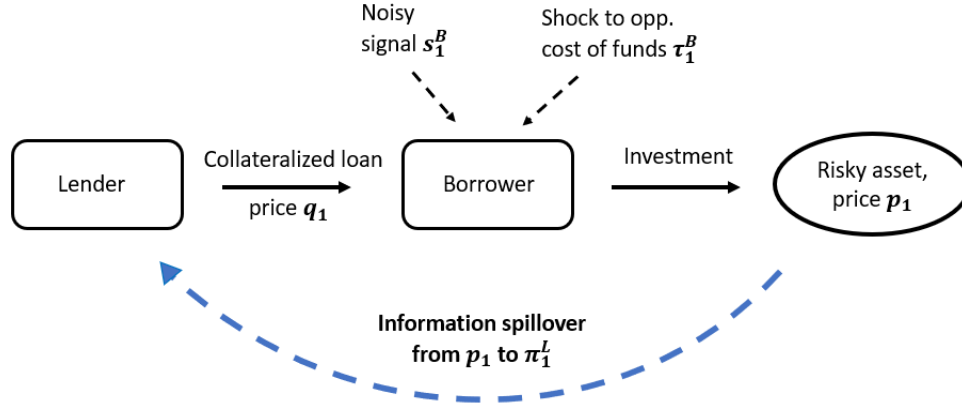


Figure 7: Alternative Case with Upstream Information Spillovers

Note: This figure stylistically illustrates the alternative case in which the borrower receives the private signal and cost shock at date 1, leading to upstream spillovers through the price of the risky asset.

be applicable to the secondary market for corporate bonds in which an asset manager finances its purchase of a corporate bond by borrowing from a dealer in an OTC market. In appendix C, we show that while the mechanism differs slightly, the fundamental insights are the same as in the baseline case. This alternative case is illustrated in figure 7.

Suppose the borrower gets a cost shock τ_1^B and private signal s_1^B , but the lender does not (i.e. τ_1^L is a fixed parameter known to both agents). Here, the price of debt q_1 is still pinned down by the lender's optimality condition and therefore conveys no information to the lender about the borrower's private information. However, the price of the risky asset p_1 is pinned down by the borrower's optimality condition and therefore conveys information to the lender about the s_1^B .

Nevertheless, the lender faces an equivalent identification problem as in our baseline case: The lender cannot separately identify the borrower's signal s_1^B and cost shock τ_1^B from the observed price p_1 .³⁶ As a result, information spillovers in this case flow upstream from the borrower to the lender rather than downstream.

In this setting, the beliefs of upstream agents are shaped in equilibrium by the availability of *market liquidity* rather than by funding liquidity, as was the case in our baseline case. Namely,

observed prices, but those prices depend on beliefs about beliefs, creating an infinite regress. While bilateral learning is tractable in some setting (e.g., global games with symmetric agents; Carlsson and Van Damme (1993)), our model features additional complexities: (i) asymmetric roles with different choice variables, (ii) two simultaneous market-clearing conditions (debt and asset markets), (iii) two-sided inference from two prices (q_1 and p_1), and (iv) discontinuities from fire sale thresholds. Models with bilateral learning in financial markets (e.g., Duffie and Manso (2007)) typically abstract from either fire sales or collateral constraints. Our one-sided information structure isolates the information spillover mechanism cleanly.

³⁶The borrower cannot credibly reveal its private information to the lender, as it has incentive to inflate the collateral value of its risky asset holdings to obtain cheaper funding.

changes in the price of the risky asset p_1 which are driven by cost shocks to the borrower affect lender's optimism or pessimism about the asset fundamental.

In this environment, the availability of market liquidity gives risk to belief-driven booms or busts in the asset market. Cost shocks to the borrower which result in a lower asset price cause the lender to become more pessimistic about the asset. As a result, the lender cuts back funding at date 1. Moreover, this reduction in funding liquidity may cause the borrower to liquidate more of the risky asset, depressing its price. Hence, there may be a feedback loop between market liquidity and funding liquidity. Thus, the nature of information spillovers and the effect of liquidity on beliefs and asset prices are essentially same as in the baseline case.

8 Conclusion

To shed light on the nature of financial stability in money markets, we developed a model of heterogeneous agents, collateralized debt, learning, and fire sales in which agents' perceptions of risk in asset fundamentals are shaped by liquidity shocks due to imperfect information spillovers. The model shows that the imperfect information spillover is a double-edged sword, serving to both improve the allocation of liquidity during normal times, but also amplify market fragility in response to some shocks through their effects on perceptions of downside risk. The model also sheds light on why, during episodes of financial distress, central banks find standard liquidity injections insufficient to restore stability to money markets and instead resort to extraordinary interventions. The model highlights the role that extraordinary interventions have on the transmission of information in the money market and its effects on market participants' perceptions of risk about asset quality.

References

- ACHARYA, V. V., D. GALE, AND T. YORULMAZER (2011): "Rollover Risk and Market Freezes," *The Journal of Finance*, 66, 1177–1209. [1](#), [1.2](#)
- ACHARYA, V. V., D. GROMB, AND T. YORULMAZER (2012): "Imperfect competition in the interbank market for liquidity as a rationale for central banking," *American Economic Journal: Macroeconomics*, 4, 184–217. [45](#)
- ACHARYA, V. V. AND O. MERROUCHE (2013): "Precautionary hoarding of liquidity and inter-bank markets: Evidence from the subprime crisis," *Review of finance*, 17, 107–160. [1](#)

- ALLEN, F., S. MORRIS, AND H. S. SHIN (2006): “Beauty Contests and Iterated Expectations in Asset Markets,” *The Review of Financial Studies*, 19, 719–752. 1.2
- ASRIYAN, V. (2021): “Balance sheet channel with information-trading frictions in secondary markets,” *The Review of Economic Studies*, 88, 44–90. 1, 1.2
- ASRIYAN, V., W. FUCHS, AND B. GREEN (2017): “Information spillovers in asset markets with correlated values,” *American Economic Review*, 107, 2007–2040. 1, 1.2
- (2021): “Aggregation and design of information in asset markets with adverse selection,” *Journal of Economic Theory*, 191, 105124. 1, 1.2
- ASRIYAN, V., L. LAEVEN, AND A. MARTIN (2022): “Collateral booms and information depletion,” *The Review of Economic Studies*, 89, 517–555. 1, 1.2
- ASRIYAN, V. AND V. VANASCO (2014): “Informed intermediation over the cycle,” *Working Paper*. 1
- BABUS, A. AND P. KONDOR (2018): “Trading and information diffusion in over-the-counter markets,” *Econometrica*, 86, 1727–1769. 1.2
- BANERJEE, S., J. DAVIS, AND N. GONDHI (2018): “When transparency improves, must prices reflect fundamentals better?” *The Review of Financial Studies*, 31, 2377–2414. 1.2, 21
- BOSTANCI, G. AND G. ORDOÑEZ (2024): “Business, Liquidity, and Information Cycles,” *NBER Working Paper*. 1, 1.2
- BOTSCH, M. AND V. VANASCO (2019): “Learning by lending,” *Journal of Financial Intermediation*, 37, 1–14. 19
- BRUNNERMEIER, M. K. AND L. H. PEDERSEN (2009): “Market Liquidity and Funding Liquidity,” *The Review of Financial Studies*, 22, 2201–2238. 1, 1.2
- CARLSSON, H. AND E. VAN DAMME (1993): “Global games and equilibrium selection,” *Econometrica*, 989–1018. 35
- CHANG, J.-W., E. KLEE, AND V. YANKOV (2025): “Rewiring repo,” *FEDS Working Paper*. 1, 2.1
- CHARI, V. V. AND R. JAGANNATHAN (1988): “Banking panics, information, and rational expectations equilibrium,” *The Journal of Finance*, 43, 749–761. 1.2

- CHEN, Y., Z. DA, AND D. HUANG (2019): “Arbitrage trading: The long and the short of it,” *The Review of Financial Studies*, 32, 1608–1646. [1](#)
- COPELAND, A., A. MARTIN, AND M. WALKER (2014): “Repo Runs: Evidence from the Tri-Party Repo Market,” *The Journal of Finance*, 69, 2343–2380. [13](#)
- DANG, T. V., G. GORTON, AND B. HOLMSTRÖM (2020): “The Information View of Financial Crises,” *Annual Review of Financial Economics*, 12, 39–65. [1.2](#)
- DÁVILA, E. AND A. KORINEK (2018): “Pecuniary Externalities in Economies with Financial Frictions,” *The Review of Economic Studies*, 85, 352–395. [1.2](#), [27](#)
- DÁVILA, E. AND C. PARLATORE (2021): “Trading costs and informational efficiency,” *The Journal of Finance*, 76, 1471–1539. [1](#), [1.2](#)
- DÁVILA, E. AND A. WALTHER (2023): “Prudential policy with distorted beliefs,” *American Economic Review*, 113, 1967–2006. [1](#), [1.2](#)
- DIAMOND, D. W. AND P. H. DYBVIG (1983): “Bank Runs, Deposit Insurance, and Liquidity,” *The Journal of Political Economy*, 401–419. [1.2](#)
- DUFFIE, D. (2010): “The Failure Mechanics of Dealer Banks,” *Journal of Economic Perspectives*, 24, 51–72. [1](#)
- DUFFIE, D. AND G. MANSO (2007): “Information percolation in large markets,” *American Economic Review*, 97, 203–209. [35](#)
- FINANCIAL CRISIS INQUIRY COMMISSION (2011): *The Financial Crisis Inquiry Report: Final Report of the National Commission on the Causes of the Financial and Economic Crisis in the United States*, U.S. Government Printing Office, chapter 15 “March 2008: The Fall of Bear Stearns and Conclusions” is especially relevant. [5](#)
- FOSTEL, A. AND J. GEANAKOPOLOS (2015): “Leverage and Default in Binomial Economies: a Complete Characterization,” *Econometrica*, 83, 2191–2229. [1.2](#)
- GALE, D. AND T. YORULMAZER (2013): “Liquidity Hoarding,” *Theoretical Economics*, 8, 291–324. [1.2](#)
- GARCIA-MACIA, D. AND A. VILLACORTA (2023): “Macroprudential policy with liquidity panics,” *The Review of Financial Studies*, 36, 2046–2090. [1](#), [1.2](#)
- GEANAKOPOLOS, J. (1997): “Promises Promises,” in *The Economy as an Evolving Complex System II*, ed. by S. D. Brian Arthur and D. Lane, CRC Press, 285–320. [1.2](#)

- (2003): “Liquidity, Default, and Crashes Endogenous Contracts in General Equilibrium,” in *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, Cambridge University Press, vol. 2, 170. 1.2
- (2010): “The Leverage Cycle,” *NBER Macroeconomics Annual*, 24, 1–65. 1.2
- GILCHRIST, S., B. WEI, V. Z. YUE, AND E. ZAKRAJŠEK (2024): “The Fed takes on corporate credit risk: An analysis of the efficacy of the SMCCF,” *Journal of Monetary Economics*, 146, 103573. 3
- GOLDSTEIN, I. AND A. PAUZNER (2005): “Demand–deposit Contracts and the Probability of Bank Runs,” *the Journal of Finance*, 60, 1293–1327. 1.2
- GORTON, G., T. LAARITS, AND A. METRICK (2020): “The run on repo and the Fed’s response,” *Journal of Financial Stability*, 48, 100744. 1
- GORTON, G. AND A. METRICK (2012): “Securitized Banking and the Run on Repo,” *Journal of Financial Economics*, 104, 425–451. 1, 13
- GORTON, G. AND G. ORDONEZ (2014): “Collateral Crises,” *American Economic Review*, 104, 343–78. 1, 1.2
- GOTTARDI, P., V. MAURIN, AND C. MONNET (2019): “A theory of repurchase agreements, collateral re-use, and repo intermediation,” *Review of Economic Dynamics*, 33, 30–56. 17
- GROSSMAN, S. J. AND J. E. STIGLITZ (1980): “On the Impossibility of Informationally Efficient Markets,” *The American Economic Review*, 70, 393–408. 1.2
- HADDAD, V., A. MOREIRA, AND T. MUIR (2021): “When selling becomes viral: Disruptions in debt markets in the COVID-19 crisis and the Fed’s response,” *The Review of Financial Studies*, 34, 5309–5351. 3
- (2025): “Whatever it takes? The impact of conditional policy promises,” *American Economic Review*, 115, 295–329. 3
- INFANTE, S. AND A. VARDOULAKIS (2021): “Collateral Runs,” *The Review of Financial Studies*, 34, 2949–2992. 1.2
- KAHN, R. J., M. MCCORMICK, V. NGUYEN, M. PADDRIK, AND H. P. YOUNG (2023): “Anatomy of the repo rate spikes in September 2019,” *Journal of Financial Crises*, 5, 1–25. 1, 2.1

- KURLAT, P. (2016): “Asset Markets with Heterogeneous Information,” *Econometrica*, 84, 33–85. 1, 1.2
- LORENZONI, G. (2008): “Inefficient Credit Booms,” *The Review of Economic Studies*, 75, 809–833. 1, 10, 1.2, 2, 3.1.3, 27
- MALHERBE, F. (2014): “Self-fulfilling liquidity dry-ups,” *The Journal of Finance*, 69, 947–970. 1, 1.2
- PARK, S. (1991): “Bank failure contagion in historical perspective,” *Journal of Monetary Economics*, 28, 271–286. 45
- PÉRIGNON, C., D. THESMAR, AND G. VUILLEMEY (2018): “Wholesale funding dry-ups,” *The Journal of Finance*, 73, 575–617. 1
- PLANTIN, G. (2009): “Learning by holding and liquidity,” *The Review of Economic Studies*, 76, 395–412. 1.2
- SIMSEK, A. (2013): “Belief Disagreements and Collateral Constraints,” *Econometrica*, 81, 1–53. 1, 1.2, 17
- SINCLAIR, A. J. (2023): “Do prime brokers intermediate capital?” *Journal of Financial Intermediation*, 53, 101004. 1
- STIGLITZ, J. E. (1981): “The allocation role of the stock market: Pareto optimality and competition,” *The Journal of Finance*, 36, 235–251. 24
- VANASCO, V. (2017): “The downside of asset screening for market liquidity,” *The Journal of Finance*, 72, 1937–1982. 1, 1.2

A Appendix: Optimization problems

A.1 Lender's problem

The representative lender solves a portfolio choice problem, deciding how much unit of collateral to accept, d_t^L for each date t , and how much cash to hold, κ_t^L for each date t , subject to its budget constraint, taking prices p_t, q_t and available contracts as given. **D.1** microfounds debt as a collateralized debt contract with c_t units of the risky asset being posted as collateral at t to be repurchased at $t + 1$.

Lender's portfolio choice problem at date 1

At date 1, the lender's portfolio choice is to decide how much to lend versus how much cash to hold. The lender's opportunity cost of lending is its date 1 marginal return of cash (which is subject to the liquidity shock) τ_1^L , since this is the date 2 return that the lender gets on its holdings of cash at date 1. If the borrower cannot finance the promised payment to the lender out of its cash holdings or with new borrowing at date 1, then either the borrower has to liquidate some of its holdings of the risky asset to the traditional sector for cash or the lender seizes the collateral and liquidates it to the traditional sector. Both of these actions are payoff-equivalent, so we ignore the confiscation of collateral by the lender and instead assume that the borrower liquidates to the traditional sector in this case.

The lender's problem at date 1 is to maximize its expected utility $E_1^L[C_2^L]$ from date 2 consumption by choosing its portfolio (cash versus a loan) subject to budget constraints at date 1 and date 2. Let d_1^L denote the lender's choice of how many units of loan to invest in (at price q_1) at date 1. (In equilibrium, the condition for market clearing for the loan will be $d_1^L = c_1$ if $c_1 > 0$. If $c_1 = 0$, then $d_1^L > c_1$ is possible because of the reason we discuss later in the lender's optimal decision. By assumption **1**, $d_1^L > 0$ is always possible even when $c_1 = 0$ as we show later.)

Lender's date 0 budget constraint At date 0, the lender allocates its cash endowment between date 0 cash holdings and the date 0 loan to the borrower.

$$\underbrace{\kappa_0^L}_{\text{date 0 cash holdings}} + \underbrace{q_0 d_0^L}_{\text{value of date 0 loan}} \leq \underbrace{e_0^L}_{\text{date 0 cash endowment}}$$

Lender's date 1 budget constraint At date 1, the flow of payments received by the lender is simply $q_0 d_0^L - q_1 d_1^L$, the difference in the value of the loan under the contract terms agreed to at date 0 minus the value of the loan under the new contract terms.

$$\underbrace{\kappa_1^L}_{\text{date 1 cash holdings}} + \underbrace{q_1 d_1^L}_{\text{value of date 1 loan}} \leq \underbrace{\kappa_0^L}_{\text{date 0 cash holdings}} + \underbrace{d_0^L}_{\text{value of date 0 loan}}$$

i.e.

$$\underbrace{\kappa_1^L - \kappa_0^L}_{\text{net increase in cash holdings}} \leq \underbrace{d_0^L - q_1 d_1^L}_{\text{net decrease in loan}}$$

Lender's date 2 budget constraint At date 2, the contract is settled or defaulted upon by the borrower, the lender earns a return from its cash holdings, and consumes. Define the date 2 proceeds from the loan as $R_2^d \equiv \min\{R, 1\}$, which accounts for the possibility of default.

$$\underbrace{C_2^L}_{\text{date 2 consumption}} = \underbrace{\tau_1^L \kappa_1^L}_{\text{proceeds from cash holdings}} + \underbrace{R_2^d d_1^L}_{\text{proceeds from date 1 loan}}$$

Lender's problem at date 1

Taking as given the date 1 contract terms and prices q_1 , and p_1 , the lender decides how much of its funds to allocate to the loan or cash. The lender's optimization problem is

$$\begin{aligned} \max_{\kappa_1^L, d_1^L} \quad & E_1^L [C_2^L] \\ \text{s.t.} \quad & C_2^L \leq \tau_1^L \kappa_1^L + R_2^d d_1^L \\ & \kappa_1^L - \kappa_0^L \leq d_0^L - q_1 d_1^L \end{aligned}$$

with non-negativity constraints:

$$\kappa_1^L \geq 0, d_1^L \geq 0.$$

Both budget constraints will bind at the optimum, so we can replace cash holdings κ_1^L using the date 1 budget constraint.

$$C_2^L = \tau_1^L (\kappa_0^L + d_0^L - q_1 d_1^L) + R_2^d d_1^L$$

Let $\xi_{d_1}^L$ and $\xi_{\kappa_1}^L$ denote the Lagrange multipliers on the respective non-negativity constraints. The Lagrangian for the lender's date 1 problem is

$$L_1^L = E_1^L \left[U_L \left(\tau_1^L (\kappa_0^L + d_0^L - q_1 d_1^L) + R_2^d d_1^L \right) \right] + \xi_{d_1}^L d_1^L + \xi_{\kappa_1}^L (\kappa_0^L + d_0^L - q_1 d_1^L)$$

The lender's first-order condition (FOC) for d_1^L is

$$q_1 = \frac{E_1^L [R_2^d] + \xi_{d_1}^L}{\tau_1^L + \xi_{\kappa_1}^L}. \quad (15)$$

This says that the lender chooses how much to lend to equalize the discounted marginal return from lending at date 1 to the discounted marginal return on its opportunity cost from holding cash at date 1.

We also have the two complementary slackness conditions:

$$\xi_{d_1}^L d_1^L = 0$$

$$\xi_{\kappa_1}^L \kappa_1^L = 0$$

Note that there is no d_1^L in (15) as the lender's problem is linear. The lender is indifferent across different values of d_1^L as long as the price of the contract is equal to the expected return of the loans, $E_1^L [R_2^d] / \tau_1^L$, the ratio of the discounted marginal return from lending at date 1 to the discounted marginal return on its opportunity cost from holding cash at date 1. Note that assumption 1 ensures that the lender always holds a strictly positive quantity of cash in equilibrium at date 1, $\kappa_1^L > 0$.³⁷ Therefore, the lender's Lagrange multiplier for the non-negativity constraint satisfies $\xi_{\kappa_1}^L = 0$ in equilibrium. Moreover, assumption 1 ensures that, lending takes place in equilibrium so that $d_1^L > 0$ and $\xi_{d_1}^L = 0$ in equilibrium.³⁸ Therefore, given these assumptions, the lender's optimality condition (15) reduces in equilibrium to

$$q_1 = \frac{E_1^L [R_2^d]}{\tau_1^L}. \quad (16)$$

Thus, in equilibrium, the competitive spot price of the contract at date 1, q_1 , will ensure that this condition holds.³⁹

³⁷The lender's optimality condition implies that, in equilibrium, the lender is just indifferent between holding cash and not, which implies that the lender's Lagrange multiplier from the non-negativity constraint satisfies $\xi_{\kappa_1}^L = 0$ in equilibrium. Otherwise the lender's expected discounted return from lending would exceed its opportunity cost of funds.

³⁸Note that assumption 1 ensures that the lender's date 0 endowment e_0^L is large enough to cover all the loans even when the lender is maximally optimistic about the quality of the risky asset and believes $R = \bar{R}$ with probability 1.

³⁹To see this intuitively, suppose that q_1 is less than the expected discounted return of a loan, $q_1 < \frac{E_1^L [R_2^d]}{\tau_1^L}$. Then the lender would want to maximize its holdings of the loans and hold no cash, so that $\kappa_1^L = 0$. But assumption 1 ensured that $\kappa_1^L > 0$, so that we have a contradiction. Suppose instead that we have $q_1 > \frac{E_1^L [R_2^d]}{\tau_1^L}$. Then, the lender would

A.2 Borrower's problem

At date 1, the representative borrower solves a portfolio choice and chooses how much to borrow, subject to its budget constraint, collateral constraint, and taking as given the price of the risky asset p_1 and the price of the contract q_1 . At date 1, borrowers cannot initiate new risky assets. Moreover, borrowers can exchange their extant holdings of the risky asset to cash by selling it to the traditional sector, at the market price p_1 and investing the proceeds in cash for total date 2 return of $p_1 \tau_1^B$. Hence, the borrower's opportunity cost of holding the risky asset is given by τ_1^B/p_1 .

At date 1, the borrower must repay its date 0 debt. We show in internet appendix D.1 that debt is microfounded as a collateralized debt contract with c_t of the risky asset being posted as collateral at t to be repurchased at $t + 1$ at a unit price of f_t , normalized to $f_t = 1$. Therefore, to repay its debt at date 1, the borrower must buy back from the lender the c_0 units of the risky asset that were posted at collateral. However, the borrower obtains no date 1 cash flow from its holdings of the risky asset at date 1. Therefore, the borrower can finance this repayment of c_0 in three ways: out of any cash holdings at date 1, by raising new debt at date 1, or by liquidating the risky asset in the date 1 spot market for the risky asset.

The borrower can raise new debt at date 1 (i.e. refinance its date-0 debt) using the date-1 contract—that is, by posting c_1 units of the risky asset as collateral in exchange for $q_1 c_1$ units of the consumption good as a loan, where q_1 is the competitive spot price of this contract. If the debt raised at date 1 $q_1 c_1$ is insufficient to cover the repayment c_0 the borrower must make at date 1, this difference must be financed either out of the borrower's cash holdings stored from date 0, or by selling some portion of its holdings of the risky asset in the spot market.

Define κ_0^B as the amount of cash the borrower carries into date 1 from date 0. Now this decision is relevant only in the date-0 optimization problem. For the date-1 problem, the borrower considers it as a predetermined amount of cash κ_0^B .⁴⁰

Collateral constraint at date 1 Let a_0^B denote the borrower's total risky asset holdings brought to date 1 from date 0, and c_0 denote the borrower's date 0 risky asset holdings used as collateral at date 0. (We will later show that, in equilibrium, $c_0 = a_0^B$.) Let a_1^B denote the borrower's total risky asset holdings chosen at date 1, and c_1 denote the amount of the borrower's date 1 asset holdings used as collateral at date 1. (Note that the amount of asset sold by the borrower at date 1 is then $a_0^B - a_1^B$.) The 'collateral constraint' is then $c_1 \leq a_1^B$.

Borrower date 0 budget constraint At date 0, the borrower has a cash endowment and receives a cash loan from the lender, which it can allocate between date-0 cash holdings and in-

want to minimize d_1^L to negative infinity without the non-negativity constraint so that the non-negativity constraint of d_1^L would bind, $d_1^L = 0$. But assumption 1 ensured that $d_1^L > 0$ in equilibrium, so that we again have a contradiction.

⁴⁰We will see in the date-0 problem that the borrowers financed their payment to purchase c_0 assets with the price p_0 at date 0 by their cash endowments e_0^B and the borrowing amount $c_0 p_0 (1 - m_0) = c_0 q_0$, where m_0 is the margin of the contract. Therefore, the leftover cash can be defined as $\kappa_0^B \equiv e_0^B + c_0 p_0 (1 - m_0) - c_0 p_0 = e_0^B - c_0 p_0 m_0$.

vestment in the risky asset.

$$\underbrace{q_0 c_0}_{\text{date 0 loan}} + \underbrace{e_0^B}_{\text{date 0 cash endowment}} \geq \underbrace{p_0 a_0^B}_{\text{value of date 0 asset holdings}} + \underbrace{\kappa_0^B}_{\text{date 0 cash holdings}}$$

Borrower date 1 budget constraint At date 1, if the borrower defaults on its date-0 debt obligations, it loses its collateral and has only its cash holdings to find its portfolio choices at date 1. In equilibrium, the borrower never defaults at date 1, because the selling all the collateral would be sufficient in repaying the debt by assumption 2. For this reason, we omit this case. In the event that the borrower does not default at date 1, its date 1 budget constraint is

$$\underbrace{q_1 c_1}_{\text{date 1 loan}} + \underbrace{p_1 a_0^B}_{\text{value of date 0 asset holdings}} + \underbrace{\kappa_0^B}_{\text{date 0 cash holdings}} \geq \underbrace{p_1 a_1^B}_{\text{value of date 1 asset holdings}} + \underbrace{c_0}_{\text{date 0 loan}} + \underbrace{\kappa_1^B}_{\text{date 1 cash holdings}}$$

i.e.

$$\underbrace{q_1 c_1 - c_0}_{\text{net increase in loan}} \geq \underbrace{p_1 (a_1^B - a_0^B)}_{\text{net increase in asset holdings}} + \underbrace{\kappa_1^B - \kappa_0^B}_{\text{net increase in cash holdings}}.$$

Borrower date 2 budget constraint The borrower's date 2 budget constraint limits date 2 consumption C_2^B by the return on the borrower's portfolio of assets held at date 1.

$$\underbrace{C_2^B}_{\text{date 2 consumption}} \leq \underbrace{\tau_1^B \kappa_1^B}_{\text{proceeds from cash holdings}} + \underbrace{a_1^B R}_{\text{proceeds from asset holdings}} - \underbrace{c_1 R_2^d}_{\text{repayment of debt}},$$

where $R_2^d \equiv \min\{R, 1\}$ denotes the realized date 2 repayment of the loan (including the possibility of default).

Each borrower takes the price q_1 of the contract at date 1 as given. In equilibrium, the price q_1 will be informative to the borrower about the lender's private information at date 1. We discuss how the borrower's beliefs evolve in section 3.2. We first characterize the borrower's optimality conditions, conditional on its information set at date 1.

Borrower's optimization problem at date 1

The borrower's optimization problem at date 1 is to make its portfolio and borrowing decision to maximize expected date 2 utility $E_1^B [C_2^B]$, conditional on its date 1 information set, taking prices as p_1 and q_1 as given.

$$\max_{c_1^B, a_1^B, \kappa_1^B} E_1^B [C_2^B] \quad (17)$$

$$\begin{aligned} \text{s.t. } & q_1 c_1 - c_0 \geq p_1 (a_1^B - a_0^B) + \kappa_1^B - \kappa_0^B, \\ & C_2^B \leq \tau_1^B \kappa_1^B + a_1^B R - c_1 R_2^d \\ & c_1 \leq a_1^B, \\ & c_1 \geq 0, a_1^B \geq 0, \kappa_1^B \geq 0 \end{aligned} \quad (18)$$

The first-order condition for c_1 is

$$(\tau_1^B + \xi_{\kappa_1}^B) q_1 - E_1^B [R_2^d] - \mu_1^B + \xi_{c_1}^B = 0, \quad (19)$$

while the first-order condition for a_1^B is

$$-(\tau_1^B + \xi_{\kappa_1}^B) p_1 + E_1^B [R] + \mu_1^B + \xi_{a_1}^B = 0. \quad (20)$$

The complementary slackness conditions are given by

$$\mu_1^B (a_1^B - c_1) = 0 \quad (21)$$

$$\xi_{c_1}^B c_1 = 0 \quad (22)$$

$$\xi_{a_1}^B a_1^B = 0 \quad (23)$$

$$\xi_{\kappa_1}^B \kappa_1^B = 0 \quad (24)$$

We can further characterize the borrower's behavior by combining the FOCs for the case with risk-neutrality. Suppose that the collateral constraint is binding in equilibrium so that $\mu_1^B > 0$. (We show in lemma 4 in appendix B.4 that this must be the case.) Then, the FOCs for c_1 and a_1^B hold with equality. Suppose also that $c_1 = a_1^B > 0$ in equilibrium. Combining these binding first order conditions yields

$$\underbrace{E_1^B [R_2^d] - q_1 \tau_1^B}_{\text{cost of financing asset holdings}} + \underbrace{\xi_{\kappa_1}^B (p_1 - q_1)}_{\text{shadow value of cash}} = \underbrace{E_1^B [R] - \tau_1^B p_1}_{\text{expected net return of risky asset}} \quad (25)$$

It is convenient to define $v_1(\pi_1^B, q_1) \equiv E_1^B[R_2^d] - q_1 \tau_1^B$ and $\eta_1(\pi_1^B; p_1) \equiv E_1^B[R] - \tau_1^B p_1$, so that (25) can be expressed as

$$\underbrace{v_1(\pi_1^B, q_1)}_{\text{cost of financing asset holdings}} + \underbrace{\xi_{\kappa_1}^B(p_1 - q_1)}_{\text{shadow value of cash}} = \underbrace{\eta_1(\pi_1^B; p_1)}_{\text{expected net return of risky asset}} \quad (26)$$

If $E_1^B[R] - \tau_1^B p_1 > E_1^B[R_2^d] - q_1 \tau_1^B$, i.e. if the net return from buying asset exceeds the net return from offering collateral (balancing the benefit from a larger loan with the cost of a larger repayment), then we must have that $\xi_{\kappa_1}^B(q_1 - p_1) < 0$, which requires that $\xi_{\kappa_1}^B > 0$ (non-negativity constraint for κ_1^B binding) and $q_1 < p_1$, which we will show in lemma 4 holds in equilibrium. In this case, the borrower is at a corner solution in its portfolio choice.

A.3 Traditional sector's problem

The traditional sector becomes relevant at date 1 only and enters the period with a new endowment of cash $\kappa_1^T = e_0^T$. It can store cash between dates 1 or 2, or buy existing risky assets at date 1 in the spot market. Existing assets need to be managed subject to some increasing costs. The date-2 return of holding a_1^T of the risky asset at date one, net of these management costs, is given by $F(a_1^T)$ units of the consumption good, where $F' > 0$ and $F'' < 0$. Without loss of generality, assume $\tau_1^T = 1$. The traditional sector gets utility only from date-2 consumption C_2^T with linear utility. The traditional sector's optimization problem is to choose how to allocate its endowment between cash or the risky asset to maximize its utility.

$$\begin{aligned} \max_{\kappa_2^T, a_1^T} \quad & C_2^T \\ \text{s.t.} \quad & C_2^T = \kappa_2^T + F(a_1^T) \\ & \kappa_1^T = \kappa_2^T + p_1 a_1^T \\ & a_1^T \geq 0 \\ & \kappa_2^T \geq 0 \end{aligned} \quad (27)$$

For simplicity, and without loss of generality, we assume the traditional sector's endowment of cash is sufficiently large that its non-negativity constraint on date-2 cash holdings is never binding, $\kappa_2^T > 0$, by assumption 1. This ensures that, in equilibrium, the traditional sector could buy all existing assets.⁴¹ As a result, the traditional sector will always enter date 2 with positive cash

⁴¹To see this, note that if the traditional sector holds all the assets, then $a_1^T = a_0^B$. So its date 1 expenditure is $p_1 a_0^B$. Note also that in the eq where the traditional sector is holding assets, we have $p_1 = F'(a_1^T)$, and so this expenditure is $p_1 a_0^B = F'(a_0^B) a_0^B$. Then the above assumption guarantees that the traditional sector has enough cash endowment at

holdings κ_2^T , so the non-negativity constraint $\kappa_2^T \geq 0$ is non-binding. Using binding date-1 and -2 budget constraints, the traditional sector's date-1 problem is to choose a_1^T to maximize $\kappa_1^T - p_1 a_1^T + F(a_1^T)$ subject to $a_1^T \geq 0$. the traditional sector's optimality condition is

$$p_1 = F'(a_1^T) + \xi_{a_1}^T. \quad (28)$$

B Appendix: Omitted proofs

B.1 Derivation of the lender's posterior beliefs

First, note that the bounds of the posterior belief of the lender is $\pi_1^L \in (0, 1)$. The bounds are derived from the fact that $s_1^L = R + \varepsilon$ with $\varepsilon \in (-\infty, \infty)$, so any realization of s_1^L is possible from both state of the world $R = \bar{R}$ and $R = \underline{R}$. The lender updates posterior beliefs as in (6)

$$\pi_1^L = \frac{\pi_0 \lambda_\varepsilon(\varepsilon_1^L = s_1^L - \underline{R})}{\pi_0 \lambda_\varepsilon(\varepsilon_1^L = s_1^L - \underline{R}) + (1 - \pi_0) \lambda_\varepsilon(\varepsilon_1^L = s_1^L - \bar{R})}.$$

This expression of π_1^L is possible by using Bayes' rule for events with a positive measure, and then taking limits. For any $\delta > 0$, the events $\{s_1^L - \underline{R} \leq \varepsilon_1^L \leq s_1^L - \underline{R} + \delta\}$ and $\{s_1^L - \bar{R} \leq \varepsilon_1^L \leq s_1^L - \bar{R} + \delta\}$ are well defined and have positive probability. Therefore, Bayes' rule implies

$$\begin{aligned} & P(R = \underline{R} | s_1^L - \underline{R} \leq \varepsilon_1^L \leq s_1^L - \underline{R} + \delta) \\ &= \frac{\pi_0 P(s_1^L - \underline{R} \leq \varepsilon_1^L \leq s_1^L - \underline{R} + \delta)}{\pi_0 P(s_1^L - \underline{R} \leq \varepsilon_1^L \leq s_1^L - \underline{R} + \delta) + (1 - \pi_0) P(s_1^L - \bar{R} \leq \varepsilon_1^L \leq s_1^L - \bar{R} + \delta)} \\ &= \frac{\pi_0 \int_{s_1^L - \underline{R}}^{s_1^L - \underline{R} + \delta} \varepsilon' d\Lambda_\varepsilon(\varepsilon')}{\pi_0 \int_{s_1^L - \underline{R}}^{s_1^L - \underline{R} + \delta} \varepsilon' d\Lambda_\varepsilon(\varepsilon') + (1 - \pi_0) \int_{s_1^L - \bar{R}}^{s_1^L - \bar{R} + \delta} \varepsilon' d\Lambda_\varepsilon(\varepsilon')}, \end{aligned} \quad (29)$$

date one to make this purchase.

where P is the probability function and both the denominator and numerator are positive. As $\delta \rightarrow 0$, we have

$$\begin{aligned} P(R = \underline{R} | s_1^L - \underline{R} \leq \varepsilon_1^L \leq s_1^L - \underline{R} + \delta) &\rightarrow P(R = \underline{R} | \varepsilon_1^L = s_1^L - \underline{R}) = \pi_1^L \\ \int_{s_1^L - \underline{R}}^{s_1^L - \underline{R} + \delta} \varepsilon' d\Lambda_\varepsilon(\varepsilon') &\rightarrow \lambda_\varepsilon(\varepsilon_1^L = s_1^L - \underline{R}) \\ \int_{s_1^L - \bar{R}}^{s_1^L - \bar{R} + \delta} \varepsilon' d\Lambda_\varepsilon(\varepsilon') &\rightarrow \lambda_\varepsilon(\varepsilon_1^L = s_1^L - \bar{R}). \end{aligned}$$

Thus, taking the limit of $\delta \rightarrow 0$ on both sides of (29) results in (6).

B.2 Proof of lemma 1

Proof.

Proof of Part (A): B 's information at date 1, I_1^B , is given by the set of all variables observable to all agents, which are prices p_1 and q_1 . In contrast, L has additional information of τ_1^L and s_1^L , denoted by I_1^L , on top of the publicly available information. By the law of iterated expectation

$$E_1^B [E_1^L [x]] = E [E [x | I_1^L] | I_1^B] = E [x | I_1^B] = E_1^B [x].$$

Applying the definition of posterior beliefs $\pi_1^i = E_1^i [\mathbb{1}\{R = \underline{R}\}]$ for $i = L, B$ leads to the statement.

Proof of Part (B): By the first statement of lemma 1 and (8), we have

$$\pi_1^B(q_1) = \int_0^1 \pi \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q_1} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q_1} < \bar{\tau}^L \right\} dG_\pi(\pi)$$

for a given q_1 .

First, note that there exists $\hat{\pi}(q)$ such that

$$\frac{1 - (1 - \underline{R})\hat{\pi}(q)}{q} = \tau_0$$

for any $q < q_0$. Also, note that $\hat{\pi}(q)$ is decreasing in q . From (1), we can rearrange the difference between the prior for τ , τ_0 , and realized τ for a given π and equilibrium price q as

$$|\tau - \tau_0| = \left| \frac{1 - (1 - \underline{R})\pi}{q} - \tau_0 \right|,$$

which can be further simplified as

$$\left| \frac{1 - (1 - \underline{R})\pi}{q} - \tau_0 \right| = \frac{1}{q} |1 - (1 - \underline{R})\pi - 1 + (1 - \underline{R})\hat{\pi}(q)| = \frac{1}{q} |(1 - \underline{R})(\pi - \hat{\pi}(q))|. \quad (30)$$

We claim that for any π and $q' > q$,

$$\left| \frac{1 - (1 - \underline{R})\pi}{q} - \tau_0 \right| < \left| \frac{1 - (1 - \underline{R})\pi}{q'} - \tau_0 \right| \quad (31)$$

and therefore,

$$\lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q} - \tau_0 \right) > \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q'} - \tau_0 \right) \quad (32)$$

by the functional form assumption on λ_T .

By equation (30), (31) is true if

$$\frac{1}{q} |(1 - \underline{R})(\pi - \hat{\pi}(q))| < \frac{1}{q'} |(1 - \underline{R})(\pi - \hat{\pi}(q'))|$$

holds. The previous equation holds if

$$\frac{q'}{q} < \frac{|\pi - \hat{\pi}(q')|}{|\pi - \hat{\pi}(q)|}$$

holds, which is equivalent to

$$\frac{\frac{1 - (1 - \underline{R})\hat{\pi}(q')}{\tau_0}}{\frac{1 - (1 - \underline{R})\hat{\pi}(q)}{\tau_0}} = \frac{\frac{1}{1 - \underline{R}} - \hat{\pi}(q')}{\frac{1}{1 - \underline{R}} - \hat{\pi}(q)} < \frac{|\pi - \hat{\pi}(q')|}{|\pi - \hat{\pi}(q)|} \quad (33)$$

by the definition of $\hat{\pi}(q)$.

First, consider the case in which π is either $\pi > \hat{\pi}(q)$ or $\pi < \hat{\pi}(q')$. Then, (33) becomes

$$\frac{\frac{1}{1 - \underline{R}} - \hat{\pi}(q')}{\frac{1}{1 - \underline{R}} - \hat{\pi}(q)} < \frac{\pi - \hat{\pi}(q')}{\pi - \hat{\pi}(q)}. \quad (34)$$

Note that for any x and $0 < b < c$, $\frac{x-b}{x-c}$ is decreasing in x , because

$$\frac{\partial \left(\frac{x-b}{x-c} \right)}{\partial x} = \frac{b-c}{(x-c)^2} < 0.$$

Thus, (34) holds because $\pi \leq 1 < \frac{1}{1-\underline{R}}$.

Finally, consider the case in which π is $\hat{\pi}(q') < \pi < \hat{\pi}(q)$. Then,

$$\frac{\pi - \hat{\pi}(q')}{\pi - \hat{\pi}(q)} < \frac{|\pi - \hat{\pi}(q')|}{|\pi - \hat{\pi}(q)|}$$

holds and by (34), (33) also holds. Therefore, (31) and (32) hold.

By the claim, the support of the conditional expectation is decreasing in q_1 for each given realization of π and for the given pdf λ_T as

$$\lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q} < \bar{\tau}^L \right\} > \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q'} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q'} < \bar{\tau}^L \right\}$$

for any $q < q'$. Thus, π_1^B is decreasing in q_1 . ■

B.3 Solving the borrower's optimization problem

$$\begin{aligned} & \max_{c_1, f_1, a_1^B, \kappa_1^B} E_1^B [C_2^B] \\ & \text{s.t. } q_1 c_1 - c_0 \geq p_1 (a_1^B - a_0^B) + \kappa_1^B - \kappa_0^B, \\ & \quad C_2^B \leq \tau_1^B \kappa_1^B + a_1^B R - c_1 R_2^d(f_1) \\ & \quad c_1 \leq a_1^B, \\ & \quad c_1 \geq 0, a_1^B \geq 0, \kappa_1^B \geq 0 \end{aligned} \tag{35}$$

Note that the date 2 budget constraint implies

$$C_2^B = \tau_1^B \kappa_1^B + a_1^B R - c_1 R_2^d(f_1),$$

while the date 1 budget constraint implies

$$\kappa_1^B = q_1 c_1 - c_0 - p_1 (a_1^B - a_0^B) + \kappa_0^B.$$

Combining these two yields

$$C_2^B = \tau_1^B (q_1 c_1 - c_0 - p_1 (a_1^B - a_0^B) + \kappa_0^B) + a_1^B R - c_1 R_2^d(f_1)$$

Therefore we can write the Lagrangian as

$$L_1^B = E_1^B [C_2^B] + \mu_1^B (a_1^B - c_1^B) + \xi_{c_1}^B c_1^B + \xi_{a_1}^B a_1^B + \xi_{\kappa_1}^B (q_1 c_1^B - c_0^B + p_1 [a_0^B - a_1^B] + \kappa_0^B) \quad (36)$$

where we have $R_2^d(f_1) \equiv \min\{R, f_1\}$, and $E[R_2^d(f_1)] = (1 - \pi_1)f_1 + \pi_1 \min\{\underline{R}, f_1\}$. Replacing q_1 yields

$$C_2^B = \tau_1^B (q_1 c_1^B - c_0^B - p_1 (a_1^B - a_0^B) + \kappa_0^B) + a_1^B R - c_1^B R_2^d(f_1).$$

Then, the expectation is given by

$$\begin{aligned} E_1^B [C_2^B] &= E_1^B \left[\tau_1^B (q_1 c_1^B - c_0^B - p_1 (a_1^B - a_0^B) + \kappa_0^B) + a_1^B R - c_1^B R_2^d(f_1) \right] \\ E_1^B [C_2^B] &= \tau_1^B (q_1 c_1^B - c_0^B - p_1 (a_1^B - a_0^B) + \kappa_0^B) + a_1^B E_1^B [R] - c_1^B E_1^B [R_2^d(f_1)] \end{aligned}$$

Hence, the Lagrangian is:

$$L_1^B = E_1^B [C_2^B] + \mu_1^B (a_1^B - c_1^B) + \xi_{c_1}^B c_1^B + \xi_{a_1}^B a_1^B + \xi_{\kappa_1}^B (c_1^B q_1 - c_0^B + p_1 [a_0^B - a_1^B] + \kappa_0^B) \quad (37)$$

where $E_1^B [C_2^B]$ is given by the above.

FOC for c_1^B :

$$\begin{aligned} \frac{dE_1^B [C_2^B]}{dc_1^B} - \mu_1^B + \xi_{c_1}^B + \xi_{\kappa_1}^B q_1 &= 0 \\ \tau_1^B q_1 - E_1^B [R_2^d(f_1)] - \mu_1^B + \xi_{c_1}^B + \xi_{\kappa_1}^B q_1 &= 0 \\ (\tau_1^B + \xi_{\kappa_1}^B) q_1 - E_1^B [R_2^d(f_1)] - \mu_1^B + \xi_{c_1}^B &= 0 \end{aligned}$$

FOC for a_1^B :

$$\begin{aligned}
\frac{dE_1^B [C_2^B]}{da_1^B} + \mu_1^B + \xi_{a_1}^B - \xi_{\kappa_1}^B p_1 &= 0 \\
-\tau_1^B p_1 + E_1^B [R] + \mu_1^B + \xi_{a_1}^B - \xi_{\kappa_1}^B p_1 &= 0 \\
-(\tau_1^B + \xi_{\kappa_1}^B) p_1 + E_1^B [R] + \mu_1^B + \xi_{a_1}^B &= 0
\end{aligned}$$

B.4 Statement and proof of binding collateral constraint

Lemma 4 (Collateral Constraint). *In equilibrium, the following holds:*

- (A) *The borrower's collateral constraint is generically binding, so that $c_1 = a_1^B$.*
- (B) *The price of the asset exceeds the price of the contract, $p_1 > q_1$.*

Proof.

Proof of Part (A):

By replacing μ_1^B in (19) with (20) and using the contract price q_1 that makes L to lend in a positive amount:

$$q_1 (\tau_1^B + \xi_{\kappa_1}^B) + E_1^B [R] - E_1^B [R^d] - \tau_1^B p_1 + \xi_{a_1}^B - \xi_{\kappa_1}^B p_1 + \xi_{c_1}^B = 0,$$

which can be rearranged as

$$E_1^B [R - R^d] + \xi_{c_1}^B + \xi_{a_1}^B = (\tau_1^B + \xi_{\kappa_1}^B) (p_1 - q_1).$$

Consider the case in which B borrows a positive amount so $a_1, c_1 > 0$. Then, the above equality can be rearranged as

$$\tau_1^B + \xi_{\kappa_1}^B = \frac{E_1^B [R - R^d]}{p_1 - q_1} = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{p_1 - q_1}, \quad (38)$$

which implies that the expected return of holding the asset with the collateralized debt net of B 's shadow value of asset equals to the sum of cash return of B and B 's shadow value of cash. The return of the leveraged asset holding on the right-hand side of (38) can be greater than return of B 's storage technology, but then B has to exhaust all the cash, $\xi_{\kappa_1}^B > 0$, and B cannot pay for additional down payment (or cash collateral or variation margin), $p_1 - q_1(f_1)$. In addition, B might want to

generate more asset because the return of the leveraged asset holding is exceedingly profitable. If this is the case, then the price will adjust to the point that the equality holds.

Now we show that the collateral constraint is generically binding. If B is borrowing zero amount—that is, $c_1 = 0$, B does not have enough cash to repay the loans to L unless B liquidates all the assets, implying $a_1^B = c_1 = 0$. B borrows a positive amount $c_1 > 0$ only if the return from rolling over the debt is greater than or equal to the cash return. Thus,

$$\tau_1^B + \xi_{\kappa_1}^B = \frac{E_1^B[R - R^d]}{p_1 - q_1} \quad (39)$$

holds in any equilibrium with $c_1 > 0$. If the collateral constraint is not binding, then by (20),

$$\tau_1^B + \xi_{\kappa_1}^B = \frac{E_1^B[R]}{p_1} \quad (40)$$

holds. (39) and (40) imply that

$$\tau_1^B + \xi_{\kappa_1}^B = \frac{E_1^B[R^d]}{q_1} = \frac{(1 - \pi_1^B) + \pi_1^B R}{q_1},$$

which holds only at non-generic realization of q_1 as π_1^B is decreasing in q_1 . The return from leveraging the asset purchase, (39), should exceed the return from the asset purchase without leverage, (40), because otherwise B will not purchase with leverage—that is, $c_1 = 0$ —which is a contradiction. Even if the previous equation holds, which is a not a generic case of parameters, B is indifferent between purchasing the asset with leverage and without leverage. Therefore, we can impose a tie-breaking rule for B , choosing to maximize c_1 when indifferent.⁴² Hence, B leverages all of its asset purchase as $c_1 = a_1^B$, and the collateral constraint is binding.

Proof of Part (B):

Intuitively, this condition $p_1 > q_1$ implies that value of the risky asset p_1 exceeds its collateral value q_1 . The borrower's optimality condition (25), which can be rearranged as

$$(\xi_{\kappa_1}^B + \tau_1^B)(p_1 - q_1) = E_1^B[R] - E_1^B[R_2^d(f_1)]$$

$$p_1 - q_1 = \frac{1}{\xi_{\kappa_1}^B + \tau_1^B} \left[E_1^B[R] - E_1^B[R_2^d(f_1)] \right].$$

⁴²This tie-breaking rule can be justified by assuming that B obtains infinitesimally small utility of holding an asset as collateral.

Since $\xi_{\kappa_1}^B \geq 0$ and $\tau_1^B > 0$, we have $p_1 > q_1$ if and only if

$$\begin{aligned} E_1^B [R] &> E_1^B [R_2^d(f_1)] \\ (1 - \pi_1^B)\bar{R} + \pi_1^B \underline{R} &> (1 - \pi_1^B)f_1 + \pi_1^B \underline{R} \\ (1 - \pi_1^B)(\bar{R} - f_1) &> 0 \end{aligned}$$

This holds since $\bar{R} > 1$ by assumption. ■

B.5 Proof of lemma 2

Proof. The proof is based on the full characterization of date-1 equilibrium in proposition 7 in appendix D.2.

As in Case 1 in proposition 7, $p_1 > F'(0)$ implies that the borrower is not selling any assets to the traditional sector, and the asset price equals the fundamental value of the asset based on the borrower's beliefs. Whenever $p_1 < F'(0)$ in equilibrium, the asset is sold to the traditional sector in a positive amount, i.e. there are fire sales, as more sales to the traditional sector depresses prices as $F'(a_1^T) < F'(0)$. ■

B.6 Proof of proposition 1

Proof. The proof is based on the full characterization of date-1 equilibrium in proposition 7 in appendix D.2.

Proof of Part (A):

By (6), π_1^L increases as s_1^L decreases. Also, q_1 is decreasing in π_1^L and τ_1^L by (1), implying the first half of the statement.

In Case 2 in proposition 7, lower q_1 results in larger a_1^T and lower p_1 , as

$$q_1 + \frac{c_0 - q_1 a_0^B - \kappa_0^B}{a_1^T}$$

is decreasing in q_1 because $a_0^B \geq a_1^T$, and

$$p_1 = F'(a_1^T) = q_1 + \frac{c_0 - q_1 a_0^B - \kappa_0^B}{a_1^T}.$$

Therefore, a decrease in q_1 will increase the liquidity shortage of the borrower, leading to a larger amount of fire sales and lower asset price.

Moreover, the borrower's posterior belief π_1^B is decreasing in q_1 by lemma 1. In Case 3 in proposition 7, higher π_1^B results in lower price through this belief channel. This is because the borrower's lower valuation of the asset should be met by the lower marginal valuation of the traditional sector through the decrease in $F'(a_1^T)$, i.e. larger a_1^T .

Proof of Part (A):

In Case 2 in proposition 7, the borrower sells the asset because the borrower has to repay the date-0 debt contract and their cash holdings are not sufficient as $q_1 a_0^B < c_0^B - \kappa_0^B$. However, the borrower still believes the fundamental value of the asset is above the marginal valuation of the traditional sector evaluated at 0, $F'(0)$, which is always above the market price as $F'(0) > F'(a_1^T) = p_1$. Therefore, we have $p_1^B > F'(0) > p_1$ in a liquidity-driven fire sale. In Cases 3 and 4 in proposition 7, the borrower's valuation of the asset is below the marginal valuation of the traditional sector evaluated at 0, as $p_1^B < F'(0)$, which is why the borrower sells the asset, i.e. belief-driven fire sale. In Case 3, the borrower has an interior solution, so that $p_1^B = F'(a_1^T) = p_1$. However, in Case 4, the borrower values the asset even less than the traditional sector does, implying that $p_1^B < F'(a_0^B) = p_1$. ■

B.7 Proof of proposition 2

Proof. From lemma 7 and lemma 8 in appendix D.6, the frontiers of the baseline economy and the common information benchmark intersects only once in the domain of $\pi_1^L \leq 1$. Denote that intersection point as $(\tau_1^{L^\dagger}, \pi_1^{L^\dagger})$. Then, for any $\tau_1^L > \tau_1^{L^\dagger}$, there exists π_1^L such that there is a fire sale in the baseline economy but not in the common information benchmark. However, there is no π_1^L for which there is a fire sale in the common information benchmark but not in the baseline economy. (This is represented by the red-shaded region in figure 5.) On the other hand, for any $\tau_1^L < \tau_1^{L^\dagger}$, there exists π_1^L such that there is a fire sale in the common information benchmark but not in the baseline economy. However, there is no π_1^L for which there is a fire sale in the baseline economy but not in the common information benchmark. (This is represented by the blue-shaded region in figure 5.)

Whether the unconditional likelihood of entering a fire sale is larger in the baseline economy versus the common information benchmark depends on the relative size of these regions of the state space and the joint distribution of the cost shocks and signal across these regions. Therefore, it is ambiguous whether the unconditional likelihood of a fire sale is larger in the baseline versus the benchmark. ■

B.8 Proof of proposition 3

Proof. Proof of Part (A): Effects of misinformation on the allocation in the normal regime

By comparing a normal regime equilibrium in baseline with the Common Information Benchmark, we can see that in both equilibria, $a_1^B = a_0^B$. So in the normal regime, beliefs have no effect on the allocation of the risky asset. What they have is an effect on the allocation of cash between the lender vs the borrower, via q_1 , and the asset price p_1 (which itself doesn't matter in this regime beyond ensuring $a_1^B = a_0^B$, since pecuniary externality doesn't lead to misallocation here): $\kappa_1^B = q_1 a_0^B - c_0^B + \kappa_0^B$. In both versions, $q_1 = \frac{E_1^L[R_2^d(f_1)]}{\tau_1^L}$ determined by beliefs of the lender, which is equivalent to that of the borrower in the common information benchmark. Hence, tighter q_1 is met with cash holdings in this regime, but belief disagreements (learning mechanism) don't affect tighter q_1 .

Belief disagreement does affect p_1 (since $p_1 = \frac{E_1^B[R] - E_1^B[R_2^d(f_1)]}{\tau_1^B} + q_1$), but this has no allocative consequences in this regime. Thus, belief disagreement only affect equilibrium allocation (of asset or cash) at date 1 only to the extent that they affect allocation of asset. (Beliefs in general affect allocation of cash, but this is always determined by q_1 which is pinned down by the lender's belief. So, the borrower's belief matters only to the extent that it affects desired a_1^B).

Proof of Part (B): Effect of misinformation on the severity of fire sale

As we outlined for the normal regime, belief disagreement affects the equilibrium allocation (of asset or cash) at date 1 only to the extent that they affect allocation of asset. (This is because in both the baseline case and the common information benchmark, q_1 is pinned down by the lender's beliefs in equilibrium $q_1 = \frac{E_1^L[R_2^d(f_1)]}{\tau_1^L}$). So at the margin, belief disagreements can affect the equilibrium allocation only to the extent that it affects a_1^B and/or p_1 . Recall that in the fire sale regime, we have (78)

$$a_1^B = a_0^B - F'^{-1} \left(\frac{E_1^B[R] - E_1^B[R_2^d(f_1)]}{\tau_1^B} + \frac{E_1^L[R_2^d(f_1)]}{\tau_1^L} \right)$$

while for the benchmark with common information, we have

$$a_1^B = a_0^B - F'^{-1} \left(\frac{E_1[R] - E_1[R_2^d(f_1)]}{\tau_1^B} + \frac{E_1^L[R_2^d(f_1)]}{\tau_1^L} \right) \quad (41)$$

Since $F'^{-1}(\cdot)$ is monotonically decreasing, a_1^B is lower in the baseline case (i.e. the identification problem makes the fire sales more severe) only if the spread $E_1^B[R] - E_1^B[R_2^d(f_1)] = (1 - \pi_1^B)(\bar{R} - f_1)$ is lower. This occurs iff $\pi_1^B > \pi_1^L$. Whether this is true in equilibrium or not depends on the actual realization of π_1^L, τ_1^L given the observed q_1 . So, for a given q_1 , the identification problem will amplify the fire sales relative to the common info benchmark (lower a_1^B) when

π_1^L is low and τ_1^L is high; while it will dampen the fire sales relative to the common info benchmark (higher a_1^B) when π_1^L is high and τ_1^L is low.

The net effect of these two forces determines the overall effect of misinformation on the severity of fire sales. ■

B.9 Proof of proposition 4

We state and prove a more formal and general proposition that supports Proposition 4. Before stating the formal proposition, we define the ex ante social welfare, the counterfactual economy with fixed contract terms, and the concept of welfare wedge between the baseline economy and the counterfactual economy.

Define the ex ante social welfare of the economy at date 1 as

$$SW_1 \equiv E_1^L(C_2^L) + E_1^B(C_2^B) + E_1^T(C_2^T). \quad (42)$$

Note that each agent's expected utility is based on its own belief.

We now ask how information spillovers affect ex ante social welfare. To answer this question, we first introduce a counterfactual economy without information spillovers. In this counterfactual economy, neither B nor L can change the contract terms (no margin calls, no renegotiation, or equivalently, long-term contracts are in place).

In this counterfactual economy, the ex ante social welfare at date 1 is

$$\hat{SW}_1 = E_1^L(C_2^L) + E_0^B(C_2^B) + \kappa_0^T, \quad (43)$$

as there are no fire sales of B to the T sector, so B 's date 1 allocation is predetermined. Note that B 's expected utility is based on date 0 information, as there is no information spillover from L to B . Therefore, the (expected) ex ante social welfare of the economy at date 0 is

$$E_0[\hat{SW}_1] \equiv E_0^L(C_2^L) + E_0^B(C_2^B) + \kappa_0^T. \quad (44)$$

Now we compare the ex ante social welfare of the baseline case with that of the counterfactual economy without information spillovers. To do so, we define the *welfare wedge* as the difference in welfare in the two economies conditional on realized shocks:

$$\Delta SW_1 \equiv SW_1 - \hat{SW}_1. \quad (45)$$

Similarly, define the *expected welfare wedge* as

$$\Delta ESW_1 \equiv E_0[SW_1] - E_0[\hat{SW}_1]. \quad (46)$$

The welfare wedge is determined by three components:

1. In the normal regime at date 1 in the baseline economy, there is a reallocation of cash from L to B .
2. In the fire sale regime at date 1 in the baseline economy, there is a reallocation of the risky asset from B to T and a reallocation of cash from T to L . This affects the welfare wedge in two ways:
 - (a) It affects the welfare wedge through pecuniary externality via convexity of F .
 - (b) Other allocative effects on welfare.

The following proposition, which is a more formal statement of Proposition 4, shows that the reallocation of cash in the normal regime (effect 1) unambiguously improves social welfare relative to the counterfactual, while the pecuniary externality in the fire sale regime (effect 2a) unambiguously reduces it. Moreover, if the effect of pecuniary externality is sufficiently strong, the expected welfare wedge can be negative, implying that the counterfactual economy has a greater expected welfare.

Proposition 6.

- (A) *In the normal regime at date 1, the reallocation of cash from L to B (effect 1) unambiguously increases the welfare wedge ΔSW_1 .*
- (B) *In the fire sale regime at date 1, pecuniary externalities from fire sales (effect 2a) unambiguously decrease the welfare wedge ΔSW_1 . Moreover, the welfare wedge is continuously decreasing in the convexity of $F(\cdot)$.*
- (C) *The sign of the expected welfare wedge ΔESW_1 is ambiguous. However, the expected welfare wedge is decreasing in the convexity of $F(\cdot)$ and can be negative.*

Proof.

Proof of Part (A):

There is no fire sale at date 1 if and only if $q_1 \geq 1$. This is because, there is no fire sale at date 1 if and only if the liquidity constraint (54) is

$$q_1 c_0 = c_0 + \kappa_1^B. \quad (47)$$

Then, $\kappa_1^B \geq 0$, i.e., B cannot hold negative cash, implies that (47) is equivalent to $q_1 \geq 1$.

This necessarily implies that $\tau_1^L \leq 1$ when there is no fire sale at date 1. Recall that

$$q_1 = \frac{1 - \pi_1^L + \pi_1^L \underline{R}}{\tau_1^L} \leq \frac{1}{\tau_1^L}$$

$$\implies \tau_1^L \leq \frac{1}{q_1} \leq 1,$$

where the first inequality holds because $\pi_1^L \in [0, 1]$ and $\underline{R} < 1$, and the last inequality holds because no fire sale at date 1 implies $q_1 \geq 1$. This implies that the reallocation of cash weakly improves ex ante social welfare when there is no fire sale at date 1 relative to the counterfactual without maturity mismatch. This is because $\tau_1^L \leq 1 = \tau_1^B$, and therefore cash is reallocated from an agent with lower opportunity cost of cash (L) to the agent with a higher opportunity cost of cash (B). The welfare improvement is strictly positive if there is a strictly positive reallocation of cash, i.e., $q_1 > 1$ and $\kappa_1^B > 0$.

Proof of Part (B):

First, we characterize the components of the ex ante social welfare.

Note that lender L 's expected utility at date 1 is invariant to equilibrium. This is because of the fact that L is indifferent (in expected sense) between lending at q_1 and holding cash to earn τ_1^L in any equilibrium. Thus, L will be indifferent between lending a lot at q_1 and not lending at all in any state. Given that Assumption 1, ensures $\kappa_1^L > 0$ in equilibrium, the lender's expected utility is invariant to the equilibrium allocation at date 1. Therefore, we can ignore the effects on $E_1^L(C_2^L)$ for social welfare comparisons.

The traditional sector T 's utility does not directly depend on the realization of the asset's return R . Therefore, T 's expected utility is simply

$$E_1^T(C_2^T) = C_2^T = \kappa_2^T + F(a_1^T),$$

and the marginal utility is

$$\frac{dC_2^T}{da_1^T} = F'(a_1^T). \quad (48)$$

Hence, the ex ante social welfare depends on how B 's expected utility changes given the non-stochastic changes in T 's expected utility.

Based on this characterization of welfare, we show that the ex ante social welfare is continuously decreasing in the severity of fire sales, i.e., the convexity of F .

First, suppose that B 's budget constraint does not require B to fire sell more than it would like

to based on its belief π_1^B . In other words,

$$\kappa_1^B > 0. \quad (49)$$

Then, the amount of fire sales is optimal, as the expected marginal rate of substitution between the seller, B , and the buyer, T are the same, i.e.

$$p_1 = \frac{E_1^B[R] + q_1 \tau_1^B - E_1^B[R_2^d]}{\tau_1^B} = F'(a_0^B - a_1^B), \quad (50)$$

where $R_2^d \equiv \min\{R, 1\}$.

Now suppose that B 's budget constraint requires B to fire sell its assets more than it would like to based on its belief π_1^B . In other words,

$$\kappa_1^B = 0, \quad (51)$$

and

$$p_1 = \frac{E_1^B[R] + q_1 \tau_1^B - E_1^B[R_2^d] + \xi_{\kappa_1}^B q_1}{(\tau_1^B + \xi_{\kappa_1}^B)} \quad (52)$$

$$< p_1^B \equiv \frac{E_1^B[R] + q_1 \tau_1^B - E_1^B[R_2^d]}{\tau_1^B}, \quad (53)$$

where the last inequality holds because $p_1 > q_1$ and

$$\begin{aligned} (\tau_1^B + \xi_{\kappa_1}^B) p_1 - \xi_{\kappa_1}^B q_1 &= E_1^B[R] + q_1 \tau_1^B - E_1^B[R_2^d] \\ \Rightarrow \tau_1^B p_1 + \xi_{\kappa_1}^B (p_1 - q_1) &= \tau_1^B p_1^B \\ \Rightarrow p_1 &< p_1^B. \end{aligned}$$

Therefore, under such states, there is a discrepancy of marginal rates of substitution between B and T because of the fire sale.

From the market clearing condition, the equilibrium price is

$$p_1 = F'(a_0^B - a_1^B).$$

Recall that by Assumption 2, B can repay its debt obligation when $a_1^T = A$, i.e., B sells all the assets to T . Then, the equilibrium fire sale amount is determined by an interior solution of B 's optimization problem and continuously changing along the F' curve. B 's binding budget constraint

and collateral constraint imply

$$q_1 a_1^B + \kappa_0^B = p_1(a_1^B - a_0^B) + c_0 + \kappa_1^B,$$

resulting in

$$(q_1 - p_1)a_1^B + p_1 a_0^B = c_0 + \kappa_1^B - \kappa_0^B. \quad (54)$$

Because $q_1 < p_1$ in equilibrium, a decrease in p_1 leads to a further decline in a_1^B , resulting in larger amount of fire sales $a_0^B - a_1^B$. In other words, B has to sell more of its assets to repay its debt when the price of the asset declines. Hence, for an equilibrium with the inverse demand function of T as $\tilde{F}$ such that $F' > \tilde{F}'$, i.e., the convexity of $\tilde{F}$ is greater than that of F , (54) becomes

$$(q_1 - \tilde{p}_1)\tilde{a}_1^B + \tilde{p}_1 a_0^B = c_0 - \kappa_0^B, \quad (55)$$

where $\tilde{a}_1^B < a_1^B$, $\tilde{p}_1 = \tilde{F}'(a_0^B - \tilde{a}_1^B) < F'(a_0^B - \tilde{a}_1^B) < F'(a_0^B - a_1^B)$. In other words, the discrepancy between the marginal utility of B and that of T , $p_1^B - p_1$ is continuously increasing in the convexity of F . Thus, the ex ante social welfare is continuously decreasing in the convexity of F . Hence, pecuniary externalities from fire sales (the convexity of F) unambiguously decrease the welfare wedge.

Proof of Part (C):

The tradeoff of adjusting the lending terms at date 1 (or making an inference from q_1) is between: **(a)** more accurate valuation and efficient asset allocation (improve ex ante social welfare through efficient sales when the actual fundamental value is low), and **(b)** inefficiency of asset allocation due to fire sale (pecuniary) externalities (selling the assets more than the expected fundamental value due to price decline and liquidity constraint). Ultimately, the final social welfare implications depend on the parameters, especially the shape of F' , as it determines the degree of fire sale externalities as shown in Part (B).

In the following, we show that the welfare wedge between the baseline economy and the counterfactual economy without information spillovers can be positive or negative depending on the convexity of F .

First, suppose that $F'(a) \approx F'(0)$ for any $a \in (0, A]$, i.e., F is close to a linear function. Then, there is essentially no negative effect of fire sale externalities **(b)**. Therefore, the baseline economy has an ex ante social welfare, which is greater than in the counterfactual economy, as only the increase in allocative efficiency **(a)** affects the ex ante social welfare.

Second, suppose that $F'(a) \approx F'(A) < F'(0)$ for any $a \in (0, A]$, i.e., F' is steeply decreasing at 0, and $F'(0) - F'(A)$ is arbitrarily large enough. In other words, even an infinitesimally small amount

of fire sales leads to a huge price decline, generating a discrete jump in pecuniary externalities. Then, B 's fire sale ends up in liquidity-driven fire sales, and the allocation is always extremely inefficient as $p_1 \approx F'(A) < p_1^B \leq F'(0)$. Therefore, the counterfactual economy has an ex ante social welfare, which is greater than that in the baseline economy.

Therefore, the welfare wedge can be either positive or negative, depending on the shape of F , and the reallocation of cash and assets in the fire sale regime at date 1 has ambiguous effect on welfare wedge.⁴³ ■

B.10 Policy interventions: General setup and proof

We first provide a general setup of standard monetary policy tools and extraordinary central bank lending facilities. We then present our formal proof of Proposition 5.

B.10.1 Standard monetary policy tools

For policy tools which affect agents' opportunity cost of lending, typical of more standard monetary policy tools such as open market operations or interest on central bank reserves, central bank interventions does not add information. By affecting the level of interest rates, monetary policy tools affect the opportunity cost of lending. We can therefore map this type of tool into our baseline model as an additional component of the lender's opportunity cost of funds τ_1^G such that the lender's total opportunity cost of funds at date 1 is $\tau_1^L + \tau_1^G$.⁴⁴ This captures the level shift in the lender's opportunity cost of funds brought about by changes in interest rates. The lender's optimality condition is still given by (15) with this minor alteration. Therefore, looser monetary policy, captured by a fall in τ_1^G , can bring about a loosening of funding liquidity at date 1. However, as long as the borrower also observes this policy intervention τ_1^G , the policy does not add information. The borrower can identify the component of the lender's opportunity cost of funds attributable to the government, but cannot infer the lender's private component τ_1^L . As a result, the borrower faces an identical identification problem as in the absence of the policy. Hence, such monetary policy interventions, while supporting funding liquidity, do not affect the formation of beliefs or affect the ability of market prices to aggregate and reveal private intervention.

⁴³More generally, the sign of ΔESW_1 depends on the interaction between: (i) the distribution of shocks (τ_1^L, s_1^L) , which determines the probability of entering the fire sale regime; (ii) the curvature of F , which determines the severity of pecuniary externalities conditional on fire sales; and (iii) the distribution of cost shocks, which affects the efficiency gains from cash reallocation in the normal regime.

⁴⁴For simplicity, we assume that the level of interest rates affects only the lender's opportunity cost of funds, although the same results would obtain if it affected both agents.

B.10.2 Extraordinary central bank lending facilities

We consider the effect of extraordinary central bank lending facilities.⁴⁵ In this extension, at the beginning of date 1, the government announces its lending program, which specifies the loan terms and the type of agents who can participate. Then, the market clearing price of loans as well as the amount of loans from the government are determined simultaneously, and the borrower updates its belief based on all the available information.

To proceed, we first modify the setup by introducing a government. The government does not observe agent's private information. At date 1, the government has access to lump-sum transfers and two distortionary taxes: the government can tax each unit revenue that the borrower obtains from liquidating the risky asset at date 1 at the rate γ^B ; and the government can tax the date-2 return that the lender receives on its cash holdings from date 1 at the rate γ^L . The government can transfer lump-sum amount of cash Γ_1 and Γ_2 at date 1 and 2, respectively. Both Γ_1 and Γ_2 can be either positive (tax) or negative (subsidy).

At date 1, the government provides loans to the market to either lender or borrower at pre-specified rate q_1^G . The promised value of payment per unit of collateral is 1, and denote the amount of collateral posted to the government as c_1^G . The price of the government loan q_1^G is publicly observable at the beginning of date 1, and then the rest of the prices q_1 and p_1 are determined by the competitive markets. The equilibrium prices will be informative to the borrower about the lender's private information.

Thus, the government has three types of tools to intervene in financial markets at date 1: the distortionary taxes γ^L, γ^B allow it to reallocate liquidity at date 1, the government loans at date 1, and the lump-sum transfers allow it to redistribute wealth.

First, we consider the case in which only borrowers can get the loan from the government. The government's budget constraint at date 1 is

$$(a_0^B - a_1^B) \gamma^B + q_1^G c_1^G + \Gamma_1 = \kappa_1^G \quad (56)$$

where γ^B is the proportional tax on the borrower's liquidation of the asset at date 1, c_1^G, q_1^G is the total quantity of collateral posted by the borrower and the price per a unit of collateral, respectively, Γ_1 are lump-sum taxes at date 1, and κ_1^G is the government's date-1 cash holdings. Thus, the

⁴⁵ Acharya et al. (2012) present a model of interbank lending and asset sales markets in which banks with surplus liquidity have market power over banks who need liquidity. In their model, frictions arise in lending due to moral hazard, and assets are bank-specific. Surplus banks ration lending and instead purchase assets from needy banks, an inefficiency more acute during financial crises. A central bank acting as a 'lender of last resort' can ameliorate this inefficiency provided it is prepared to extend potentially loss-making loans or is better informed than outside markets. This rationale for central banking finds support in historical episodes, e.g., Park (1991). Our results suggest that the interaction between the identification problem arising from worsening funding liquidity and deterioration in market liquidity due to costly fire sales can be better dealt with the central bank's role of a 'dealer of last resort'.

government's date-1 cash holdings consist of the taxes collected from the borrower's liquidation of the risky asset and lump-sum taxes minus the cash used to purchase the debt from borrower at q_1^G . Note that κ_1^G can be negative and the government can balance its budget by its date-2 income.

The government's date-2 budget constraint is

$$\kappa_1^G + \Gamma_2 + \gamma^L \kappa_1^L = c_1^G R_2^d. \quad (57)$$

Thus, the government's date-2 income must be met with its negative holdings from date 1, any lump-sum taxes at date 2, or tax revenue on the lender's cash holdings from date 1.

Under this setting, providing cheap government loans when there is already sufficient funding liquidity is inefficient, as the intervention only generates distortions from taxation. Therefore, we focus on the case in which the government provides the loan contract whose price is sufficiently low (or has a sufficiently high interest rate), so the government loans is used only if the loan price q_1 is sufficiently low that induces fire sales, as $p_1 < F'(0)$. This provision of cheap additional funds would decrease 'liquidity-driven fire sales' in the case of $p_1^B > F'(0) > p_1$, as the amount of fire sales that borrower has to make to repay its debt would be lower in this case. Nevertheless, this provision of cheap additional funds would not decrease belief-driven fire sales, which occurs when $p_1^B < F'(0)$. This is because the borrower's information set remains the same as that in the baseline case without the government intervention. Hence, the provision of loans to borrowers does not ameliorate the identification problem and information spillovers issue.

Finally, we consider the case in which only lenders can get the loan from the government. The government's budget constraints at date 1 and date 2 are the same as (56) and (57), respectively. Again, providing cheap government loans when there is already sufficient funding liquidity is inefficient, as the intervention only generates distortions from taxation. Therefore, we focus on the case in which the government provides the loan contract whose price is sufficiently low (or has a sufficiently high interest rate), so the government loans are used only if the loan price q_1 is sufficiently low that induces fire sales, as $p_1 < F'(0)$. This provision of cheap additional funds would be informative only if the lender has enough cash to increase its market rate of lending q_1 .

In this case, the lender receives collateral from borrower after purchasing the collateralized debt contract and then reuses the collateral to borrow from the government. Thus, the lender receives

intermediation rent through this chain of transactions. The lender's optimization problem becomes

$$\begin{aligned}
& \max_{\kappa_1^L, d_1^L, c_1^G} E_1^L [C_2^L] \\
& s.t. C_2^L = (\tau_1^L - \gamma^L) \kappa_1^L + R_2^d d_1^L - R_2^d c_1^G \\
& \quad \kappa_1^L - \kappa_0^L \leq d_0^L - \hat{q}_1 d_1^L + q_1^G c_1^G \\
& \quad c_1^G \leq d_1^L,
\end{aligned}$$

with the non-negativity constraints, where the third constraint is the lender's collateral constraint, and $\hat{q}_1$ is the price of loans with the intervention. As previously mentioned, we focus on the case in which lender has incentives to borrow from the government, that is, $q_1 < q_1^G$, where q_1 is the equilibrium loan price without the government intervention. Hence, the FOC for d_1^L is

$$\hat{q}_1 = \frac{E_1^L [R_2^d] + \mu_1^L}{\tau_1^L - \gamma^L}, \quad (58)$$

where μ_1^L is the Lagrange multiplier for the collateral constraint. The FOC for c_1^G is

$$q_1^G = \frac{E_1^L [R_2^d] + \mu_1^L}{\tau_1^L - \gamma^L}, \quad (59)$$

therefore, the equilibrium loan price becomes $\hat{q}_1 = q_1^G$ by competition. Again, the intervention can decrease 'liquidity-driven fire sales' in the case of $p_1^B > F'(0) > p_1$, as the amount of fire sales that borrower has to make to repay its debt would be lower in this case.

Importantly, this provision of cheap additional funds to lenders reduce the information spillovers to the borrower. This is because the exogenously given government loan price q_1^G substitutes the loan price q_1 determined by the lender's private information on τ_1^L and π_1^L . Thus, the extraordinary government intervention destroys information in the market. As a result, the severity of a belief-driven fire sale is lower than that in the baseline case. Even though the provision of loans to lenders does not ameliorate the identification problem, it reduces the information spillovers and resulting inefficient fire sales due to excessive pessimism.

B.10.3 Proof of proposition 5

Proof. Proof of Part (A): As discussed in the previous subsections, standard monetary policy tools (e.g., lowering short-term interest rates, discount window lending) can be modeled as reducing the lender's opportunity cost of funds τ_1^L . By the lender's optimality condition (1), this increases $q_1 = \frac{E_1^L [R_2^d]}{\tau_1^L}$, thereby increasing funding liquidity and reducing fire sales. However, be-

cause q_1 still depends on the lender's beliefs $E_1^L[R_2^d]$, the price of debt remains informative about the lender's private signal about asset quality. Therefore, the borrower still faces the same identification problem: it cannot distinguish whether changes in q_1 reflect the policy intervention τ_1^G versus the lender's private news. By lemma 1, the borrower's beliefs π_1^B continue to depend on q_1 in the same manner, so the information spillover mechanism remains unchanged.

Proof of Part (B): When the central bank lends directly to the lender at a government-administered rate q_1^G backed by risky assets, competition ensures the equilibrium loan price satisfies $q_1 = q_1^G$. This breaks the link between q_1 and the lender's private information $(E_1^L[R_2^d], \tau_1^L)$ that would otherwise obtain from condition (1). Since q_1 is now pinned down by policy rather than reflecting the lender's beliefs, the price becomes uninformative about asset quality. This eliminates the information spillover channel through which tight funding liquidity increases the borrower's perceived downside risk π_1^B . By severing the link between funding conditions and beliefs about fundamentals, such facilities reduce both liquidity-driven and belief-driven fire sales. ■

C Appendix: Alternative case of upstream information spillovers

Thus far, we have focused on the case in which the lender gets the cost shock and private signal, but the borrower does not. This implies that information spillovers flow downstream from the lender to the borrower. Moreover, in this case we saw how liquidity conditions in funding markets (measured by the price of debt) affect the borrower's beliefs about fundamentals.

In this section, we consider the alternative case in which the borrower gets a cost shock and private signal, but the lender does not.⁴⁶ Because the mechanisms and insights are so similar to our baseline case, we do not solve the entire model under this alternative case. Rather, we show that, while the mechanism differs slightly, the fundamental insights are the same as in the baseline case. In particular, information spillovers in this case will flow upstream from the borrower to the lender rather than downstream. And, as a result the availability of market liquidity (measured by cost shocks to the borrower which affect the price of the risky asset) will shape the lender's beliefs about fundamentals. Thus, the nature of information spillovers and the effect of liquidity on beliefs and asset prices are essentially same as in the baseline case.

This alternative case is illustrated in figure 7. Suppose the borrower gets a cost shock τ_1^B and

⁴⁶For simplicity, we assume that only one agent receives both a cost shock and a private signal, while the other agent receives neither. We make this assumption, rather than allowing both agents to receive private signals, to prevent the model from becoming analytically intractable. If, in contrast, both agents received private signals, then the equilibrium would depend not only on how one agent, e.g. the borrower, updates its beliefs in response to the actions of the other, but also on an infinite feedback loop in which also the borrower's actions affect the lender's beliefs, etc. This would make an analytical characterization of equilibrium belief formation extremely difficult.

private signal s_1^B , but the lender does not (i.e. τ_1^L is a fixed parameter known to both agents). How do the equilibrium conditions differ from our baseline case? In short, all of the equations constituting the real block of the economy take the same form. The only equations that change are those characterizing the evolution of beliefs at date 1.

How would the borrower's choices reflect its private information τ_1^B, s_1^B ? In this case q_1 still only reflects the lender's first-order condition according to (16), and is therefore not informative about the borrower's private information. However, the asset price p_1 will reflect the borrower's private information through the equilibrium price (26). Using this equation, the spread between the borrower's valuation and the price of debt at date 1 is

$$\underbrace{p_1 - q_1}_{spread} = \frac{E_1^B[R] - E_1^B[R_2^d]}{\xi_{\kappa_1}^B + \tau_1^B} \quad (60)$$

(Note also that the quantity of the borrower's investment in the risky asset $a_1^B = c_1$ is just determined residually from this optimality condition and the market clearing condition, and hence contains no additional information about borrower's private information over the spread.)

Identification problem Therefore, the lender uses the observation of $p_1 - q_1$ to form its posterior belief π_1^L , but cannot separately infer π_1^B and τ_1^B due to a similar identification problem as what we had in the case with downstream information spillovers. (Indeed, because q_1 is still pinned down by the lender's first order condition, the asset price p_1 is the variable which conveys the borrower's private information. But our analytical expressions are made more simple by characterizing information based on observed spread $p_1 - q_1$.)

(Since the lender has only one observable which is informative about the borrower's private information, it cannot separately identify the borrower's information set at date 1 (i.e. its private signal) and its date 1 cost shock τ_1^B . Put differently, the lender has only one equation (6) to infer two unobservables s_1^B, τ_1^B . Indeed, there is a continuum of pairs (τ_1^B, π_1^B) that satisfy the relation above for the observed p_1 .)

Belief formation Following the same steps to prove lemma 1, we can symmetrically characterize the evolution of the lender's beliefs in this environment. In particular, we have that

1. Lender's belief is the same as the expected posterior of the borrower, i.e. $\pi_1^L = E_1^L[\pi_1^B]$.
2. π_1^L is decreasing in $p_1 - q_1$.

Information spillovers operate upstream In this environment, information spillovers operate *upstream* from the borrower to the lender, rather than downstream as in our baseline case. Suppose that the borrower receives a cost shock (high τ_1^B) or bad private signal so that the asset price is low. The lender then observes that the spread $p_1 - q_1$ is low and will update its belief, which becomes more pessimistic.

Market liquidity affects beliefs Thus, the beliefs of upstream agents are shaped in equilibrium by the availability of *market liquidity* rather than by funding liquidity, as was the case in our baseline case. More precisely, changes in the price of the risky asset p_1 which are driven by cost shocks to the borrower affect lender's optimism or pessimism about the asset fundamental.

Belief-driven booms and busts In this environment, the availability of market liquidity can give rise to belief-driven booms or busts in the asset market. Suppose again that the borrower receives an adverse cost shock, leading to drop in the spread $p_1 - q_1$ and causing the lender to become more pessimistic due to the information spillover (π_1^L goes up). As can be seen from the equilibrium expression for q_1 , the equilibrium price of date 1 debt will fall as a result of the higher π_1^L —that is, there will be a tightening of funding liquidity. This worsens the borrower's liquidity position, and forces it to liquidate more of the risky asset at the margin, further lowering the asset price. Hence, market illiquidity gives rise to funding illiquidity through the lender's endogenous pessimism. This feedback from market illiquidity to funding illiquidity gives rise to asset price busts. Symmetrically, in response to a positive cost shock to the borrower, the upstream information spillover would give rise to an asset price boom through the feedback from market liquidity to funding liquidity.

Thus, in this case, same insights hold, with two differences in the mechanism. First, information spillovers propagate upstream from borrowers to lenders. Second, the beliefs of the lender are shaped by the degree of market liquidity, rather than funding liquidity – cost shocks to downstream firms shape the beliefs of upstream firms. As a result, the model still produces belief-driven booms or busts triggered by shocks which affect market liquidity. Moreover, these booms or busts are characterized by a feedback loop between market liquidity and funding liquidity.

D Online Appendix: Omitted proofs and results

D.1 Formalizing the financial contracts

Here, we provide a formal contracting environment in terms of a competitive market for repo contracts. At the end of this section, we summarize the environment and show that it is payoff-equivalent to the version in the main text.

Contracting environment

Market segmentation implies that there are gains to trading financial contracts. More precisely, L cannot hold the risky asset at any point, but can gain exposure to the risky asset by lending to B through the use of financial contracts. We assume that, although the risky asset matures after two periods, the borrower finances its holdings of the risky asset using one-period collateralized debt contracts. By financing long-term risky assets with short-term debt, the borrower is exposed to liquidity risk at date 1. We formalize this environment and the nature of liquidity risk below.

Financial contracts

Formally, a date t contract is given by (c_t, f_t) , which stipulates a transfer at date t of the risky asset of size c_t from the borrower to the lender as collateral with a promise for the borrower to repurchase the collateral at date $t + 1$ at a unit price of f_t , normalized to $f_t = 1$. (That is, the borrower repays the lender $f_t c_t = c_t$ units of the consumption good at date $t + 1$.) We assume that the collateral on the loan is held by some outside custodian, not modeled explicitly, who either returns the collateral to the borrower if the contract is honored or sells it and pays the proceeds to the lender if the borrower defaults. This ensures that the lender accepts the asset as collateral for the loan despite not being able to hold the asset itself.

These contracts are traded in a competitive market, where q_t denotes the market price of the contract at $t = 0, 1$. Therefore, the size of the loan from the lender to the borrower at t —that is, the quantity of the consumption good loaned at date t —is given by $q_t c_t$. At date $t + 1$, the borrower decides whether to default on the contract and forgo the collateral or not. If the borrower chooses to default on the contract at date $t + 1$, the value of this collateral $c_t p_t$ is transferred to the lender.

Note that both date 0 and date 1 contracts are one-period contracts and involve a repayment of c_t at date $t + 1$. One may wonder why the borrower would be willing to enter into the date 0 contract in the first place, given that this will require the borrower to repay the lender c_0 units of the consumption good at date 1 despite the fact that the borrower has no income at date 1. In equilibrium, the borrower will be willing to borrow at date 0 because it anticipates that it can finance this repayment c_0 (at least partially) by refinancing (i.e. rolling over) its debt using the date

1 contract, albeit under different terms. One can interpret this contracting environment as involving long-term (two-period) debt which is renegotiated in the intermediate period.

Note that the lender has full-recourse on the debt at date 1. Thus, the borrower will try to pay the debt in full at date 1 unless it exhausts all of cash and asset holdings. However, at date 2, there is no further recourse to the borrower's balance sheet as the borrower can simply walk away from the collateral. Under the date 1 terms of the contract, the borrower will decide whether to default on the contract and forgo the collateral or not, thus, the payment from the borrower to the lender at date 2 will be determined endogenously as $\min\{p_2, 1\}$, where 1 is the promised payment amount.

Summary and payoff equivalence

More precisely, at date $t = 0, 1$, the borrower can sell the lender a repo contract which stipulates that the borrower sells a unit of the risky asset (i.e., collateral) to the lender at date t and repurchases the collateral at $t+1$ for a unit price of f_t normalized to $f_t = 1$. Hence, the contract is defined at the level of the units of collateral exchanged, c_t . d_t^L is the number of units of the contract demanded by the lender. Therefore, market clearing requires that $c_t = d_t^L$.

What does the flow of payments look like? At date t , the lender buys d_t^L units of the contract (which therefore involves buying d_t^L units of the collateral from the borrower) at a price of q_t per contract. Hence, there is a payment of $q_t d_t^L$ units of the consumption good from the lender to the borrower. At date $t + 1$, the borrower either repurchases the collateral from the lender according to the contract or defaults. In the good state, there is no default, so the borrower buys back c_t at a predetermined price $f_t = 1$. Hence the lender receives $c_t f_t = c_t = d_t^L$ units of the consumption good. In the bad state, the borrower prefers to default. In that case, the borrower forgoes its collateral. The lender keeps the c_t units of collateral which yields a gross interest rate of R . Thus, we can express the lender's payoff from loan as $d_1^L R^d$, where $R^d = \min\{R, 1\}$.

Hence, this repo contracting environment is payoff-equivalent to one in which the borrower issues a bond backed by its holdings of the risky asset. In this case, the lender buys d_t^L units of a bond issued by the borrower at unit price of q_t and which promises a gross interest rate of 1. Hence the loan involves $q_t d_t^L$ unit of consumption good. If there is no default at $t + 1$, the borrower pays back d_t^L at date 2. If the borrower defaults, it forgoes collateral, and so the lender gets the return of the risky asset, i.e. it receives a payoff of $R^d = \min\{R, 1\}$ per unit of the bond.

D.2 Full characterization of date-1 equilibrium

In the normal regime, we have

$$a_1^T, \xi_{d_1}^L, \xi_{c_1}^B, \xi_{a_1}^B, \xi_{\kappa_1}^L, \xi_{\kappa_1}^B = 0$$

$$q_1 = \frac{E_1^L [R_2^d(f_1)]}{\tau_1^L}$$

$$p_1 = \frac{E_1^B [R] - E_1^B [R_2^d(f_1)]}{\tau_1^B} + q_1$$

$$a_1^B = c_1^B = d_1^L = a_0^B$$

$$\kappa_1^B = q_1 a_0^B - c_0^B + \kappa_0^B$$

$$\kappa_1^L = \kappa_0^L + d_0^L - q_1 a_0^B$$

$$\kappa_1^T = \kappa_0^T - p_1 a_1^T$$

$$C_2^B = \tau_1^B \kappa_1^B + a_1^B R - a_1^B R_2^d(f_1)$$

$$C_2^L = \tau_1^L \kappa_1^L + R_2^d d_1^L$$

$$C_2^T = \kappa_1^T + F(a_1^T)$$

$$\mu_1^B = (\tau_1^B + \xi_{\kappa_1}^B) p_1 - E_1^B [R] - \xi_{a_1}^B$$

$$\mu_1^T = F'(0) - p_1$$

$$\pi_1^L = Pr(R = \underline{R} | s_1^L, I_0) = \frac{\pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R})}{(1 - \pi_0) \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \bar{R}) + \pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R})}$$

$$\pi_1^B = \int_0^1 \pi \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q_1} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q_1} < \bar{\tau}^L \right\} dG_\pi(\pi)$$

In the fire sale regime, we have

$$\mu_1^T, \xi_{c_1}^B, \xi_{a_1}^B, \xi_{d_1}^L, \xi_{\kappa_1}^L, \xi_{\kappa_1}^B = 0$$

$$q_1 = \frac{E_1^L [R_2^d(f_1)]}{\tau_1^L}$$

$$p_1 = F'(a_1^T)$$

$$a_1^B = a_0^B - F'^{-1} \left(\frac{E_1^B [R] - E_1^B [R_2^d(f_1)]}{\tau_1^B} + \frac{E_1^L [R_2^d(f_1)]}{\tau_1^L} \right)$$

$$a_1^T = a_0^B - a_1^B$$

$$c_1^B = d_1^L = a_1^B$$

$$\kappa_1^B = q_1 a_1^B - c_0^B + \kappa_0^B - p_1 (a_1^B - a_0^B)$$

$$\kappa_1^L = d_0^L - q_1 d_1^L + \kappa_0^L$$

$$\kappa_1^T = \kappa_0^T - p_1 a_1^T$$

$$C_2^B = \tau_1^B \kappa_1^B + a_1^B R - a_1^B R_2^d(f_1)$$

$$C_2^L = \tau_1^L \kappa_1^L + R_2^d d_1^L$$

$$C_2^T = \kappa_1^T + F(a_1^T)$$

$$\mu_1^B = \tau_1^B p_1 - E_1^B [R]$$

$$\pi_1^L = Pr(R = \underline{R} | s_1^L, I_0) = \frac{\pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R})}{(1 - \pi_0) \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \bar{R}) + \pi_0 \lambda_\varepsilon (\varepsilon_1^L = s_1^L - \underline{R})}$$

$$\pi_1^B = \int_0^1 \pi \lambda_T \left(\frac{1 - (1 - \underline{R})\pi}{q_1} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1 - \underline{R})\pi}{q_1} < \bar{\tau}^L \right\} dG_\pi(\pi)$$

Below is the full solution of the date-1 equilibrium and derivation.

Proposition 7. *The equilibrium at date 1 can be characterized as the following:*

1. **Normal Regime:** If $F'(0) < \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L}$ and $q_1 a_0^B \geq c_0^B - \kappa_0^B$, then $c_1 = a_1^B = a_0^B = d_1^L$, $a_1^T = 0$, $\kappa_1^B = q_1 a_0^B - c_0^B + \kappa_0^B$, and there will be no fire-sales in the market and the asset price is

$$p_1 = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L}$$

or any number less than or equal to p_1 if $q_1 a_0^B - c_0^B = \kappa_0^B$, which implies $\kappa_1^B = 0$.

2. **Fire Sale Regime with the borrower holding only the risky asset:** If $F'(0) < \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L}$ but $q_1 a_0^B < c_0^B - \kappa_0^B$, then $c_1 = a_1^B = a_0^B - a_1^T = d_1^L$, a_1^T is determined by

$$F'(a_1^T) = q_1 + \frac{c_0 - q_1 a_0^B - \kappa_0^B}{a_1^T},$$

$\kappa_1^B = 0$, and the asset price is $p_1 = F'(a_1^T)$.

3. **Fire Sale Regime with the borrower at interior solution in portfolio choice:** If $F'(0) > \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L} > F'(a_0^B)$ holds, then $c_1 = a_1^B = d_1^L$ is determined by

$$F'(a_0^B - c_1) = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L},$$

with $a_1^T = a_0^B - c_1$, $p_1 = F'(a_0^B - c_1) = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L}$ and $\kappa_1^B = q_1 a_1^B - c_0^B + \kappa_0^B$.

4. **Fire Sale Regime with market collapse (B holds only cash):** If $F'(a_0^B) > \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \underline{R}}{\tau_1^L}$, then $c_1 = a_1^B = d_1^L = 0$, $a_1^T = a_0^B$, and $\kappa_1^B = p_1 a_0^B - c_0^B + \kappa_0^B$ with $p_1 = F'(a_0^B)$, so all the assets are sold to T and there will be no debt contract between B and L.

Proof. From lemma 4 in appendix B.4, we know that the collateral constraint is binding, and the asset price is greater than the contract price. Now we solve for the equilibrium price p_1 .

Case 1. Consider the case in which B holds a positive amount of assets, $a_1 = c_1 > 0$. The reservation asset price p_1^B for B that makes B indifferent between purchasing the asset and holding

cash is

$$p_1^B = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \bar{R}}{\tau_1^L}. \quad (61)$$

Case 1.1. If B is not selling any of its asset due to budget constraint, which is possible only if

$$q_1 a_0^B + \kappa_0^B \geq c_0,$$

then equilibrium price becomes $p_1 = p_1^B$, because otherwise T will be the only marginal buyer of the asset, which implies $a_1 = 0$, a contradiction.

Case 1.2. If B lacks cash to repay the debt as

$$q_1 a_0^B + \kappa_0^B < c_0,$$

then the asset price can be lower than p_1^B and $\xi_{\kappa_1}^B > 0$ as

$$p_1^B > p_1 = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B + \xi_{\kappa_1}^B} + \frac{(1 - \pi_1^L) + \pi_1^L \bar{R}}{\tau_1^L} = F'(a_1^T),$$

because B liquidates some of the assets to T in order to pay the debt to L even though B prefers not to do so. Therefore, the asset price could be lower than the marginal return of the asset in such equilibrium and determined by the traditional sector's inverse demand. B will liquidate the assets up to the necessary amount needed to match the budget constraint with $\kappa_1^B = 0$,

$$a_1^T F'(a_1^T) = c_0 - q_1(a_0^B - a_1^T) - \kappa_0^B$$

in such equilibrium. Rearranging the terms yields

$$F'(a_1^T) = q_1 + \frac{c_0 - q_1 a_0^B - \kappa_0^B}{a_1^T}, \quad (62)$$

and $p_1 = F'(a_1^T)$. Note that there exists a unique a_1^T in this case, because F' is strictly decreasing.

Finally, the budget constraint determines the cash holdings as

$$\kappa_1^B = q_1 a_0^B - c_0^B + \kappa_0^B,$$

that pins down the optimal decision vector of B if $a_1^T = 0$. If $a_1^T > 0$, the marginal buyer will be T ,

and the price of the asset will be

$$p_1 = F'(a_0^B - c_1). \quad (63)$$

Combining (63) with (38) yields

$$F'(a_0^B - c_1) = \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \bar{R}}{\tau_1^L}. \quad (64)$$

Therefore, if B is selling a positive amount of asset, the traditional sector's inverse demand function will pin down the asset price as well as the fire-sale amount. If either π_1^L is high or τ_1^L is high, then the right-hand side of (64) decreases and $a_0^B - c_1$ increases, meaning B sells more asset to T .

In the case that $F'(a_0^B) \geq \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \bar{R}}{\tau_1^L}$, then B sells all the asset holdings to T and B does not borrow from L .

On the contrary, if

$$F'(0) < \frac{(1 - \pi_1^B)(\bar{R} - 1)}{\tau_1^B} + \frac{(1 - \pi_1^L) + \pi_1^L \bar{R}}{\tau_1^L},$$

then $c_1 = a_1^B = a_0^B$ assuming $c_0 = a_0^B$, and asset price p_1 is indeterminant as no agent is buying or selling in a positive amount and any value above $F'(0)$ is possible. ■

D.3 Proof of lemma 3

Given the date-1 equilibrium, we can derive equilibrium at date 0. We assume $F'(a) = \alpha$ for all $a \geq 0$ from now on. This assumption is for tractability of the solution because otherwise the equilibrium is determined by implicit functions. The only case that disappears with this assumption is case 3 in proposition 7, which is an intermediate equilibrium between full fire sales and zero fire sales.

D.3.1 Lender's problem at date 0

Proof. In section 3.3, we showed that when the lender, L , tightens funding liquidity, L 's return for each unit of cash invested (either as cash or lending) is always τ_1^L regardless of the realization of (s_1^L, τ_1^L) , because L always holds extra cash at date 1.

Taking date 1 equilibrium outcomes as given, L 's problem at date 0 is

$$\max_{\kappa_0^L, d_0^L} E_0^L [\tau_1^L (\kappa_0^L + d_0^L)]$$

$$s.t. \quad \kappa_0^L + q_0 d_0^L \leq e_0^L$$

with non-negativity constraints,

$$\kappa_0^L \geq 0, d_0^L \geq 0.$$

Because L 's budget constraint binds at the optimum, substitute $\kappa_0^L = e_0^L - q_0 d_0^L$. The first-order condition with respect to the only choice variable, d_0^L , is

$$E_0^L [\tau_1^L (1 - q_0)] + \xi_{d_0}^L = 0. \quad (65)$$

By assumption 1, $d_0^L > 0$ in equilibrium. Therefore, the contract price that makes L indifferent across any amount of d_0^L is

$$q_0 = 1. \quad (66)$$

D.3.2 Borrower's problem at date 0

For each realization of q_1 at date 1, the borrower, B , takes the equilibrium market outcome, including B 's own optimal decisions, as given. As we have seen in proposition 7, there are different regimes under different realization of q_1 . B accounts for that and solves the following optimization problem:

$$\begin{aligned} \max_{c_0, a_0, \kappa_0^B} & \quad \underbrace{E_0^B [E_1 [C_2^B | q_1]]}_{\text{expected utility for each realized } q_1} \\ & = \underbrace{E_0^B [\tau_1^B (\kappa_0^B + q_1 a_0 - c_0) + (1 - \pi_1^B(q_1)) a_0 (\bar{R} - 1) | q_1 \geq \bar{q}_1]}_{\text{no liquidation case}} \Pr_0^B(q_1 \geq \bar{q}_1) \\ & \quad + \underbrace{E_0^B \left[\left(a_0 - \frac{c_0 - \kappa_0^B - q_1 a_0}{\alpha - q_1} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) \middle| \underline{q}_1 \leq q_1 < \bar{q}_1 \right]}_{\text{sells the asset to pay debt but no sales due to pessimism}} \Pr_0^B(\underline{q}_1 \leq q_1 < \bar{q}_1) \\ & \quad + \underbrace{E_0^B [\tau_1^B (\kappa_0^B - c_0 + \alpha a_0^B) | q_1 < \underline{q}_1]}_{\text{liquidates all the assets}} \Pr_0^B(q_1 < \underline{q}_1) \end{aligned} \quad (67)$$

$$\begin{aligned}
& \text{s.t. } e_0^B \geq p_0 a_0^B - q_0 c_0 + \kappa_0^B, \\
& a_0^B \geq c_0, \\
& \pi_1^B(q_1) = \int_0^1 \pi \lambda_T \left(\frac{1 - (1-R)\pi}{q_1} \right) \mathbb{1} \left\{ \underline{\tau}^L < \frac{1 - (1-R)\pi}{q_1} < \bar{\tau}^L \right\} dG_\pi(\pi), \\
& \underbrace{\bar{q}_1 = \frac{c_0 - \kappa_0^B}{a_0}}_{\text{cutoff for liquidity induced fire sales}}, \\
& \underbrace{\alpha = \frac{(1 - \pi_1^B(\underline{q}_1))(\bar{R} - 1)}{\tau_1^B} + \underline{q}_1}_{\text{cutoff for belief induced fire sales}},
\end{aligned}$$

with non-negativity constraints,

$$c_0 \geq 0, a_0^B \geq 0, \kappa_0^B \geq 0.$$

There are three different regimes for B depending on the realization of q_1 , which corresponds to each expected utility representation in the above optimization problem:

1. If $q_1 \geq \bar{q}_1$, then B will be able to roll over the debt from date 0 and keep all the assets.
2. If $\underline{q}_1 \leq q_1 < \bar{q}_1$, then B does not have enough cash to pay the promised debt amount but still optimistic enough to hold the assets as much as possible for the given prices. Therefore, B does liquidity-induced fire sales but sells no more than the necessary amount.
3. If $q_1 < \underline{q}_1$, then B is pessimistic about the asset payoff, so T values the asset more than B . Therefore, there will be belief-induced fire sales. B will sell all the assets to T .

We first confirm that the cutoff for belief-induced fire sales is above the level of q_1 that requires full sales of assets due to liquidity. This is because B can always sell all assets and repay the debt by assumption 2, and thus, any q_1 higher than $\underline{q}_1$ will be enough for B to repay the debt while holding a positive amount of asset.

Lemma 5. *If $q_1 \geq \underline{q}_1$, B can hold a positive amount of assets $c_1 > 0$ while repaying the debt to L in full.*

Proof. Recall that the amount of fire sales is

$$\frac{c_0 - \kappa_0^B - q_1 a_0}{\alpha - q_1},$$

which is decreasing in q_1 because

$$\frac{\partial \left(\frac{c_0 - \kappa_0^B - q_1 a_0}{\alpha - q_1} \right)}{\partial q_1} = \frac{-a_0(\alpha - q_1) + c_0 - \kappa_0^B - q_1 a_0}{(\alpha - q_1)^2} = \frac{-a_0 \alpha + c_0 - \kappa_0^B}{(\alpha - q_1)^2} < 0,$$

where the last inequality comes from assumption 2. ■

Denote the Lagrangian multiplier for the collateral constraint as μ . Substituting out κ_0^B and $\bar{q}_1$ using the binding budget constraint and the cutoff equation yields

$$\begin{aligned} L_0^B = & \int_{\frac{1}{\tau^L}}^1 \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left[\tau_1^B (e_0^B + q_0 c_0 - p_0 a_0 + q_1 a_0 - c_0) + (1 - \pi_1^B(q_1)) a_0 (\bar{R} - 1) \right] dH(q_1) \\ & + \int_{\underline{q}_1}^{\frac{1}{\tau^L}} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left(a_0 - \frac{c_0 - (e_0^B + q_0 c_0 - p_0 a_0) - q_1 a_0}{\alpha - q_1} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\ & + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \tau_1^B (e_0^B + q_0 c_0 - p_0 a_0 - c_0 + \alpha a_0^B) dH(q_1) \\ & + \mu(a_0 - c_0) + \xi_{c_0} c_0 + \xi_{a_0} a_0 + \xi_{\kappa_0^B} (e_0^B + q_0 c_0 - p_0 a_0), \end{aligned} \tag{68}$$

where $H(\cdot)$ is B 's subjective distribution function of q_1 . Then, we can apply the Leibniz integral rule for derivatives of the Lagrangian function. The first-order conditions of B 's optimization problem are

FOC for c_0 :

$$\begin{aligned}
& -\frac{1-q_0}{a_0} \left[\tau_1^B \left(e_0^B + q_0 c_0 - p_0 a_0 + \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} a_0 - c_0 \right) \right. \\
& + \left. \left(1 - \pi_1^B \left(\frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \right] \\
& - \int_{\frac{1/\tau}{a_0}}^{1/\tau} \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \tau_1^B (1-q_0) dH(q_1) \\
& + \frac{1-q_0}{a_0} \left(a_0 - \frac{c_0 - (e_0^B + q_0 c_0 - p_0 a_0) - \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} a_0}{\alpha - \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0}} \right) \\
& \times \left(1 - \pi_1^B \left(\frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) (\bar{R} - 1) \\
& - \int_{\underline{q}_1}^{\frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0}} \frac{(1-q_0)}{\alpha - q_1} (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\
& - \int_{\underline{R}/\tau}^{\underline{q}_1} \tau_1^B (1-q_0) dH(q_1) - \mu + \xi_{c_0} + \xi_{\kappa_0^B} q_0 = 0,
\end{aligned} \tag{69}$$

FOC for a_0 :

$$\begin{aligned}
& - \frac{p_0 a_0 - ((1 - q_0)c_0 - e_0^B + p_0 a_0)}{a_0^2} \left[\tau_1^B \left(e_0^B + q_0 c_0 - p_0 a_0 + \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} a_0 - c_0 \right) \right. \\
& \quad \left. + \left(1 - \pi_1^B \left(\frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \right] \\
& + \int_{\frac{1}{\tau}} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} [\tau_1^B(q_1 - p_0) + (1 - \pi_1^B(q_1))(\bar{R} - 1)] dH(q_1) \\
& + \frac{p_0 a_0 - ((1 - q_0)c_0 - e_0^B + p_0 a_0)}{a_0^2} \left(a_0 - \frac{c_0 - (e_0^B + q_0 c_0 - p_0 a_0) - \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} a_0}{\alpha - \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0}} \right) \\
& \times \left(1 - \pi_1^B \left(\frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) (\bar{R} - 1) \\
& + \int_{\underline{q}_1} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left(1 - \frac{p_0 - q_1}{\alpha - q_1} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\
& + \int_{\underline{R}/\tau}^{\underline{q}_1} \tau_1^B(\alpha - p_0) dH(q_1) + \mu + \xi_{a_0} - \xi_{\kappa_0^B} p_0 = 0.
\end{aligned} \tag{70}$$

D.3.3 Equilibrium trade and prices

FOC for c_0 , (69), can be further simplified as

$$\begin{aligned}
& -\frac{1-q_0}{a_0} \left(1 - \pi_1^B \left(\frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \\
& - \int_{\underline{R}/\bar{\tau}}^{\underline{1}/\bar{\tau}} \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \tau_1^B (1-q_0) dH(q_1) - \int_{\underline{R}/\bar{\tau}}^{q_1} \tau_1^B (1-q_0) dH(q_1) \\
& + \frac{1-q_0}{a_0} \left(1 - \pi_1^B \left(\frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \\
& - \int_{\underline{q}_1}^{\underline{1}/\bar{\tau}} \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \frac{(1-q_0)}{\alpha - q_1} (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) - \mu + \xi_{c_0} + \xi_{\kappa_0^B} q_0 \\
& = - \int_{\underline{q}_1}^{\underline{1}/\bar{\tau}} \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \tau_1^B (1-q_0) dH(q_1) - \int_0^{q_1} \tau_1^B (1-q_0) dH(q_1) \\
& - \int_{\underline{q}_1}^{\underline{1}/\bar{\tau}} \frac{(1-q_0)c_0 - e_0^B + p_0 a_0}{a_0} \frac{(1-q_0)}{\alpha - q_1} (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) - \mu + \xi_{c_0} + \xi_{\kappa_0^B} q_0 \\
& = -\mu + \xi_{c_0} + \xi_{\kappa_0^B} = 0,
\end{aligned}$$

and the second to last inequality holds because of $q_0 = 1$, which is from (66).

If B does not borrow at all $c_0 = 0$, then the collateral constraint should be binding, implying $a_0 = c_0 = 0$. If B is borrowing a positive amount as $c_0 > 0$, then $\xi_{c_0} = 0$ and there can be two different cases. Consider the first case in which $\xi_{\kappa_0^B}$ positive, implying $\kappa_0^B = 0$ and $\mu = \xi_{\kappa_0^B} q_0 > 0$. Thus, the collateral constraint is also binding and $c_0 = a_0$. In the second case in which $\xi_{\kappa_0^B}$ is zero, μ should also be zero, implying that the collateral constraint is not binding and B purchases some assets without leverage as $a_0 > c_0$.

The result holds because of the linearity of B 's utility. If B holds a positive amount of cash, B should be indifferent between holding more or less cash. However, if the return of purchasing more assets by borrowing more from L at date 0 at the cost of less amount of cash at date 1 exceeds the cash return, B borrows with full capacity and spends all the cash to purchase the asset.

FOC for a_0 , (70), can be further simplified as

$$\begin{aligned}
& - \frac{p_0 a_0 - ((1 - q_0)c_0 - e_0^B + p_0 a_0)}{a_0^2} \left[\left(1 - \pi_1^B \left(\frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \right] \\
& + \int_{\underline{R}/\bar{\tau}}^{1/\bar{\tau}} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} [\tau_1^B(q_1 - p_0) + (1 - \pi_1^B(q_1)) (\bar{R} - 1)] dH(q_1) \\
& + \frac{p_0 a_0 - ((1 - q_0)c_0 - e_0^B + p_0 a_0)}{a_0^2} \left(1 - \pi_1^B \left(\frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \right) \right) a_0 (\bar{R} - 1) \\
& + \int_{\underline{q}_1} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left(\frac{\alpha - p_0}{\alpha - q_1} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\
& + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \tau_1^B(\alpha - p_0) dH(q_1) + \mu + \xi_{a_0} - \xi_{\kappa_0^B} p_0 \\
& = \int_{\underline{R}/\bar{\tau}}^{1/\bar{\tau}} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} [\tau_1^B(q_1 - p_0) + (1 - \pi_1^B(q_1)) (\bar{R} - 1)] dH(q_1) \\
& + \int_{\underline{q}_1} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left(\frac{p_0 - \alpha}{q_1 - \alpha} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\
& + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \tau_1^B(\alpha - p_0) dH(q_1) + \mu + \xi_{a_0} - \xi_{\kappa_0^B} p_0 = 0.
\end{aligned}$$

We finally derive the date-0 equilibrium allocation and prices.

From the FOCs for c_0 and a_0 ,

$$\begin{aligned}
& \int_{\underline{R}/\bar{\tau}}^{1/\bar{\tau}} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} [\tau_1^B(q_1 - p_0) + (1 - \pi_1^B(q_1)) (\bar{R} - 1)] dH(q_1) \\
& + \int_{\underline{q}_1} \frac{(1 - q_0)c_0 - e_0^B + p_0 a_0}{a_0} \left(\frac{p_0 - \alpha}{q_1 - \alpha} \right) (1 - \pi_1^B(q_1)) (\bar{R} - 1) dH(q_1) \\
& + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \tau_1^B(\alpha - p_0) dH(q_1) + \xi_{c_0} + \xi_{a_0} - \xi_{\kappa_0^B} (p_0 - 1) = 0.
\end{aligned}$$

From the borrower's optimality condition under $a_0 = c_0 > 0$, B holds zero amount of cash, $\kappa_0^B = 0$.

Under this case, B 's return from this leveraged asset purchase becomes

$$\begin{aligned} \xi_{\kappa_0^B} = & \left[\int_1^{1/\bar{\tau}} [\tau_1^B(q_1 - p_0) + (1 - \pi_1^B(q_1))(\bar{R} - 1)] dH(q_1) \right. \\ & + \int_{\underline{q}_1}^1 \left(\frac{p_0 - \alpha}{q_1 - \alpha} \right) (1 - \pi_1^B(q_1))(\bar{R} - 1) dH(q_1) \\ & \left. + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \tau_1^B(\alpha - p_0) dH(q_1) \right] / (p_0 - 1). \end{aligned} \quad (71)$$

It is sufficient to show that $\xi_{\kappa_0^B} > \tau_1^B$, so B prefers to purchase the asset with leverage to holding cash.

If B uses up all the cash endowments to purchase the asset with leverage, the asset price is

$$p_0 = \frac{A + e_0^B}{A}.$$

The price should be higher than T 's willingness-to-pay for the asset, which is also above the debt payment amount at date 1, so B is able to repay the debt in full if B wants to liquidate the assets. Therefore,

$$p_0 = \frac{A + e_0^B}{A} > \alpha \geq 1. \quad (72)$$

From assumption 1,

$$\begin{aligned} e_0^B \tau_1^B & < A(E_0^B[R] - 1) \\ \Rightarrow \tau_1^B & < \frac{A}{e_0^B}(E_0^B[R] - 1) = \frac{E_0^B[R] - 1}{p_0 - 1}, \end{aligned} \quad (73)$$

where the last equality comes from (72). Hence, if the marginal return of purchasing the asset with leverage, $\xi_{\kappa_0^B}$, is greater than or equal to $\frac{E_0^B[R] - 1}{p_0 - 1}$, then $a_0 = c_0 = A$ and $\kappa_0^B = 0$ is the equilibrium portfolio of B . The difference between the two returns is

$$\begin{aligned} \xi_{\kappa_0^B} - (E_0^B[R] - 1) = & \int_1^{1/\bar{\tau}} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1) \\ & + \int_{\underline{q}_1}^1 \left[\pi_1^B(q_1)(1 - \underline{R}) - \left(\frac{p_0 - q_1}{\alpha - q_1} \right) (1 - \pi_1^B(q_1))(\bar{R} - 1) \right] dH(q_1) \\ & + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(\alpha - p_0) - (1 - \pi_1^B(q_1))(\bar{R} - 1)] dH(q_1). \end{aligned} \quad (74)$$

We will show that this difference is positive. Because $\frac{(1 - \pi_1^B(q))(\bar{R} - 1)}{\tau_1^B} + q$ is increasing in q and $\alpha = \frac{(1 - \pi_1^B(\underline{q}_1))(\bar{R} - 1)}{\tau_1^B} + \underline{q}_1 \geq 1$,

$$\alpha - q < \frac{(1 - \pi_1^B(q))(\bar{R} - 1)}{\tau_1^B}$$

for any $q \in [\underline{q}_1, 1]$. Thus, we have

$$\begin{aligned} & \int_{\underline{q}_1}^1 \left[\pi_1^B(q_1)(1 - \underline{R}) - \left(\frac{p_0 - q_1}{\alpha - q_1} \right) (1 - \pi_1^B(q_1))(\bar{R} - 1) \right] dH(q_1) \\ & > \int_{\underline{q}_1}^1 \left[\pi_1^B(q_1)(1 - \underline{R}) - \left(\tau_1^B \frac{p_0 - q_1}{(1 - \pi_1^B(q_1))(\bar{R} - 1)} \right) (1 - \pi_1^B(q_1))(\bar{R} - 1) \right] dH(q_1) \\ & = \int_{\underline{q}_1}^1 [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1). \end{aligned} \quad (75)$$

Again, because $\alpha = \frac{(1 - \pi_1^B(\underline{q}_1))(\bar{R} - 1)}{\tau_1^B} + \underline{q}_1$ and $(1 - \pi_1^B(q))$ is increasing in q ,

$$\begin{aligned} & \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(\alpha - p_0) - (1 - \pi_1^B(q_1))(\bar{R} - 1)] dH(q_1) \\ & > \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} \left[\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B \left(\frac{(1 - \pi_1^B(\underline{q}_1))(\bar{R} - 1)}{\tau_1^B} + \underline{q}_1 - p_0 \right) - (1 - \pi_1^B(\underline{q}_1))(\bar{R} - 1) \right] dH(q_1) \\ & > \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1). \end{aligned} \quad (76)$$

Combining (74), (75), and (76) implies

$$\begin{aligned} \xi_{\kappa_0^B} - (E_0^B[R] - 1) & > \int_1^{1/\bar{\tau}} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1) \\ & \quad + \int_{\underline{q}_1}^1 [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1) \\ & \quad + \int_{\underline{R}/\bar{\tau}}^{\underline{q}_1} [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B(q_1 - p_0)] dH(q_1) \\ & = \int [\pi_1^B(q_1)(1 - \underline{R}) + \tau_1^B q_1 - \tau_1^B p_0] dH(q_1) \\ & = \pi_0(1 - \underline{R}) + \tau_1^B E_0^B[q_1] - \tau_1^B p_0. \end{aligned} \quad (77)$$

Recall that $q_1 = \frac{(1 - \pi_1^L) + \pi_1^L R}{\tau_1^L}$ from (1), which implies

$$E_0^B[q_1] = E_0^B \left[\frac{(1 - \pi_1^L) + \pi_1^L R}{\tau_1^L} \right] = E_0^B [(1 - \pi_1^L) + \pi_1^L R] E_0^B \left[\frac{1}{\tau_1^L} \right] = [(1 - \pi_0) + \pi_1^0 R] E_0^B \left[\frac{1}{\tau_1^L} \right],$$

where the last two equalities hold because of the independence between τ_1^L and s_1^L and the law of iterated expectations, respectively. Because $f(x) = 1/x$ is a convex function,

$$E_0^B \left[\frac{1}{\tau_1^L} \right] > \frac{1}{E_0^B [\tau_1^L]} = \frac{1}{\tau_0^L}$$

by Jensen's inequality. Finally, from (73), $E_0^B[R] > \tau_1^B p_0$.

Therefore, (77) becomes

$$\begin{aligned} \xi_{\kappa_0^B} - (E_0^B[R] - 1) &> \pi_0(1 - \underline{R}) + \tau_1^B E_0^B[q_1] - \tau_1^B p_0 \\ &> \pi_0(1 - \underline{R}) + (1 - \pi_0) + \pi_0 \underline{R} - (1 - \pi_0) \bar{R} - \pi_0 \underline{R} = 0. \end{aligned}$$

Therefore, the return from investing in the asset with leverage $\xi_{\kappa_0^B}$ is greater than the return from cash holdings τ_1^B . Finally, this also shows that the equilibrium with $\kappa_0^B = 0$ is the only equilibrium. If $\kappa_0^B > 0$, then the price of the asset p_0 will be even lower, increasing the asset return even higher.

■

D.4 Counterfactual on learning

To understand how learning interacts with funding and market liquidity, we first compare the response of our baseline economy to shocks to the response of an alternative benchmark economy in which beliefs are stale—that is, we assume agents' date-0 beliefs are never updated in response to new information. We measure the severity of a fire sale by the extent to which the borrower fire sells its holdings of the risky asset to the traditional sector: the greater (and, equivalently, the lower the asset price p_1), the more severe the fire sale at date 1. This result is summarized in lemma 6 below.

Lemma 6 (Learning amplifies the severity of fire sales). *For any set of shocks (π_1^L, τ_1^L) such that the equilibrium is in the fire sale regime at date 1, the price of the risky asset p_1 is lower and the extent of misallocation of the asset a_1^T is higher in equilibrium compared to an alternative benchmark economy in which agents' posterior beliefs are exogenously set to their date-0 priors.*

Proof. We show how learning amplifies misallocation during a fire sale for any q_1 (i.e. shock pair (π_1^L, τ_1^L)), relative to a benchmark in which beliefs don't change—that is, agents have private

information but do not update their beliefs in response to new information. Note that in the fire sale regime, we have

$$a_1^B = a_0^B - F'^{-1} \left(\frac{E_1^B[R] - E_1^B[R_2^d]}{\tau_1^B} + \frac{E_1^L[R_2^d]}{\tau_1^L} \right) \quad (78)$$

$$p_1 = F'(a_1^T)$$

Since $F'(\cdot)$ is monotonically decreasing, $F'^{-1}(\cdot)$ is also monotonically decreasing. Therefore, holding the borrower's beliefs constant for a moment, a lower $q_1 = \frac{E_1^L[R_2^d]}{\tau_1^L}$ implies that a_1^B is lower. Note that the spread $E_1^B[R] - E_1^B[R_2^d] = (1 - \pi_1^B)(\bar{R} - 1)$ is decreasing in π_1^B (since $\bar{R} > 1$ by assumption). A lower q_1 implies that π_1^B is higher (B is more pessimistic) by lemma 1, which also makes a_1^B lower. Thus, a lower q_1 leads to a lower a_1^B in two ways: the direct effect of a lower q_1 on a_1^B and the effect of lower q_1 on a_1^B through greater pessimism and a lower spread.

Since a larger decrease in funding liquidity leads to lower a_1^B , this also implies that a_1^T is higher and hence p_1 is lower. Hence, the fire sale is more severe in the fire sale regime when the funding illiquidity is more severe. Hence, pessimism/the information externality is associated with greater misallocation/worse fire sales during a fire sale. ■

We showed in section 3.3 that pessimism and optimism on the fundamental value of the asset arises endogenously in our model as a result of funding liquidity. Lemma 6 above shows that this increased pessimism contributes to market illiquidity, by depressing asset prices and making it more costly to raise funds through liquidation. Therefore, the model yields an adverse feedback between pessimistic beliefs about fundamentals, funding liquidity, and market illiquidity—a dynamic, which exacerbates crises.

D.5 Date-1 equilibrium under common information benchmark

In the normal regime under the Common Information Benchmark we have

$$a_1^T, \xi_{d_1}^L, \xi_{c_1}^B, \xi_{a_1}^B, \xi_{\kappa_1}^L, \xi_{\kappa_1}^B = 0$$

$$q_1 = \frac{E_1[R_2^d(f_1)]}{\tau_1^L}$$

$$p_1 = \frac{E_1[R] - E_1[R_2^d(f_1)]}{\tau_1^B} + q_1$$

$$a_1^B = c_1^B = d_1^L = a_0^B$$

$$\kappa_1^B = q_1 a_0^B - c_0^B + \kappa_0^B$$

$$\kappa_1^L = \kappa_0^L + d_0^L - q_1 a_0^B$$

$$\kappa_1^T = \kappa_0^T - p_1 a_1^T$$

$$C_2^B = \tau_1^B \kappa_1^B + a_1^B R - a_1^B R_2^d(f_1)$$

$$C_2^L = \tau_1^L \kappa_1^L + R_2^d d_1^L$$

$$C_2^T = \kappa_1^T + F(a_1^T)$$

$$\mu_1^B = (\tau_1^B + \xi_{\kappa_1}^B) p_1 - E_1[R] - \xi_{a_1}^B$$

$$\mu_1^T = F'(0) - p_1$$

$$\pi_1 := \pi_1^L = Pr(R = \underline{R} | s_1^L, I_0) = \frac{\pi_0 \lambda_\epsilon (\epsilon_1^L = s_1^L - \underline{R})}{(1 - \pi_0) \lambda_\epsilon (\epsilon_1^L = s_1^L - \overline{R}) + \pi_0 \lambda_\epsilon (\epsilon_1^L = s_1^L - \underline{R})}$$

$$\pi_1^B = \pi_1.$$

In the fire sale regime under the Common Information Benchmark, we have

$$\mu_1^T, \xi_{c_1}^B, \xi_{a_1}^B, \xi_{d_1}^L, \xi_{\kappa_1}^L, \xi_{\kappa_1}^B = 0$$

$$q_1 = \frac{E_1[R_2^d(f_1)]}{\tau_1^L}$$

$$p_1 = F'(a_1^T)$$

$$a_1^B = a_0^B - F'^{-1} \left(\frac{E_1[R] - E_1[R_2^d(f_1)]}{\tau_1^B} + \frac{E_1[R_2^d(f_1)]}{\tau_1^L} \right)$$

$$a_1^T = a_0^B - a_1^B$$

$$c_1^B = d_1^L = a_1^B$$

$$\kappa_1^B = q_1 a_1^B - c_0^B + \kappa_0^B - p_1 (a_1^B - a_0^B)$$

$$\kappa_1^L = d_0^L - q_1 d_1^L + \kappa_0^L$$

$$\kappa_1^T = \kappa_0^T - p_1 a_1^T$$

$$C_2^B = \tau_1^B \kappa_1^B + a_1^B R - a_1^B R_2^d(f_1)$$

$$C_2^L = \tau_1^L \kappa_1^L + R_2^d d_1^L$$

$$C_2^T = \kappa_1^T + F(a_1^T)$$

$$\mu_1^B = \tau_1^B p_1 - E_1[R]$$

$$\pi_1 := \pi_1^L = \Pr(R = \underline{R} | s_1^L, I_0) = \frac{\pi_0 \lambda_\epsilon (\epsilon_1^L = s_1^L - \underline{R})}{(1 - \pi_0) \lambda_\epsilon (\epsilon_1^L = s_1^L - \bar{R}) + \pi_0 \lambda_\epsilon (\epsilon_1^L = s_1^L - \underline{R})}$$

$$\pi_1^B = \pi_1.$$

D.6 Defining the partition in the common information benchmark

The following lemma shows how this frontier compares to that under the Baseline Economy.

Lemma 7 (Frontier in the Common Information Benchmark). *The frontier $(\tau_1^L, \pi_1^L) | \tilde{p}_1$ in the Common Information Benchmark is strictly convex and has a negative slope in the domain for τ_1^L . Moreover, the y-intercept $\frac{1}{1-\bar{R}}$ of the frontier $(\tau_1^L, \pi_1^L) | \tilde{p}_1$ is the same as the y-intercept for*

the frontier $(\tau_1^L, \pi_1^L) | \hat{p}_1$ in the Baseline Economy, and the x -intercept $\frac{\tau_1^B}{\tau_1^B F'(0) - \bar{R} + 1}$ in the Common Information Benchmark exceeds that in the Baseline Economy $\frac{1}{\hat{q}_1}$.

Proof. See appendix D.7. ■

For the given $\hat{q}_1$ of the frontier in the Baseline Case, there exists a point $\hat{\tau}_1^L$ such that the corresponding $\pi_1^L = 1$ as

$$\pi_1^L = 1 = \frac{1 - \hat{\tau}_1^L \hat{q}_1}{1 - \underline{R}}.$$

Similarly, there exists a point $\tilde{\tau}_1^L$ such that the corresponding $\pi_1^L = 1$ as

$$\pi_1^L = 1 = \frac{\tau_1^B f_1 + \tilde{\tau}_1^L (\bar{R} - f_1 - \tau_1^B F'(0))}{\tilde{\tau}_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})}.$$

Some of the results, which follow, depend on whether the frontier partitioning the state space into the normal and fire sale regimes in the baseline economy intersects the frontier in the common information benchmark. We define the following condition related to the y -intercepts:

Condition 1.

$$\hat{\tau}_1^L > \tilde{\tau}_1^L \tag{79}$$

If condition 1, which is $\hat{\tau}_1^L > \tilde{\tau}_1^L$ holds, then the frontier in the baseline economy is above the frontier under the common information benchmark up to some point $(\tau_1^{L*}, \pi_1^{L*})$ such that $\pi_1^{L*} < 1$, where the two frontiers meet. The following lemma shows that condition 1 is necessary and sufficient for the frontiers in the baseline and benchmark to meet in the relevant domain, and that this condition always holds in equilibrium.

Lemma 8.

- A. Condition 1 is a necessary and sufficient condition for the frontiers in the baseline case and the common information benchmark to meet in the domain such that $\pi_1^L \leq 1$.
- B. Condition 1 always hold in equilibrium.

Proof. See appendix D.8. ■ Condition 1 implies that the frontier in the baseline case starts declining linearly at a point $(\hat{\tau}_1^L, 1)$, whereas the frontier in the common information benchmark goes through the point $(\tilde{\tau}_1^L, 1)$. Because both frontiers started at the same y -intercept, the average slope of the frontier in the common information benchmark is above that in the baseline case.

Because the slope of the frontier in the common information benchmark is increasing (by strict convexity shown in lemma 7), the frontier in the baseline economy is always below the frontier in the common information benchmark. The proof shows that this condition is always satisfied in equilibrium.

D.7 Proof of lemma 7

Here we characterize the threshold between the normal and fire sale regimes in the Common Information Benchmark.

Proof. Frontier in Baseline Economy

Recall the equilibrium asset price.

$$p_1 = \frac{E_1^B[R] - E_1^B[R_2^d(f_1)]}{\tau_1^B} + q_1.$$

The threshold price is defined by asset price at threshold of Normal and fire sale regimes. This threshold asset price is $\hat{p}_1 = F'(0)$ satisfying

$$\begin{aligned} \frac{E_1^B[R] - E_1^B[R_2^d(f_1)]}{\tau_1^B} + \hat{q}_1 &= F'(0) \\ \hat{q}_1 + \frac{(1 - \pi_1^B(\hat{q}_1))(\bar{R} - f_1)}{\tau_1^B} &= F'(0) \\ (1 - \pi_1^B(\hat{q}_1))(\bar{R} - f_1) + \tau_1^B \hat{q}_1 &= \tau_1^B F'(0). \end{aligned} \tag{80}$$

This defines the threshold value $\hat{q}_1$. Given this, the set of (τ_1^L, π_1^L) consistent with $\hat{q}_1$ is given by

$$\frac{(1 - \pi_1^L)f_1 + \pi_1^L \bar{R}}{\tau_1^L} = \hat{q}_1.$$

Solve for π_1^L :

$$\pi_1^L = \frac{f_1 - \tau_1^L \hat{q}_1}{f_1 - \bar{R}} = \frac{1 - \tau_1^L \hat{q}_1}{1 - \bar{R}}.$$

This defines the curve of the frontier partitioning the state space into the normal and fire sale regimes.

The y-intercept of the curve (when $\tau_1^L = 0$, though never occurs) is $\frac{1}{1 - \bar{R}} > 1$, while the x-intercept (when $\pi_1^L = 0$) is $\frac{1}{\hat{q}_1} > 0$.

The slope of the curve is negative and constant:

$$\frac{d\pi_1^L}{d\tau_1^L} = \frac{-\hat{q}_1}{1 - \bar{R}} < 0.$$

Frontier in Common Information Benchmark

When $\pi_1^B = \pi_1^L$, the threshold in benchmark is defined by

$$(1 - \pi_1^L) (\bar{R} - f_1) + \tau_1^B \hat{q}_1 = \tau_1^B F'(0)$$

i.e.

$$(\bar{R} - f_1) - \left[(\bar{R} - f_1) + \frac{\tau_1^B}{\tau_1^L} (f_1 - \bar{R}) \right] \pi_1^L + \frac{\tau_1^B}{\tau_1^L} f_1 = \tau_1^B F'(0) \quad (81)$$

We now trace out the frontier of all (τ_1^L, π_1^L) such that this is satisfied. First, we derive the x-intercept of the frontier by supposing that $\pi_1^L = 0$:

$$\begin{aligned} (\bar{R} - f_1) + \frac{\tau_1^B}{\tau_1^L} f_1 &= \tau_1^B F'(0) \\ (\bar{R} - f_1) \tau_1^L + \tau_1^B f_1 &= \tau_1^L \tau_1^B F'(0) \\ \tau_1^B f_1 &= \tau_1^L (\tau_1^B F'(0) - \bar{R} + f_1) \\ \tau_1^L &= \frac{\tau_1^B f_1}{\tau_1^B F'(0) - \bar{R} + f_1}. \end{aligned}$$

Claim 1. *This x-intercept is greater than the x-intercept in the Baseline case.*

Proof. We want to show that the x-intercept of the frontier in the Common Info benchmark is larger than that in the Baseline economy, i.e.

$$\begin{aligned} \frac{\tau_1^B f_1}{\tau_1^B F'(0) - \bar{R} + f_1} &> \frac{f_1}{\hat{q}_1} \\ \hat{q}_1 &> \frac{\tau_1^B F'(0) - (\bar{R} - f_1)}{\tau_1^B} \end{aligned}$$

where $\hat{q}_1$ is defined by (80) in the Baseline case. Recall that $\hat{q}_1$ is defined by

$$\hat{q}_1 = F'(0) - \frac{(1 - \pi_1^B(\hat{q}_1)) (\bar{R} - f_1)}{\tau_1^B}$$

So we want to show that

$$F'(0) - \frac{(1 - \pi_1^B(\hat{q}_1)) (\bar{R} - f_1)}{\tau_1^B} > \frac{\tau_1^B F'(0) - (\bar{R} - f_1)}{\tau_1^B}.$$

The inequality is equivalent to

$$\begin{aligned} F'(0) - \frac{(1 - \pi_1^B(\hat{q}_1)) (\bar{R} - f_1)}{\tau_1^B} &> F'(0) - \frac{(\bar{R} - f_1)}{\tau_1^B} \\ (1 - \pi_1^B(\hat{q}_1)) (\bar{R} - f_1) &< (\bar{R} - f_1) \\ (1 - \pi_1^B(\hat{q}_1)) &< 1, \end{aligned}$$

which holds because $\pi_1^B > 0$. ■ Hence, we have that the x-intercept is larger in the Common Info Benchmark compared to the Baseline case. (Note that this automatically implies that the x-intercept is positive, since it's obviously positive in the Baseline case.)

What is the y-intercept of the frontier in the Common Info Benchmark? (i.e. what happens to π_1^L as τ_1^L approaches zero?): The frontier equation (81) can be rearranged as

$$\begin{aligned} (\bar{R} - f_1) - \left[(\bar{R} - f_1) + \frac{\tau_1^B}{\tau_1^L} (f_1 - \underline{R}) \right] \pi_1^L + \frac{\tau_1^B}{\tau_1^L} f_1 &= \tau_1^B F'(0) \\ - \left[(\bar{R} - f_1) + \frac{\tau_1^B}{\tau_1^L} (f_1 - \underline{R}) \right] \pi_1^L &= \tau_1^B F'(0) - \frac{\tau_1^B}{\tau_1^L} f_1 - (\bar{R} - f_1) \\ \pi_1^L &= \frac{\frac{\tau_1^B}{\tau_1^L} f_1 + (\bar{R} - f_1) - \tau_1^B F'(0)}{(\bar{R} - f_1) + \frac{\tau_1^B}{\tau_1^L} (f_1 - \underline{R})} \\ \pi_1^L &= \frac{\tau_1^B f_1 + \tau_1^L (\bar{R} - f_1) - \tau_1^L \tau_1^B F'(0)}{\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})} \\ \pi_1^L &= \frac{\tau_1^B f_1 + \tau_1^L (\bar{R} - f_1 - \tau_1^B F'(0))}{\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})}. \end{aligned}$$

The y-intercept (when $\tau_1^L = 0$) is $\frac{f_1}{f_1 - \underline{R}}$. Hence, this is the same y-intercept as in the Baseline case.

The slope of this frontier is defined by taking the derivative of π_1^L with respect to τ_1^L :

$$\begin{aligned} \frac{\partial \pi_1^L}{\partial \tau_1^L} &= \frac{\partial \left(\frac{\tau_1^B f_1 + \tau_1^L (\bar{R} - f_1 - \tau_1^B F'(0))}{\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})} \right)}{\partial \tau_1^L} \\ &= \frac{(\bar{R} - f_1 - \tau_1^B F'(0)) (\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})) - (\bar{R} - f_1) (\tau_1^B f_1 + \tau_1^L (\bar{R} - f_1 - \tau_1^B F'(0)))}{(\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R}))^2} \\ &= \frac{(\bar{R} - f_1 - \tau_1^B F'(0)) (\tau_1^B (f_1 - \underline{R})) - (\bar{R} - f_1) (\tau_1^B f_1)}{(\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R}))^2} < 0, \end{aligned}$$

which holds because $(\bar{R} - f_1 - \tau_1^B F'(0)) < (\bar{R} - f_1)$ and $\tau_1^B f_1 > \tau_1^B (f_1 - \underline{R})$. Denote $\Psi \equiv (\bar{R} - f_1 - \tau_1^B F'(0)) (\tau_1^B (f_1 - \underline{R})) - (\bar{R} - f_1) (\tau_1^B f_1)$. Also, the second derivative becomes

$$\frac{\partial^2 \pi_1^L}{\partial (\tau_1^L)^2} = \frac{-2 (\bar{R} - f_1) (\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R})) \Psi}{(\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R}))^4} > 0,$$

where the last inequality holds by $\bar{R} > f_1 > \underline{R}$ and $\Psi < 0$. Therefore, the frontier is strictly convex with a negative slope (in our relevant domain). ■

D.8 Proof of condition for intersection of the frontiers

Using the results from the proof of lemma 7 in appendix D.7, we derive the necessary and sufficient condition for the existence of the intersection of the frontiers in the Baseline Economy and the Common Information Benchmark.

Proof. Proof of Part (A): In the Baseline Economy, the slope is always $\frac{d\pi_1^L}{d\tau_1^L} = \frac{-\hat{q}_1}{f_1 - \underline{R}} < 0$. In the Common Info Benchmark, the slope is

$$\frac{(\bar{R} - f_1 - \tau_1^B F'(0)) (\tau_1^B (f_1 - \underline{R})) - (\bar{R} - f_1) (\tau_1^B f_1)}{(\tau_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - \underline{R}))^2}$$

which is increasing as shown in lemma 7. Therefore, once the frontier in the Common Information Benchmark is at the same point $(\tau_1^{L*}, \pi_1^{L*})$ or above that point as (τ_1^{L*}, π_1^L) with $\pi_1^L \geq \pi_1^{L*}$ and has the same slope as the frontier in the Baseline Case at τ_1^{L*} , the two frontiers would never meet for any $\tau_1^L > \tau_1^{L*}$ (see figure 5 for a graphical example).

First, suppose that $\hat{\tau}_1^L \leq \tilde{\tau}_1^L$. We claim that the two frontiers will never meet for $\pi_1^L \leq 1$. Since the two frontiers have the same y-intercept, $\frac{1}{1 - \underline{R}}$, the average slope of the frontier in the Common Information Benchmark within the interval $[0, \hat{\tau}_1^L]$ has to be greater than that of the frontier in the Baseline Case within the same interval. Because of strict convexity, the slope of the frontier in the Common Information Benchmark at $\hat{\tau}_1^L$ has to be greater than that of the frontier in the Baseline Case as well. Thus, the two frontiers will never meet for the states with $\pi_1^L \leq 1$.

Now suppose that the two frontiers never meet for the states with $\pi_1^L \leq 1$. We claim that $\hat{\tau}_1^L \leq \tilde{\tau}_1^L$ should hold and prove this by contradiction. Suppose the contrary, $\hat{\tau}_1^L > \tilde{\tau}_1^L$. Because the x-intercept in the Common Information Benchmark is greater than the x-intercept in the Baseline case, the frontier in the Baseline case continuously goes from $\pi_1^L = 1$ to $\pi_1^L = 0$ within an interval that is contained in the interval between $\tilde{\tau}_1^L$ and the x-intercept in the Common Information Benchmark. Since the frontier is continuously decreasing and convex, there exists a point that the two frontiers meet by the intermediate value theorem.

Proof of Part (B): Recall from appendix D.6 that the intercept $\hat{\tau}_1^L$ is defined by $\pi_1^L = 1 = \frac{1 - \hat{\tau}_1^L \hat{q}_1}{1 - R}$, which we can rewrite as $\hat{\tau}_1^L = \frac{R}{\hat{q}_1}$. Also, the intercept $\tilde{\tau}_1^L$ is defined by $\pi_1^L = 1 = \frac{\tau_1^B f_1 + \tilde{\tau}_1^L (\bar{R} - f_1 - \tau_1^B F'(0))}{\tilde{\tau}_1^L (\bar{R} - f_1) + \tau_1^B (f_1 - R)}$, which we can rewrite as $\tilde{\tau}_1^L = \frac{R}{F'(0)}$. Therefore, condition 1, $\hat{\tau}_1^L > \tilde{\tau}_1^L$, holds if and only if $F'(0) > \hat{q}_1$. We showed in section 4.2 that $F'(0) = \hat{p}_1$, where $\hat{p}_1$ is the threshold asset price below which the fire sale regime obtains at date 1. Therefore, condition 1 is identical to $\hat{p}_1 > \hat{q}_1$. Recall also from lemma 4 that, for each realization of shocks (τ_1^L, s_1^L) , we have $q_1 < p_1$ in equilibrium. This implies that $\hat{p}_1 > \hat{q}_1$, and hence, condition 1 always holds in equilibrium. ■